A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions

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We present a complete set of rewrite rules for multi-qudit Clifford circuits in all prime dimensions. Completeness implies that any two Clifford circuits representing the same linear map can be rewritten into each other using these rules. There are 19 rewrite rules in total, each involving no more than three qudits and admitting an intuitive interpretation. We use the isomorphism between the symplectic group $\mathrm{Sp}(2n, \mathbb{Z}_d)$ and the quotient of the Clifford group by the Pauli group to first derive a complete set of symplectic relations for $\mathrm{Sp}(2n, \mathbb{Z}_d)$. We then lift these to Clifford relations by incorporating Pauli corrections. The derivation uses a circuit normal form that captures the stabiliser tableau of a Clifford operator and is unique up to Pauli correction. The reduction of a circuit to this symplectic normal form requires 66 relations, which we then reduce to 18 symplectic relations. Our computations in $\mathrm{Sp}(2n, \mathbb{Z}_d)$ are formalised in the Agda proof assistant, providing a machine-verified proof of correctness.

Contents

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1 Introduction

Many physical platforms for qubits naturally possess higher energy levels that can encode qudits, enabling greater information density at the cost of increased control complexity [19, 26, 49, 50, 54]. In particular, when d is an odd prime, qudit systems exhibit rich algebraic structures that result in unique error correction capabilities [16, 52], lower overhead magic state distillation [17, 48], stronger quantum correlations [15, 22], and improvements to various quantum algorithms [11, 45]. Ultimately, there is a trade-off between the engineering cost of accessing higher dimensions and the computational advantages they provide. By characterising qudit circuits in a systematic manner, we can better understand this balance and exploit the potential of higher-dimensional quantum information processing.

Recently, algebraic approaches to quantum circuits have gained significant attention. An important problem in this line of work is to develop sound and complete equational theories for circuit families, meaning any two circuits representing the same linear map can be transformed into each other using a finite set of rules. Such presentations enable syntactic reasoning about circuits and underpin applications ranging from circuit optimisation to formal verification [14, 18, 33, 35, 56, 57]. While there has been substantial progress in the qubit setting [2, 3, 7–9, 29, 40, 41, 51], analogous results for qudit circuits remain comparatively underdeveloped.

Among quantum circuit fragments, the Clifford group plays a particularly prominent role. Clifford unitaries are central to the stabiliser formalism [1, 24], fault-tolerant quantum computing [27, 28, 30], efficient classical simulation [31, 43], and practical circuit optimisation techniques [13, 14]. As a result, obtaining a complete and finite equational theory for qudit Clifford circuits is a natural and foundational problem.

Note that there is a complete equational theory for all (including non-unitary) qudit Clifford maps in the form of the qudit Clifford ZX-calculus [12, 46, 53]. One might hope that we could easily adapt this into a complete calculus for unitary circuits, just by restricting the allowed maps. However, as ZX-diagrams represent general linear maps this does not work in any straightforward way. In general, completeness of a graphical calculus for a broad class of linear maps does not yield a calculus restricted to unitary maps. Consider for instance that a complete ZX-calculus for universal qubit linear maps was found in 2017 [34, 44], but that it was not until 2022 that a complete calculus of universal quantum circuits appeared (which used very different methods) [20]. Moreover, there is no complete calculus for, for instance, Toffoli-Hadamard circuits, even though there is one for the corresponding set of linear maps [5].

1.1 Main Results

Building on the characterisation of the qutrit Clifford group [39], where a qutrit is a three-dimensional ($d = 3$) qudit, we present a complete set of rewrite rules for multi-qudit Clifford circuits in all odd prime dimensions. There are 19 *Clifford relations* in total, each involving at most three qudits and admitting an intuitive interpretation.

Theorem 4.10. *The rewrite rules in Figure 1 are* complete *for n-qudit Clifford circuits over any qudit dimension d that is an odd prime. That is, if two n-qudit unitary Clifford circuits C_1 and C_2 implement the same linear map, then the rewrite rules in Figure 1 can be used to derive their equality.*

To prove Theorem 4.10, we introduce a symplectic normal form that corresponds to the stabiliser tableau of a Clifford operator up to Pauli corrections. This normal form is substantially easier to work with compared to the more involved normal form for qutrit Clifford operators. We then find a set of 66 parametrised relations which suffice to reduce any Clifford circuit to this normal form. Finally, we show how these relations can be further reduced to a small set of 19 Clifford relations in Figure 1. In addition to the Clifford generators H, S and CZ, we also use "derived generators" listed in Figure 2, such as SWAP, CX, and the *multiplier* M_g. M_g is defined by its action on the computational basis states as $M_g|x\rangle = |gx\rangle$, where multiplication is over $\mathbb{Z}_d$. Using these derived generators, we can present the rewrite rules as intuitive gate-level interactions, mostly in the form of commutation and quasi-commutation rules.

Interpretation of the rewrite rules The rewrite rules of Figure 1 mostly correspond to equations that should look familiar, at least for those equations involving at least 2 qudits. For three qudits we only have equations corresponding to commuting a CZ gate through another CZ, CX or set of SWAPs, together with the Reidemeister-style equation

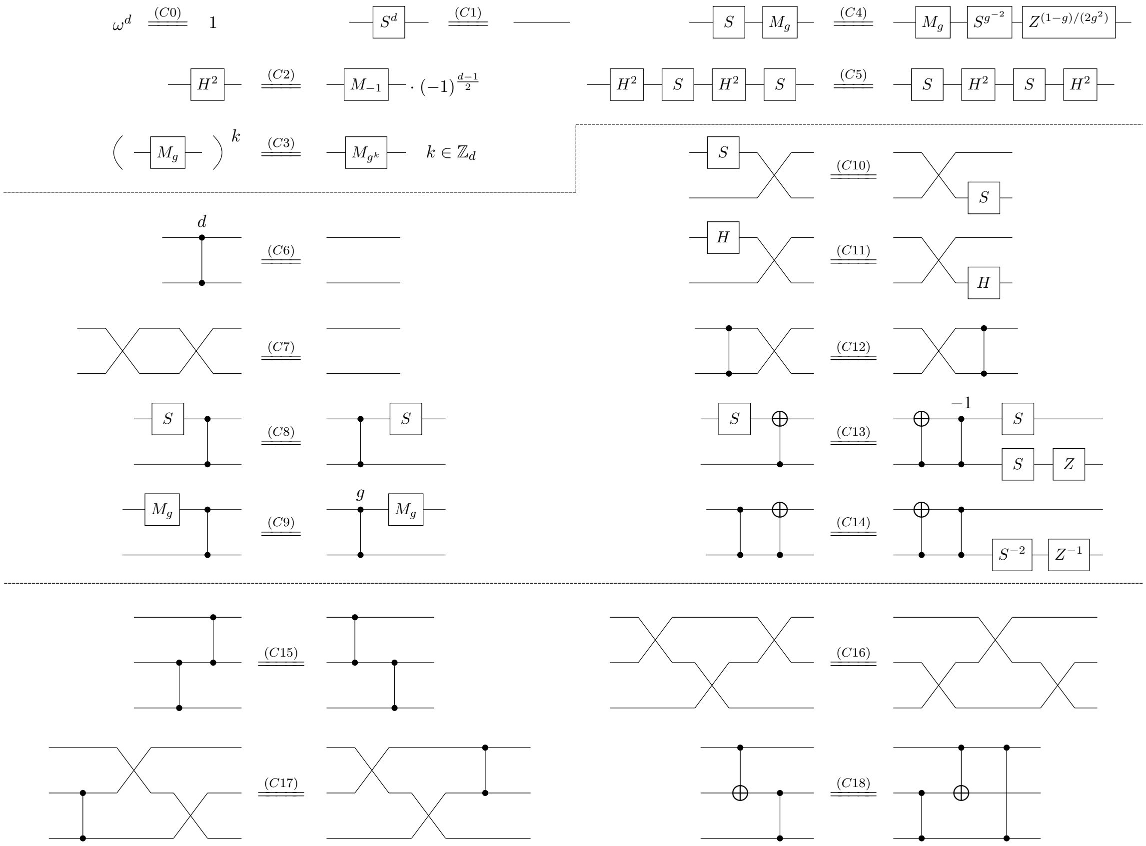


Figure 1: A complete set of rewrite rules for n-qudit Clifford circuits where $\omega = e^{2\pi i/d}$, d is an odd prime, and $n \in \mathbb{N}$. Derived generators such as M_g, X, Z, SWAP, and CX are defined in Figure 2.

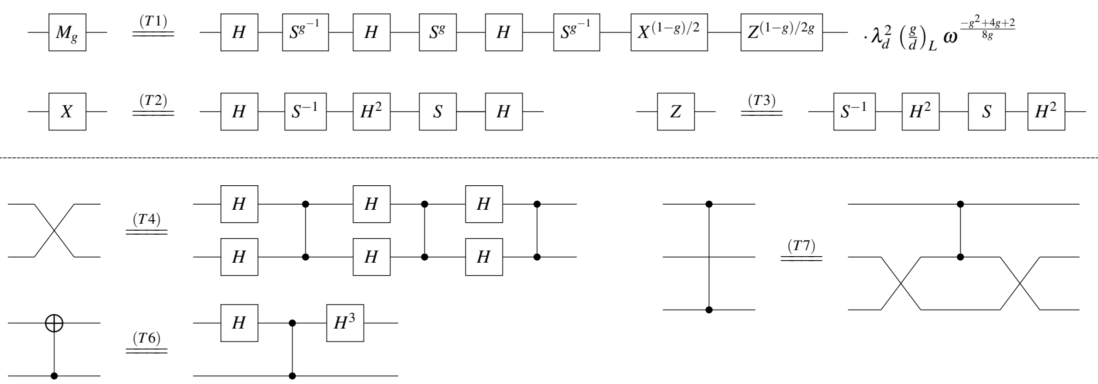


Figure 2: A subset of the derived Clifford generators, expressed as circuits over the generating set $\{H, S, \mathsf{CZ}\}$. These generators are formally defined in Figure 4. Here, g is a chosen generator of the multiplicative group $\mathbb{Z}_d^*$, $\lambda_d = (-i)^{\frac{d-1}{2}}$, and $\left(\frac{g}{d}\right)_L$ denotes the Legendre symbol in number theory [47, Appendix A].

(C16). For two qudits we again have the rules for commuting generators through the SWAP (C10)-(C12), the order of the CZ gate (C6) and SWAP (C7) and some more commutation relations for some of the generators (C8),(C9),(C13),(C14).

The equations for a single qudit are perhaps less familiar. (C0) and (C1) are simply the order of the Clifford phase ω and S gate. The equation of (C2), which relates H^2 to M_{-1} can be used to derive an Euler decomposition of the Hadamard that relates it to a sequence of Z and X rotations. Equation (C3) relates doing iterated multiplication with a multiplier to doing the multiplication in one go. This rule could be replaced by an equation $M_a M_b = M_{ab}$ for all $a, b \in \mathbb{Z}_d^*$ which would give us $(d-1)^2$ equations instead of just the d equations that (C3) represents. In particular, all equations involving g in Figure 1, (C4) and (C9), continue to hold if we replace g by any element of $\mathbb{Z}_d^*$, and so we could equally state these rules without choosing a g, at the cost of having more rules that are families of equations instead of a single equation. Finally, note that (C5) states that the gate $H^2; S; H^2$ commutes with S. Such a rule was also needed for qutrits, and was shown to be necessary in that setting [10].

Some interesting observations follow from these rewrite rules. First, there is no complicated dependency on the dimension of the qudit d. The only dependencies are in the order of the S and CZ gates, and the calculation of modular inverses in (C4) and (C14). One might have for instance expected more dependencies on $d \mod 4$, such as in the scalar factor in the definition of M_g in Figure 2. Second, the rewrites only act on three qudits at most. This number does not increase with d or n. This is in contrast to the rewrite rules for universal qubit circuits where rewrite rules on an arbitrary number of qubits must be included [20].

Technical Highlights Although the qutrit case [39] is contained in the all-odd-prime statement, our proof is not obtained by simply extending a dimension-specific analysis of $d = 3$. Instead, the argument is dimension-independent from the outset: the normal forms, box relations, and rewriting procedures are formulated parametrically over $\mathbb{Z}_d$, for an arbitrary odd prime d. This creates technical difficulties that are absent from the qutrit case.

When $d = 3$, the algebraic properties of the field $\mathbb{Z}_3$ simplify many calculations. For example, there are only two non-zero elements, each of which is its own inverse. Additionally, global phases can be handled by direct numerical calculation. Moreover, there is only one non-identity Clifford multiplier, $H^2 = M_{-1}$ up to a global phase. For arbitrary odd prime d, these coincidences disappear, so the rewrite system must keep track of field parameters, inverses, multipliers, and Pauli corrections throughout the normalisation procedure. A key challenge is therefore to present the large family of relations compactly, for instance through parametrised equations. This keeps the search space manageable and allows the relations to be compressed into a small rewrite system.

This compact parametrisation also enables the proof of completeness to be checked systematically. All computations in the symplectic group $\mathrm{Sp}(2n, \mathbb{Z}_d)$ are formalised in the Agda proof assistant, providing a machine-verified proof of correctness. The corresponding code is available in the GitHub repository [6]. Our results not only bridge group theory with syntactic quantum circuit reasoning, but also lay the foundation for formal verification of quantum protocols in qudit architectures, and automated circuit optimisation and synthesis.

1.2 Proof Overview

A qudit is a d-dimensional quantum system. Here we consider the case in which d is an odd prime. In what follows, we define unitary operators by their actions on the computational basis states. The single-qudit Pauli operators are defined as $X|j\rangle = |j+1\rangle$, $Z|j\rangle = \omega^j|j\rangle$, where $\omega = e^{2\pi i/d}$ and addition is over $\mathbb{Z}_d$. The n-qudit Pauli group $\mathcal{P}_n$ is generated by tensor products of X and Z, together with phases ω^t for $t \in \mathbb{Z}_d$. The qudit H, S, *and* CZ *gates* [23, 25, 27, 37, 38, 47] are defined below, where $i = e^{2\pi i/4}$ is the imaginary unit. The choice of global phases ensures that each matrix determinant is equal to 1.

$$H : |j\rangle \mapsto \frac{1}{\lambda_d \sqrt{d}} \sum_{\ell=0}^{d-1} \omega^{j\ell} |\ell\rangle, \;\; \lambda_d = (-i)^{\frac{d-1}{2}}, \quad S : |j\rangle \mapsto \omega^{\frac{j(j-1)}{2}} |j\rangle, \quad CZ : |j\rangle|\ell\rangle \mapsto \omega^{j\ell} |j\rangle|\ell\rangle.$$

The n-qudit Clifford group $\mathcal{C}_n$ is the normaliser of $\mathcal{P}_n$ in $\mathcal{U}(d^n)$. It is generated by H, S, and CZ gates via matrix multiplication and tensor product up to global phases that we will ignore. It is well known that the qudit Clifford group modulo Paulis is isomorphic to the symplectic group $\mathrm{Sp}(2n, \mathbb{Z}_d)$ [4]. This correspondence allows Clifford generators to be represented by symplectic matrices over the finite field $\mathbb{Z}_d$, forming the basis of stabiliser tableau representation of Clifford operators [1, 27]. Building on this correspondence, we generalise the prior Clifford completeness results [39, 41, 51] for qubits and qutrits to all odd primes.

A Unique Normal Form for Multi-Qudit Clifford Circuits To find a complete set of rewrite rules, we first define a unique normal form for Clifford circuits. We introduce a normal form with two components: a *symplectic normal form* characterising the stabiliser tableau of the Clifford circuit, followed by a *Pauli normal form* giving the necessary Pauli correction. This decomposition reflects the semidirect product structure of the projective Clifford group $\widehat{\mathcal{C}}_n$, where $\widehat{\mathcal{C}}_n \cong \mathrm{Sp}(2n, \mathbb{Z}_d) \ltimes (\mathbb{Z}_d)^{2n}$ [55]. We show that symplectic normal forms $N^{(n)}$ are in one-to-one correspondence with symplectic matrices $M \in \mathrm{Sp}(2n, \mathbb{Z}_d)$. The normal form $N^{(n)}$ is defined inductively using structured layers of circuit fragments, which we call *normal boxes*. These are labelled A, B, D, and E, each implementing a specific action on Pauli generators (these names match the corresponding boxes for qutrits and qubits [51] which also included a C and F box which are now absorbed into our Pauli normal form). An illustration of the multi-qudit Clifford normal form is given in Figure 3.

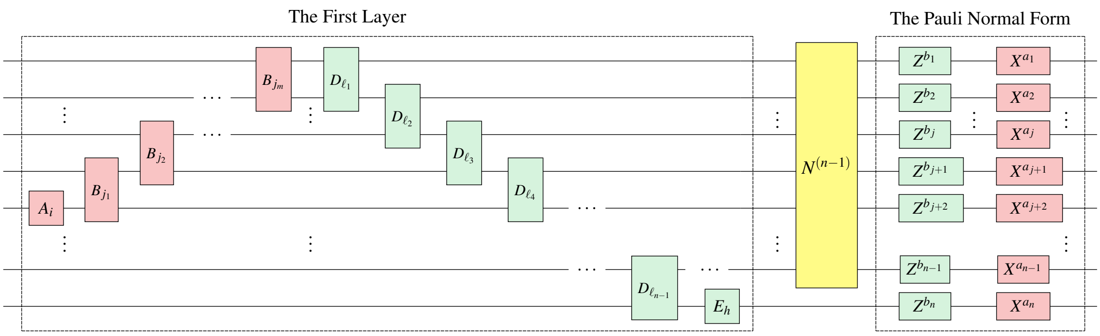


Figure 3: The inductive construction of the normal form for any $C \in \mathcal{C}_n$, up to a global phase. Each layer of the normal form alternates the Z-normal (red) and X-normal (green) circuits, progressively eliminating nontrivial entries in the stabiliser tableau. The first layer implements the conjugation action of C on the n-th qudit up to Pauli correction. By induction, the normal form $N^{(n-1)}$ implements the remaining symplectic action on the first $n-1$ qudits. Finally, the Pauli normal form carries out the Pauli correction.

Establishing Qudit Clifford Completeness To obtain a complete set of Clifford relations, it suffices to find a set of rewrite rules that transforms any Clifford circuit into its unique normal form. We proceed in three steps. First, we show that if a Clifford generator (H, S, or CZ) is applied before a normal form, the resulting circuit can rewritten into a new normal form. This is achieved by showing how each generator can be "pushed" through each distinct normal box, yielding equations of the form illustrated below. We refer to them as the *box relations*.

(1)

Here, dir denotes the *residual dirty gates*, which consist of Clifford generators and whose precise form depends on the parameters a, b, and the dimension d. These dirty gates are then pushed further into the remaining part of the normal form. By grouping these relations together in a systematic way, we obtain 66 box relations, each parametrised by the indices of the corresponding normal boxes.

We then reduce these relations to a smaller set of 18 symplectic relations by showing, first by hand and then in Agda how to prove the 66 box relations from the 18 relations. These 18 equations then form a compact and natural presentation of the multi-qudit symplectic Clifford group. Finally, we update these relations with Pauli corrections, and show that they imply all the relations governing Paulis and their interaction with Cliffords. This gives Figure 1, a complete set of multi-qudit Clifford relations.

1.3 Paper Organisation

The remainder of this paper is organised as follows. In Section 2, we introduce the necessary algebraic and circuit-theoretic preliminaries, including qudit Pauli and Clifford operators, circuit interpretations, and the rewriting framework used throughout the paper. Section 3 designs a unique normal form for multi-qudit Clifford circuits, based on a symplectic normal form together with a Pauli correction. In Section 4, we derive a complete set of rewrite rules for multi-qudit Clifford circuits: we first establish completeness at the symplectic level, and then lift these results to unitary Clifford circuits by accounting for Pauli corrections. Section 5 concludes with a brief summary of our results and a discussion of future directions. For readability, most technical proofs are deferred to the appendices.

2 Preliminaries

Here, we introduce the algebraic background and conventions used throughout the paper. In Section 2.1, we introduce the rings and roots of unity underlying our odd-prime qudit formalism. In Section 2.2, we define the Pauli and Clifford operators, review basic identities, and recall the symplectic structure of the Clifford group. In Section 2.3, we move from operators to circuits by specifying Clifford circuit families, their interpretation maps (unitary, projective, and symplectic), as well as the notions of soundness and completeness for rewriting systems that will be used in later sections. Most of the technical proofs can be found in Section A.

2.1 Rings

We write $\mathbb{N}$ for the collection of non-negative integers, $\mathbb{Z}$ for the ring of integers, $\mathbb{Z}_m$ for the ring of integers modulo $m \in \mathbb{N}$, and $\mathbb{Z}_m^*$ for the collection of non-zero elements of $\mathbb{Z}_m$. If a and b are two integers, we write $a \equiv_m b$ if a and b are equivalent modulo m (i.e., equal in $\mathbb{Z}_m$). If $a \in \mathbb{Z}_m$ is invertible, we sometimes denote its multiplicative inverse by $1/a$. Throughout the paper, d denotes an arbitrary positive odd prime of $\mathbb{Z}$.

Recall that a complex number $c \in \mathbb{C}$ is a *root of unity* if $c^n = 1$ for some $n \in \mathbb{N}$. We will be especially interested in d-th roots of unity.

Definition 2.1. *We write* ω_d *for the* primitive d-th root of unity $\omega_d = e^{2\pi i/d}$.

In what follows, we drop the subscript and write ω for ω_d, leaving d implicit. Note that $\omega^d = 1$, $\omega^\dagger = \omega^{-1} = \omega^{d-1}$, and $\sum_{j=0}^{d-1} \omega^j = 0$.

Definition 2.2. *The ring* $\mathbb{Z}[\omega]$ *of* cyclotomic integers of degree d *is the smallest subring of* $\mathbb{C}$ *containing* ω.

Since $\omega^\dagger = \omega^{d-1}$, the ring $\mathbb{Z}[\omega]$ is closed under complex conjugation. Every element of $\mathbb{Z}[\omega]$ can be written as a linear combinations of $\omega^j, 0 \leq j \leq d-1$, so that

$$\mathbb{Z}[\omega] = \left\{ \sum_{j=0}^{d-1} a_j \omega^j \mid a_j \in \mathbb{Z} \right\}. \tag{2}$$

Definition 2.3. *The ring* $\mathbb{Z}[1/d, \omega]$ *is the smallest subring of* $\mathbb{C}$ *containing* $1/d$ *and* ω.

The ring $\mathbb{Z}[1/d, \omega]$ is the localisation of $\mathbb{Z}[\omega]$ by at the multiplicative set $\{1, d, d^2, \ldots\}$.

$$\mathbb{Z}[1/d, \omega] = \left\{ \frac{u}{d^\ell} \mid u \in \mathbb{Z}[\omega], \ell \in \mathbb{Z}_{\geq 0} \right\}. \tag{3}$$

Definition 2.4. *We define* $\lambda_d := (-i)^{\frac{d-1}{2}}$ *where* $i = e^{2\pi i/4}$ *is the imaginary unit.*

The ring $\mathbb{Z}[1/d, \omega]$ has two properties that will be useful later. Propositions 2.5 and 2.6 are proved in Section A.

Proposition 2.5. $\frac{1}{\lambda_d \sqrt{d}} \in \mathbb{Z}[1/d, \omega]$.

Proposition 2.6. *The roots of unity in* $\mathbb{Z}[1/d, \omega]$ *are* $\pm\omega^t$, $t \in \mathbb{Z}_d$.

2.2 Operators

We now introduce the operators of primary interest in this work, namely the *Pauli* operators [27] and the *Clifford* operators [23, 25, 27, 37, 38, 47].

In what follows, we write I_m for the $m \times m$ identity matrix, and we omit the subscript m when it is not relevant or can be inferred from the context. We say that an operator U is a *scalar*, if it is a scalar multiple of the identity operator, i.e., if $U = \lambda I$, for some $\lambda \in \mathbb{C}$. For simplicity, we denote U by λ in such a case.

We write $\mathrm{Sp}(2n, \mathbb{Z}_d)$ for the *symplectic group* and $\mathcal{U}_m$ for the *unitary group*. The collection $\{\alpha I_m \mid \alpha \in \mathbb{C}\}$ of scalar multiples of the identity matrix forms a subgroup of $\mathcal{U}_m$ called the *group of global phases*. By a slight abuse of notation, we denote it by $\mathcal{U}_1$.

Definition 2.7. *Let* $m \in \mathbb{N}$. $\mathcal{U}_m(\mathbb{Z}[1/d, \omega])$ *is the group of* $m \times m$ *unitaries with entries in* $\mathbb{Z}[1/d, \omega]$.

Throughout the paper, we define unitary operators by specifying their action on the computational basis. This uniquely determines their action on all states. Unless stated otherwise, juxtaposition AB denotes matrix multiplication of the unitaries A and B.

2.2.1 Pauli Operators

Definition 2.8. *The single-qudit* Pauli X and Z gates *are defined as follows.*

$$X : |j\rangle \mapsto |j+1\rangle \qquad \text{and} \qquad Z : |j\rangle \mapsto \omega^j |j\rangle, \tag{4}$$

where $0 \leq j \leq d-1$ *and addition is performed modulo* d*.*

We record several properties of the Pauli X and Z gates. The corresponding derivations can be found in Section A. We have

$$X^d = Z^d = 1. \tag{5}$$

It follows from Equation (5) that $X^\dagger = X^{d-1}$ and $Z^\dagger = Z^{d-1}$. In addition, we have $ZXZ^\dagger X^\dagger = \omega$, which implies that

$$ZX = \omega XZ, \qquad ZXZ^\dagger = \omega X, \qquad \text{and} \qquad XZX^\dagger = \omega^{-1} Z. \tag{6}$$

Definition 2.9. *For* $n \in \mathbb{N}$*, the* n*-qudit* Pauli group $\mathcal{P}_n$ *is the group of* n*-qudit unitary operators generated by* X *and* Z *under composition and tensor product.*

Note that $\omega \in \mathcal{P}_1$ because $ZXZ^\dagger X^\dagger = \omega$.

The group $\mathcal{P}_n$ is not closed under tensor product. Definition 2.9 states that among all of the operators that can be constructed using X and Z through composition and tensor product, $\mathcal{P}_n$ contains exactly those that act on n qudits. Using Equations (5) and (6) and the usual properties of the tensor product, one can show that

$$\mathcal{P}_n = \left\{ \omega^c \left(X^{a_1} Z^{b_1} \otimes \cdots \otimes X^{a_n} Z^{b_n} \right) \mid c, a_j, b_j \in \mathbb{Z}_d \text{ for all } 1 \leq j \leq n \right\}.$$

As a consequence, we have $|\mathcal{P}_n| = d^{2n+1}$.

2.2.2 Clifford Operators

Definition 2.10. *The single-qudit* H *and* S *gates and the two-qudit* CZ *gate are defined as follows, where* λ_d *is given as in Definition 2.4.*

$$H : |j\rangle \mapsto \frac{1}{\lambda_d \sqrt{d}} \sum_{\ell=0}^{d-1} \omega^{j\ell} |\ell\rangle, \qquad S : |j\rangle \mapsto \omega^{\frac{j(j-1)}{2}} |j\rangle, \qquad \text{and} \qquad CZ : |j\rangle|\ell\rangle \mapsto \omega^{j\ell} |j\rangle|\ell\rangle,$$

where $0 \leq j \leq d-1$ *and multiplication is performed modulo* d*.*

The gates H, S, and CZ are called the *Hadamard* gate, the *phase* gate, and the *controlled-Z* gate, respectively. We use these gates to define the *Clifford* group, similarly to how the X and Z gates were used to define the Pauli group in Definition 2.9.

Definition 2.11. *For* $n \in \mathbb{N}$*, the* n*-qudit* Clifford group $\mathcal{C}_n$ *is the group of* n*-qudit unitary operators generated by* H*,* S*, and* CZ *under composition and tensor product.*

We record a few properties of Clifford operators. In particular, we show that the Clifford phases are exactly the integer powers of ω.

Proposition 2.12. *For* $0 \leq j \leq d-1$*,* $H^2|j\rangle = \lambda_d^{-2} |-j\rangle$*. Here the negation* $-j$ *is performed modulo* d*.*

Note that H^2 is a special case of a *multiplier* since up to a global phase, $H^2|j\rangle = |-j\rangle$. Multipliers admit simple commutation rules with the Clifford generators, which help streamline the relation reduction in [6].

Definition 2.13. *For every $g \in \mathbb{Z}_d^*$, we define the* multiplier by g *as the Clifford unitary M_g that acts on computational basis states as $M_g|x\rangle = |gx\rangle$.*

By direct computation, we can verify that $M_g^{-1} = M_{g^{-1}}$ where $g^{-1} \in \mathbb{Z}_d^*$ is the inverse of g modulo d. Since it sends computational basis states to computational basis states, it must be unitary so that $M_g^\dagger = M_g^{-1}$. In Lemma 2.14, we show how to construct a multiplier using H, S, X, and Z gates, so that it is indeed Clifford. Here, $\left(\frac{g}{d}\right)_L$ denotes the Legendre symbol in number theory [47, Appendix A].

Lemma 2.14. *For every $g \in \mathbb{Z}_d^*$, M_g defined below satisfies $M_g|x\rangle = |gx\rangle$ for all $x \in \mathbb{Z}_d$.*

$$-\boxed{M_g}- \quad = \quad -\boxed{H}-\boxed{S^{g^{-1}}}-\boxed{H}-\boxed{S^{g}}-\boxed{H}-\boxed{S^{g^{-1}}}-\boxed{X^{(1-g)/2}}-\boxed{Z^{(1-g)/2g}}- \quad \cdot \lambda_d^2 \left(\frac{g}{d}\right)_L \omega^{\frac{-g^2+4g+2}{8g}}$$

Lastly, we record some important properties of Clifford operators and Clifford scalars.

Corollary 2.15. $H^4 = 1$.

Lemma 2.16. $S^d = 1$.

Lemma 2.17. $\mathcal{C}_n \subseteq \mathcal{U}_{d^n}(\mathbb{Z}[1/d, \omega])$.

Proof. This follows from the fact that the matrices representing H, S, and CZ in the computational basis have entries in $\mathbb{Z}[1/d, \omega]$. In the case of H, this follows from Proposition 2.5. □

Lemma 2.18. $\det(S) = \det(H) = \det(CZ) = 1$.

Proposition 2.19. *The Clifford phases are exactly the integer powers of ω. That is, for all $n \in \mathbb{N}$, we have $\mathcal{C}_n \cap U(1) = \{\omega^t \mid t \in \mathbb{Z}_d\}$.*

Proof. All zero-qudit Clifford operators are of the form ω^t for some $t \in \mathbb{Z}_d$, so that the statement holds when $n = 0$. When $n \geq 1$, the scalar $\omega = \omega I_{d^n}$ has determinant 1. It then follows from properties of the determinant and Proposition 2.19, that every element of $\mathcal{C}_n$ has determinant 1. Now consider a phase $cI_{d^n} \in \mathcal{C}_n$. Then $c \in \mathbb{Z}[1/d, \omega]$ and

$$c^{d^n} = \det(cI) = 1. \tag{7}$$

Hence, c is a root of unity in $\mathbb{Z}[1/d, \omega]$ and therefore, by Proposition 2.6, $c = \pm\omega^t$ for some $t \in \mathbb{Z}_d$. Now suppose that $c = -\omega^t$, for some $t \in \mathbb{Z}_d$. Then $\left(-\omega^t\right)^{d^n} = -\left(\omega^d\right)^{t \cdot d^{n-1}} = -1$, which contradicts Equation (7). Hence, c must be of the form $c = \omega^t$, for some $t \in \mathbb{Z}_d$. □

2.2.3 Clifford Conjugation of Pauli Generators

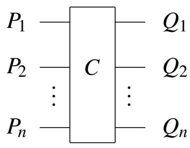
Another way to define the Clifford unitaries is as those unitaries that send the Pauli operators to Pauli operators under conjugation.

Definition 2.20. *For $n \in \mathbb{N}$, the* normaliser of the Pauli group $\mathcal{P}_n$ *is denoted by $\widetilde{\mathcal{C}_n}$,*

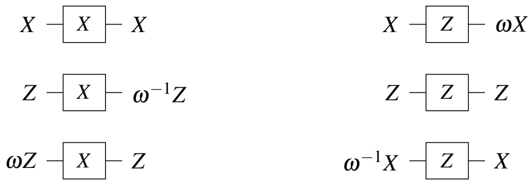
$$\widetilde{\mathcal{C}_n} \coloneqq \{U \in \mathcal{U}_{d^n} \mid UPU^\dagger \in \mathcal{P}_n \;\; \textit{for all } P \in \mathcal{P}_n\}.$$

We will denote the action of $\widetilde{\mathcal{C}_n}$ by conjugation on the Pauli group $\mathcal{P}_n$ as $C \bullet P := CPC^\dagger$ for $C \in \widetilde{\mathcal{C}_n}$. Note that if we write $Q := C \bullet P$, we then also have $CP = QC$. Then we can interpret this as "pushing P through C gives us Q". It will be useful to introduce a circuit notation for this.

Definition 2.21. *Let $C \in \widetilde{\mathcal{C}_n}$ and $P, Q \in \mathcal{P}_n$ with $Q = C \bullet P$. Write $P := P_1 \otimes \ldots \otimes P_n$ and $Q := Q_1 \otimes \ldots \otimes Q_n$ for $P_j, Q_j \in \mathcal{P}_1$. We write this in circuit notation as follows:*



Example 2.22. *By* Equation (6), *$X, Z \in \widetilde{\mathcal{C}_1}$. Diagrammatically, we have*



(8)

A Clifford unitary is fully determined up to global phase by its action of conjugating Pauli operators.

Proposition 2.23. *If $C \bullet P = P$ for all $P \in \mathcal{P}_n$, then C is a scalar (i.e. proportional to the identity).*

Proof. Any complex $d \times d$-matrix can be written in the form $\sum_{i=1}^{d^2} c_i P_i,\ c_i \in \mathbb{C},\ P_i \in \{X^a Z^b \mid a, b \in \mathbb{Z}_d\}$. It follows that $\mathcal{P}_n$ spans the vector space formed by $d^n \times d^n$ complex matrices. Since $C \bullet P = P$ for all $P \in \mathcal{P}_n$, $C \bullet M = CMC^\dagger = M$ for all operators M. Hence $CM = MC$ so that C must commute with all matrices. The only matrices with this property are the scalars. $\square$

Then note that because conjugation is a group homomorphism, we have that $C \bullet P = D \bullet P$ for all P implies $(CD^\dagger) \bullet P = P$ so that $CD^\dagger = \lambda I$ and hence $C = \lambda D$ for some scalar λ.

Since this conjugation is a group homomorphism, it suffices to know its action on a set of generators of $\mathcal{P}_n$. A useful set of generators are those with only one Z or X gate acting on a single qudit, and identities everywhere else.

Definition 2.24. *Let Z_j denote the n-qudit Pauli $P_1 \otimes \cdots \otimes P_n$ with $P_\ell = I$ if $\ell \neq j$, and $P_j = Z$. We write X_j for the analogous n-qudit Pauli by substituting X with Z in the above definition. Let $\mathcal{B}_n := \{X_j, Z_j \mid 1 \leq j \leq n\}$ be the n-qudit* Pauli basis *for $\mathcal{P}_n$. An element in $\mathcal{B}_n$ is called a* Pauli generator.

Note that $\mathcal{B}_n$ generates $\mathcal{P}_n$. Given any n-qudit Clifford operator C, the collection of operators $P_j = C \bullet Z_j, Q_j = C \bullet X_j, 1 \leq j \leq n$ fully determines C (up to a global phase). Moreover, P_j and Q_j inherit the commuting and non-commuting relations from Z_j and X_j. By Equation (6), $P_j Q_j = \omega Q_j P_j$. When $j \neq \ell$, $P_j Q_\ell = Q_\ell P_j$.

Lemmas 2.25 and 2.26 show that the Clifford generators H, S, and CZ are indeed in $\widetilde{\mathcal{C}_n}$ by specifying their action on the Pauli generators under conjugation. Full proofs are given in Section A.

Lemma 2.25. $H, S \in \widetilde{\mathcal{C}_1}$. *Specifically,* $HXH^\dagger = Z$, $HZH^\dagger = X^{-1}$, $SXS^\dagger = XZ$, $SZS^\dagger = Z$.

$$\begin{array}{ccc} X & -\boxed{H}- & Z \\ Z & -\boxed{H}- & X^{-1} \\ Z^{-1} & -\boxed{H}- & X \end{array} \qquad \begin{array}{ccc} X & -\boxed{S}- & XZ \\ Z & -\boxed{S}- & Z \\ XZ^{-1} & -\boxed{S}- & X \end{array} \tag{9}$$

Lemma 2.26. $CZ \in \widetilde{\mathcal{C}_2}$. *Specifically,*

$$\begin{array}{ccc} X & \bullet & X \\ I & \bullet & Z \end{array} \quad \begin{array}{ccc} Z & \bullet & Z \\ I & \bullet & I \end{array} \quad \begin{array}{ccc} I & \bullet & Z \\ X & \bullet & X \end{array} \quad \begin{array}{ccc} I & \bullet & I \\ Z & \bullet & Z \end{array} \quad \begin{array}{ccc} X & \bullet & X \\ Z^{-1} & \bullet & I \end{array} \quad \begin{array}{ccc} Z^{-1} & \bullet & I \\ X & \bullet & X \end{array} \tag{10}$$

We can also directly verify that multipliers are Pauli normalising by checking their action on the Paulis: $M_g Z M_{g^{-1}} = Z^{g^{-1}}$ and $M_g X M_{g^{-1}} = X^g$.

These results show that $\mathcal{C}_n \subseteq \widetilde{\mathcal{C}_n}$. In fact, these groups are the same up to the inclusion of global phases. This fact is well-known, but will also follow from our construction of an explicit normal form for any operator in $\widetilde{\mathcal{C}_n}$ in Section 3.

Note that Definition 2.20 includes all possible global phases $e^{i\alpha}$, making $\widetilde{\mathcal{C}_n}$ uncountably infinite for all $n \in \mathbb{N}$. In contrast, the global phases in $\mathcal{C}_n$ are exactly powers of ω. That is, $\mathcal{C}_n \cap \mathcal{U}_1 = \langle \omega \rangle$. The groups $\widetilde{\mathcal{C}_n}$ and $\mathcal{C}_n$ are then related as follows. First, note that since global phases commute with everything, we have that the phases form normal subgroups $\mathcal{U}_1 \trianglelefteq \widetilde{\mathcal{C}_n}$ and $\langle \omega \rangle \trianglelefteq \mathcal{C}_n$. The quotients $\widetilde{\mathcal{C}_n}/\mathcal{U}_1$ and $\mathcal{C}_n/\langle \omega \rangle$ are then well-defined. It is a fact that

$$\widetilde{\mathcal{C}_n}/\mathcal{U}_1 \;\cong\; \mathcal{C}_n/\langle \omega \rangle.$$

In other words, the Clifford group (up to global phases) is the normaliser of the Pauli group (up to global phases). Because $\mathcal{C}_n \leq \widetilde{\mathcal{C}_n}$, we have $\mathcal{P}_n \trianglelefteq \mathcal{C}_n$, and the quotient $\mathcal{C}_n/\mathcal{P}_n$ is also well-defined.

2.2.4 Symplectic Structure of the Clifford Group

Proposition 2.23 shows that two Cliffords that have the same action on the Paulis are the same up to global phase. We can consider a slightly weaker notion where $C \bullet P = \lambda_P (D \bullet P)$ for all Paulis P where λ_P is some phase that is allowed to vary depending on P. In this case $C = \lambda Q D$ for some global phase λ and a Pauli Q that is uniquely defined by $Q \bullet P = \lambda_P P$.

That is, if we ignore phases in the conjugation map, this means that we are ignoring Pauli operators on the Cliffords. We can then view conjugation as a map $\bullet : \widetilde{\mathcal{C}_n}/\mathcal{P}_n \to \mathrm{Hom}(\mathcal{P}_n/U_1, \mathcal{P}_n/U_1)$. The group $\mathcal{P}_n/U_1$ is isomorphic to $\mathbb{Z}_d^{2n}$ since modding out the phases means that Z and X now commute and hence multiplication of Paulis is just $Z^{a_1} X^{b_1} Z^{a_2} X^{b_2} = Z^{a_1+a_2} X^{b_1+b_2}$. Hence, we can view the conjugation action $\bullet$ as mapping elements of $\widetilde{\mathcal{C}_n}/\mathcal{P}_n$ to matrices of size $2n \times 2n$ over $\mathbb{Z}_d$.

However, since conjugation preserves the commutation relations between Paulis, $\bullet$ cannot map a Clifford to an arbitrary $2n \times 2n$ matrix. Instead it can only produce *symplectic* matrices.

Definition 2.27. *Let* $\mathbb{Z}_d^{2n}$ *be an even-dimensional vector space over* $\mathbb{Z}_d$*. The* symplectic inner product $\omega(\cdot,\cdot)$ *is defined by* $\omega((\vec{a},\vec{b}),(\vec{c},\vec{d})) = \vec{a}\cdot\vec{d} - \vec{b}\cdot\vec{c}$ *where* $(\vec{a},\vec{b}),(\vec{c},\vec{d}) \in \mathbb{Z}_d^{2n}$ *are arbitrary vectors, and* $\cdot$ *is the regular dot-product of vectors. We say a matrix* $M : \mathbb{Z}_d^{2n} \to \mathbb{Z}_d^{2n}$ *is* symplectic *when* $\omega(M\vec{v}, M\vec{w}) = \omega(\vec{v},\vec{w})$ *for all* $\vec{v},\vec{w} \in \mathbb{Z}_d^{2n}$*. We denote the group of symplectic matrices by* $\mathrm{Sp}(2n,\mathbb{Z}_d)$*.*

In this definition a vector $\vec{v} = (\vec{a},\vec{b})$ corresponds to an n-qubit Pauli where the Z powers of the Pauli correspond to $\vec{a}$, and the X powers to $\vec{b}$. The value of the symplectic inner product $\omega(\vec{v},\vec{w}) = k$ then corresponds to the Paulis defined by $\vec{v}$ and $\vec{w}$ commuting up to a phase ω^k. Since this commutation and hence this value ω^k is preserved by conjugation, conjugation corresponds to a symplectic matrix.

Proposition 2.28. $\mathcal{C}_n/\mathcal{P}_n \cong \mathrm{Sp}(2n,\mathbb{Z}_d)$

By "bringing the Paulis to the other side of the equation" we can then see that the Clifford group is the semidirect product of the symplectic group and the Pauli group.

Theorem 2.29 ([55])**.** $\widehat{\mathcal{C}}_n \cong \mathrm{Sp}(2n,\mathbb{Z}_d) \ltimes (\mathbb{Z}_d)^{2n}$.

Instead of directly modding out the Paulis, we could first mod out the phases.

Definition 2.30. *Let* $\widehat{\mathcal{P}}_n = \mathcal{P}_n/\langle\omega\rangle$ *be the* n*-qudit* projective Pauli group *and* $\widehat{\mathcal{C}}_n = \mathcal{C}_n/\langle\omega\rangle$ *be the* n*-qudit* projctive Clifford group*.*

The correspondence of the symplectic group to the Clifford group modulo Paulis also works when restricting to the projective groups. That is, $\widehat{\mathcal{P}}_n \trianglelefteq \widehat{\mathcal{C}}_n$ and $\widehat{\mathcal{C}}_n/\widehat{\mathcal{P}}_n \cong Sp(2n,\mathbb{Z}_d)$.

2.2.5 Derived Generators

For convenience, Figure 4 defines a set of derived Clifford operators. They will be used to compactly present the qudit Clifford relations. The soundness of the phases, the Pauli X and Z gates, and well as the SWAP gate is established in Lemmas B.8 to B.11.

2.3 Circuits

In Section 2.2, the Clifford generators H, S, and CZ, as well as the derived generators listed in Figure 4 were treated as operators. In this section, we consider them as generators for circuits. The difference between these two perspectives is that when X is viewed as an operator, we have for instance, $X^d = 1$. However, the circuit X^d is not equal to the circuit of an identity matrix (the former contains d gates, while the latter contains none).

We write $\mathfrak{C}$ for the collection of all circuits over the gate set $\{\omega, X, Z, H, S, \mathrm{CZ}\}$ and $\mathfrak{C}_n$ for the subset of $\mathfrak{C}$ containing only the circuits on exactly n qudit wires. We similarly define $\mathfrak{P}$ and $\mathfrak{P}_n$ as the collection of circuits over the gate set $\{\omega, X, Z\}$ and the collection of circuits in $\mathfrak{P}$ on exactly n qudit wires. Note that $\mathfrak{P} \subseteq \mathfrak{C}$ and $\mathfrak{P}_n \subseteq \mathfrak{C}_n$.

2.3.1 Interpretations

An *interpretation map* for a family of circuits is a function which assigns an operator to every circuit. The standard interpretation is the *unitary interpretation*, in which a circuit is interpreted as the unitary operator it represents.

Definition 2.31. *Let* $C \in \mathfrak{C}_n$*. We define the* unitary interpretation $[\![C]\!]_u \in \mathcal{C}_n$ *of* C *to be the unitary operator obtained by interpreting the gates in* $\{\omega, X, Z, H, S, CZ\}$ *as the corresponding operators, the parallel composition of circuits as the tensor product of operators, and the sequential composition of circuits as the matrix multiplication of operators.*

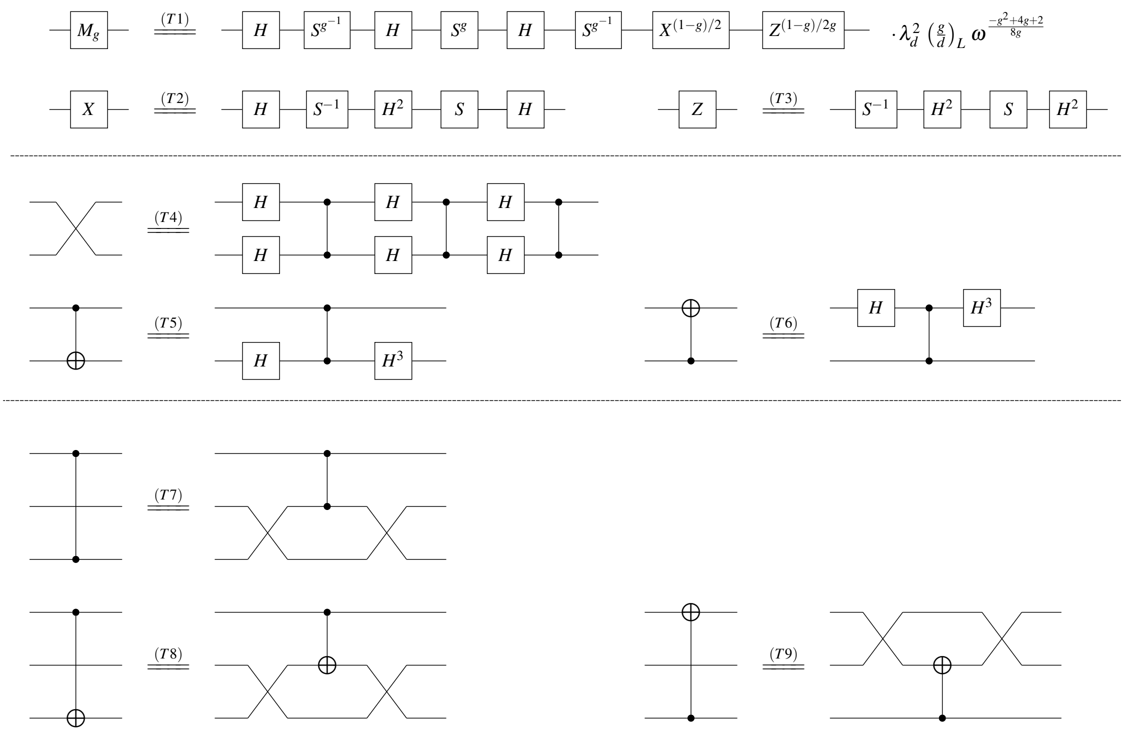


Figure 4: A set of derived Clifford generators, expressed as circuits over the generating set $\{H, S, \mathsf{CZ}\}$. Here, g generates the multiplicative group $\mathbb{Z}_d^*$, and $\left(\frac{g}{d}\right)_L$ denotes the Legendre symbol in number theory [47, Appendix A].

Now recall that the quotients $\mathcal{C}_n/\mathcal{P}_n$ and $\mathcal{C}_n/\langle\omega\rangle$ are well-defined and let $\pi_s : \mathcal{C}_n \to \mathcal{C}_n/\mathcal{P}_n \cong \mathrm{Sp}(2n, \mathbb{Z}_d)$ and $\pi_p : \mathcal{C}_n \to \mathcal{C}_n/\langle\omega\rangle$ be the corresponding canonical projections. We use these projections to define two more interpretation maps.

Definition 2.32. *Let $C \in \mathfrak{C}_n$. We define:*

- *the* interpretation of C up to Paulis $[\![C]\!]_s \in \mathcal{C}_n/\mathcal{P}_n$ *as* $[\![C]\!]_s = \pi_s([\![C]\!]_u)$ *and*
- *the* interpretation of C up to global phases $[\![C]\!]_p \in \mathcal{C}_n/\langle\omega\rangle$ *of* C *as* $[\![C]\!]_p = \pi_p([\![C]\!]_u)$.

We sometimes refer to $[\![\cdot]\!]_s$ as the *symplectic interpretation* (since $[\![C]\!]_s$ belongs to the symplectic group) and to $[\![\cdot]\!]_p$ as the *projective interpretation* (since $[\![C]\!]_s$ belongs to the projective group). The three interpretations introduced so far organise themselves into the diagram below.

$$\begin{tikzcd}
 & \mathfrak{C}_n \arrow[dl, "{[\![\cdot]\!]_s}"'] \arrow[d, "{[\![\cdot]\!]_u}"] \arrow[dr, "{[\![\cdot]\!]_p}"] & \\
\mathcal{C}_n/\mathcal{P}_n & \mathcal{C}_n \arrow[l, "\pi_s"] \arrow[r, "\pi_p"'] & \mathcal{C}_n/\langle\omega\rangle
\end{tikzcd}$$

The interpretation maps $[\![\cdot]\!]_u$, $[\![\cdot]\!]_s$, and $[\![\cdot]\!]_p$ correspond to three ways to interpret Clifford circuits, and each interpretation comes with a level of granularity. We can now use these interpretation maps to define different types of equality among the elements of $\mathfrak{C}$.

Definition 2.33. *Let $C_1, C_2 \in \mathfrak{C}_n$. We write*

- $C_1 \equiv C_2$ *if C_1 and C_2 are the same circuit.*
- $C_1 \equiv_u C_2$ *if* $[\![C_1]\!]_u = [\![C_2]\!]_u$.
- $C_1 \equiv_p C_2$ *if* $[\![C_1]\!]_p = [\![C_2]\!]_p$.
- $C_1 \equiv_s C_2$ *if* $[\![C_1]\!]_s = [\![C_2]\!]_s$.

As Definition 2.33 states: $C_1 \equiv C_2$ when C_1 and C_2 are identical circuits; $C_1 \equiv_u C_2$ when C_1 and C_2 represent the same unitary; $C_1 \equiv_p C_2$ when C_1 and C_2 represent the same unitary up to a global phase; $C_1 \equiv_s C_2$ when C_1 and C_2 represent the same unitary up to a Pauli. Note that we have the following implications

$$C_1 \equiv C_2 \Rightarrow C_1 \equiv_u C_2 \Rightarrow C_1 \equiv_p C_2 \Rightarrow C_1 \equiv_s C_2,$$

None of the above implications can be reversed in general. However, we do know, for example, that if $C_1 \equiv_s C_2$, then there exists a unique $P \in \mathcal{P}_n$ such that $C_1 \equiv_u C_2 P$.

Since $\mathfrak{P} \subseteq \mathfrak{C}$, all of the above interpretations restrict to $\mathfrak{P}$, giving well-defined interpretations of Pauli circuits.

2.3.2 Relations and Rewriting

A *relation* on $\mathfrak{C}$ (or $\mathfrak{P}$) is a pair (C, C') of elements of $\mathfrak{C}$ (or $\mathfrak{P}$). If $\mathcal{R}$ is a set of relations on $\mathfrak{C}$, we write $\sim_{\mathcal{R}}$ for the smallest *congruence* on $\mathfrak{C}$ that contains all of the relations in $\mathcal{R}$. By congruence, we mean an equivalence relation on $\mathfrak{C}$ that is compatible with composition and tensor product. For two circuits D and D', we have $D \sim_{\mathcal{R}} D'$ if and only if D can be rewritten into D' by applying the relations from $\mathcal{R}$. That is, $D \sim_{\mathcal{R}} D'$ if and only if there is a sequence of rewrites

$$D \to D_1 \to \cdots \to D_k \to D'$$

such that each rewrite is an application of a relation in $\mathcal{R}$ that is applied to some local part of the circuits D_i.

Let $[\![\cdot]\!]_x$ be one of the three interpretations defined above (i.e., one of $[\![\cdot]\!]_u$, $[\![\cdot]\!]_p$, and $[\![\cdot]\!]_s$) and let $\equiv_x$ be the corresponding equivalence of circuits. We say that a collection $\mathcal{R}$ of relations is

- *sound with respect to* $[\![\cdot]\!]_x$ if $C \sim_{\mathcal{R}} D$ implies $C \equiv_x D$ for all $C, D \in \mathfrak{C}$, and
- *complete with respect to* $[\![\cdot]\!]_x$ if $C \equiv_x D$ implies $C \sim_{\mathcal{R}} D$ for all $C, D \in \mathfrak{C}$.

3 A Unique Normal Form for Multi-Qudit Clifford Circuits

In this section, we define a normal form in $\mathfrak{C}_n$ for elements in $\mathcal{C}_n$. Similar to the Clifford normal form for qubits and qutrits [39, 41, 51], our normal form for a qudit Clifford operator is based on uniquely representing its stabiliser tableau. However, our construction follows a conceptually different route. In Section 3.1, we first define the *symplectic normal form*, as a normal form for elements in $\mathcal{C}_n$ up to Pauli corrections, which is based on the isomorphism $\mathrm{Sp}(2n, \mathbb{Z}_d) \cong \mathcal{C}_n/\mathcal{P}_n \cong \widehat{\mathcal{C}}_n/\widehat{\mathcal{P}}_n$. The unitary implementations of symplectic normal circuits form a complete set of representatives for the cosets of $\mathcal{P}_n$ in $\mathcal{C}_n$. In other words, for every symplectic matrix $M \in \mathrm{Sp}(2n, \mathbb{Z}_d)$, there is a unique circuit $C \in \mathfrak{C}_n$ in the symplectic normal form such that $[\![C]\!]_s = M$. In Section 3.2, we then define a normal form for elements in $\widehat{\mathcal{P}}_n$, and call this the *Pauli normal form.*

In Section 3.3 we derive the normal form of elements in $\widehat{\mathcal{C}}_n$ by composing the symplectic and Pauli normal forms, using the semi-direct product structure of the Cliffords of Theorem 2.29. The normal form for $\mathcal{C}_n$ follows by then adding back in global phases.

To define the symplectic normal form, our first goal is to map the Pauli generators to the desired outcomes (up to global phases) by using some elementary building blocks in $\mathfrak{C}_1$ and $\mathfrak{C}_2$, which we call *normal boxes.* To ensure uniqueness we need then also need to make sure we include precisely one unique to construct using these elementary building blocks. We will consider four *types* of normal boxes which we label as A, B, D, and E; see Figure 5. Each type of normal box has a subscript denoting which *variant* it is.

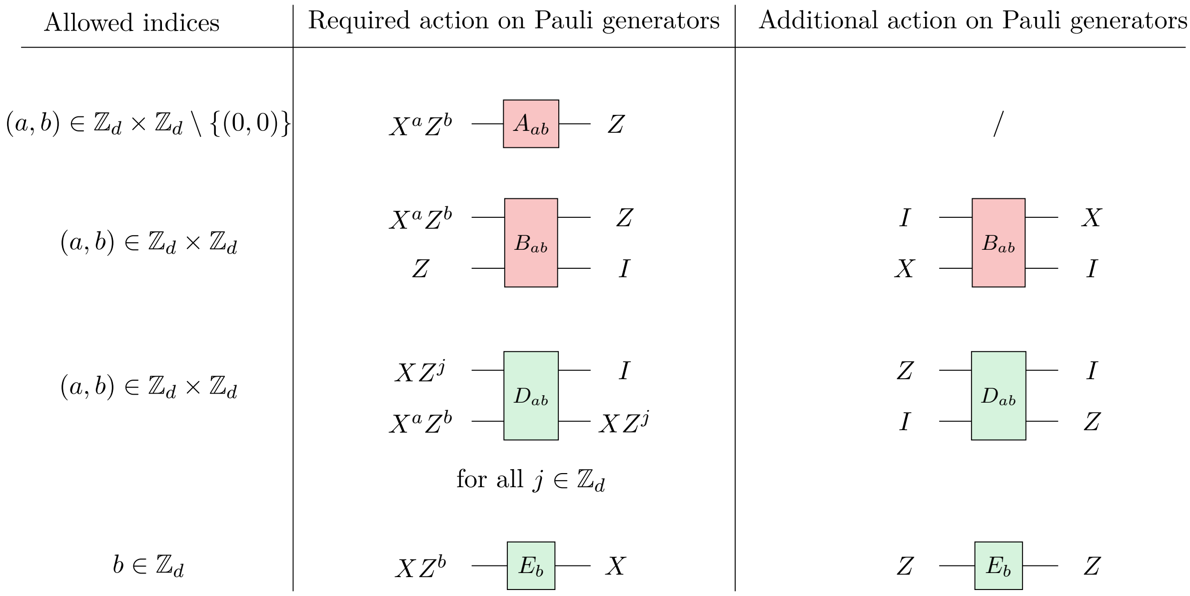


Figure 5: The types of normal boxes, and the specified action they must have on Paulis (up to phase). The colour of the box (red or green) specifies whether these are part of the Z-part, respectively the X-part, of the normal form.

The "required action" the boxes need to satisfy in Figure 5 are needed for the normal form construction to work. However, these don't uniquely define these boxes. We identify some "additional action" that helps simplify some of the arguments later. But even so the symplectic maps of the boxes are not uniquely determined and care must be taken to find concrete implementations that give the simplest possible relations later in the completeness proof.

We give concrete implementations of the normal boxes in Figure 6. Note that as opposed to the normal boxes used in the proofs of qubit and qutrit completeness [39, 41, 51], these definitions are entirely parametric in the labels of the variants with no case distinctions needed. This feature plays an important role in simplifying the search for box relations later.

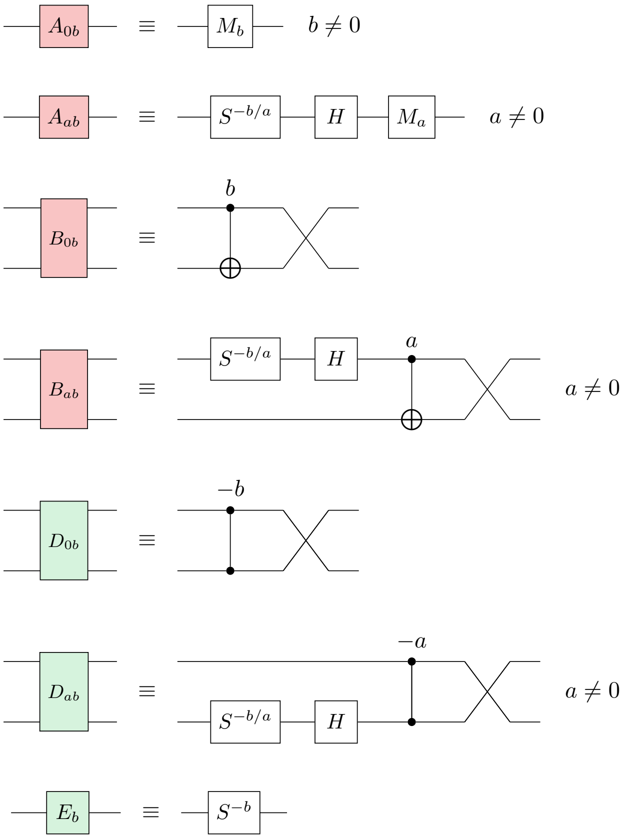


Figure 6: The concrete implementations of the Z and X normal boxes. Here $a, b \in \mathbb{Z}_d$ and $a \neq 0$.

3.1 The Symplectic Normal Form for Multi-Qudit Clifford Circuits

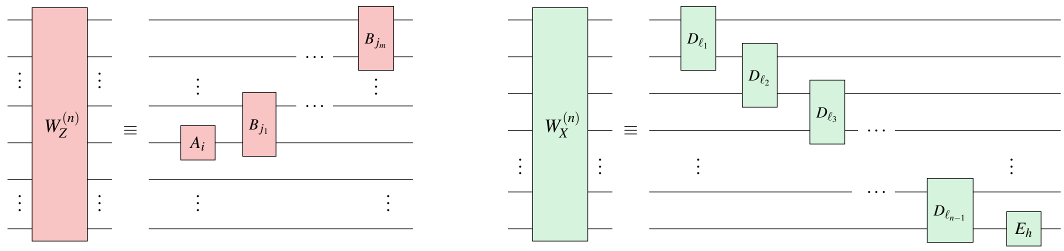
We want to use these normal boxes to define a unique normal form for symplectic matrices, i.e. stabiliser tableaus. This requires us to show the normal form is *universal*, meaning it can represent an arbitrary symplectic matrix M, and *unique*, meaning that each $M \in \mathrm{Sp}(2n, \mathbb{Z}_d)$ is uniquely represented by one specific instantiation of this normal form.

We will show universality by combining together the normal boxes into layers of circuits that we call Z-normal and X-normal. A single layer will "fix up" the action of the Paulis on a single fixed qudit. By then inductively doing this for all qudits we reduce it to to have the identity action on all the Paulis. This procedure is analogous to Gaussian elimination.

In a bit more detail, let C be a Clifford circuit that we will consider to be an automorphism ϕ of $\mathcal{P}_n$ using the conjugation action $\phi(P) = C \bullet P$. We claim that we can then construct a Z-normal circuit W_Z and an X-normal circuit W_X (to be defined later) such that $(W_X W_Z)^{-1}$ acts as ϕ^{-1} on Z_n and X_n. That is, if we define $C' = C\,(W_X W_Z)^{-1}$ and $\phi'(P) = C' \bullet P$, then $\phi'(Z_n) = Z_n$ and $\phi'(X_n) = X_n$ (up to phase). By then finding other W_Z and W_X we can keep updating C using some normal circuit N to make CN^{-1} act as the identity on all Z_j and X_j. Then by Proposition 2.23 N must implement the same linear map as C.

Definition 3.1. *Let the A, B, D, and E normal boxes be the concrete circuits defined in*

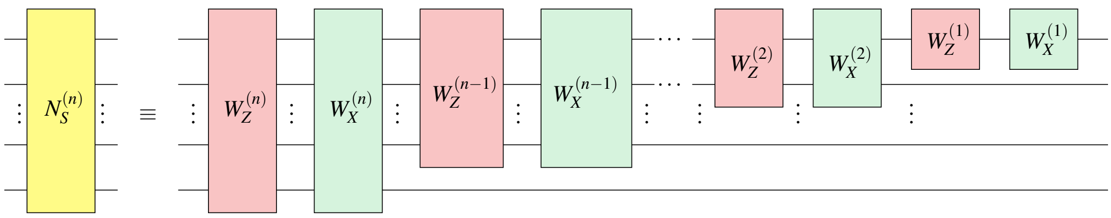
Figure 6. *We then define the following types of normal circuits:*



We call the circuits on the left Z-normal *and on the right* X-normal. *In these circuits, each normal box can have arbitrary labels.*

The Z- and X-normal circuits are the building blocks of the symplectic normal form for an n-qudit Clifford circuit. A pair of Z- and X-normal circuits suffices to rewrite a pair of rows and columns of a symplectic matrix to the identity.

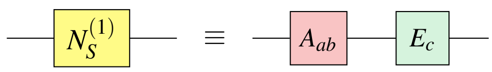
Definition 3.2. *An n-qudit Clifford circuit is* symplectically normal *if it is of the following form:*



That is, we alternate layers of Z- and X-normal circuits, with each subsequent layer acting on one fewer qudit.

Note that we can view the symplectically normal circuits as being defined inductively, with the n-qudit symplectic normal form adding a layer of Z- and X-normal circuits to the left of an existing $(n-1)$-qudit symplectic normal form.

Remark 3.3. *When $n = 1$, a symplectic normal form is composed of only A and E boxes:*



Here $(a, b) \in \mathbb{Z}_d \times \mathbb{Z}_d \setminus \{(0, 0)\}$ and $c \in \mathbb{Z}_d$ and this normal form can represent a single-qudit Clifford uniquely up to Pauli correction.

The proofs of the statements below, which establish the universality and uniqueness of the normal form are analogous to those for the qutrit case in [39], so we postpone them to Section E.

Lemma 3.4. *For every non-scalar n-qudit Pauli operator $P \in \mathcal{P}_n \setminus \{\omega^t I;\ t \in \mathbb{Z}_d\}$, there exists a unique Z-normal circuit W_Z such that $[\![W_Z]\!]_u \bullet P \equiv_p Z \otimes I \otimes \cdots \otimes I$.*

Lemma 3.5. *For every $Q \in \mathcal{P}_n$ such that $(Z \otimes I \otimes \cdots \otimes I)Q = \omega Q(Z \otimes I \otimes \cdots \otimes I)$, there exists a unique X-normal circuit W_X such that $[\![W_X]\!]_u \bullet Q \equiv_p I \otimes \cdots \otimes I \otimes X$.*

Lemma 3.6. *Every X-normal circuit W_X satisfies $[\![W_X]\!]_u \bullet (Z \otimes I \otimes \cdots \otimes I \otimes I) \equiv_p I \otimes I \otimes \cdots \otimes I \otimes Z$.*

Lemma 3.7. *Let $P, Q \in \mathcal{P}_n$ such that $PQ = \omega QP$. Then there exist unique normal circuits W_Z and W_X such that $[\![W_X W_Z]\!]_u \bullet P \equiv_p I \otimes \cdots \otimes I \otimes Z$ and $[\![W_X W_Z]\!]_u \bullet Q \equiv_p I \otimes \cdots \otimes I \otimes X$.*

Proposition 3.8. *Let $\phi : \mathcal{P}_n \to \mathcal{P}_n$ be an automorphism of the Pauli group such that ϕ preserves scalars ($\phi(\omega) = \omega$). There exists a unique Clifford circuit $C \in \mathfrak{C}_n$ in symplectic normal form such that for all $P \in \mathcal{P}_n$, $[\![C]\!]_u \bullet P \equiv_p \phi(P)$.*

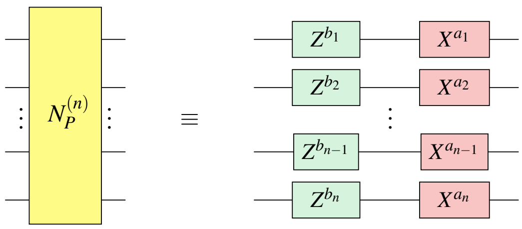
Since there is a bijection between the symplectic matrices and the normal forms, we can count the number of n-qudit normal forms to compute the cardinality of $\mathrm{Sp}(2n, \mathbb{Z}_d)$. The proof of Lemma 3.9 can be found in Section E.

Lemma 3.9. $|\mathrm{Sp}(2n, \mathbb{Z}_d)| = \prod_{k=1}^{n} \left(d^{2k} - 1\right) \cdot \left(d^{2k-1}\right)$.

3.2 The Normal Form for Pauli Operators

Here, we introduce a normal form for multi-qudit Pauli operators, which will serve as a basic building block in Section 3.3. This normal form is unique up to a global phase.

Definition 3.10. *An n-qudit circuit in $\mathfrak{C}_n$ is said to be a* Pauli normal form *if it is of the following form:*



Lemma 3.11. *For every element $P \in \mathcal{P}_n$, there exists a unique Pauli normal form $N_P^{(n)}$ such that $[\![N_P^{(n)}]\!]_u \equiv_p P$.*

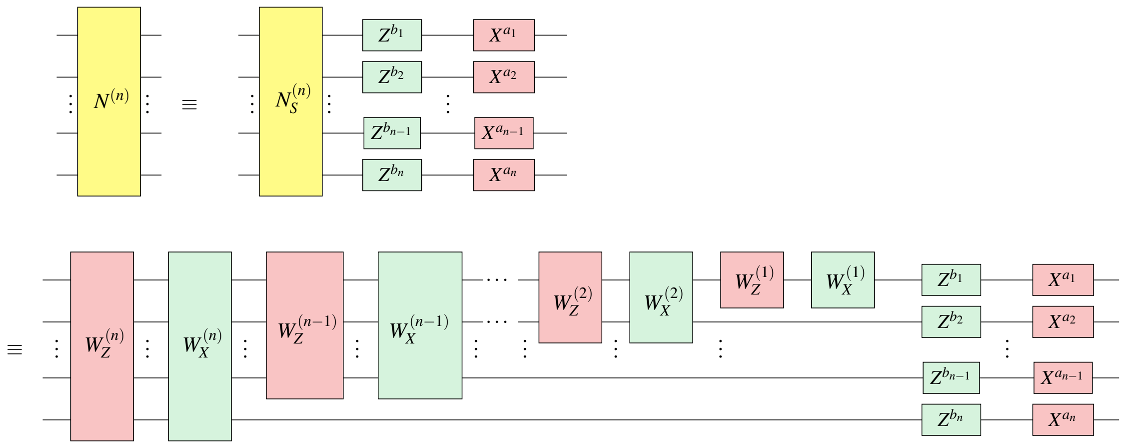
Proof. This follows from the fact that every $P \in \mathcal{P}_n$ can be uniquely written as $P = \omega^t \bigotimes_{j=1}^{n} X^{a_j} Z^{b_j}$ for $t, a_j, b_j \in \mathbb{Z}_d$. $\square$

3.3 Composing the Symplectic and Pauli Normal Forms

We now combine the symplectic and Pauli normal forms to obtain a canonical normal form for arbitrary multi-qudit Clifford circuits. Using the semi-direct product structure of the Clifford group, this composition separates the symplectic action from the residual Pauli correction, yielding a unique representation up to global phase. In what follows, $A; B$ denotes the sequential composition of gates in circuit (diagrammatic) order, meaning that A is applied first and then B. This corresponds to the matrix product BA.

Definition 3.12. *An n-qudit Clifford circuit $N^{(n)}$ is a* Clifford norm form *if $N^{(n)} \equiv$*

$N_S^{(n)}; N_P^{(n)}$ *where* $N_P^{(n)}$ *is a Pauli normal form and* $N_S^{(n)}$ *is a symplectic normal form:*



Proposition 3.13. *For every n-qudit Clifford operator $C \in \mathcal{C}_n$, there exists a unique Clifford normal form circuit $N \in \mathfrak{C}_n$ such that $[\![N]\!]_u \equiv_p C$.*

Proof. We first prove the existence. By Proposition 3.8, there is a unique circuit $N_S \in \mathfrak{C}_n$ in the symmetric normal form such that $[\![N_S]\!]_u \bullet P \equiv_p C \bullet P$. Hence $[\![N_S]\!]_s = \pi_s(C)$. This implies there is a unique Pauli $P \in \mathcal{P}_n$ such that $P \cdot [\![N_S]\!]_u = C$. Then by Lemma 3.11, there is a unique circuit $N_P \in \mathfrak{C}_n$ such that $[\![N_P]\!]_u [\![N_S]\!]_u \equiv_p C$. It follows that $N \equiv N_S; N_P$ is a circuit in Clifford normal form such that $[\![N]\!]_u \equiv_p C$.

Now we prove the uniqueness. Suppose $N' \equiv N'_P N'_S$ is another circuit in $\mathfrak{C}_n$ in Clifford normal form such that $[\![N]\!]_u \equiv_p C$, where N'_S and N'_P are symplectic normal form circuit and Pauli normal form circuit respectively. It follows that $[\![N'_S]\!]_s = [\![N']\!]_s = \pi_s(C) = [\![N_S]\!]_s$. By the uniqueness of symplectic normal form, we must have $N'_S = N_S$. This also implies $[\![N'_S]\!]_u = [\![N_S]\!]_u$. Since $[\![N']\!]_u \equiv_p [\![N]\!]_u$, we must then have $[\![N'_P]\!]_u \equiv_p [\![N_P]\!]_u$ as Pauli operators in $\mathcal{P}_n$. By the uniqueness of Pauli normal form, we must have $N'_P \equiv N_P$. We conclude that $N' \equiv N$. □

Since each element of $\widehat{\mathcal{C}}_n$ is in bijection with a unique Clifford normal form as shown in Definition 3.12, counting these normal forms yields the cardinality of $\widehat{\mathcal{C}}_n$. Using Proposition 2.12, which characterises the Clifford phases as the integer powers of ω, we then obtain the cardinality of the Clifford group $\mathcal{C}_n$.

Remark 3.14. *By Lemmas 3.9 and 3.11 and proposition 3.13,*

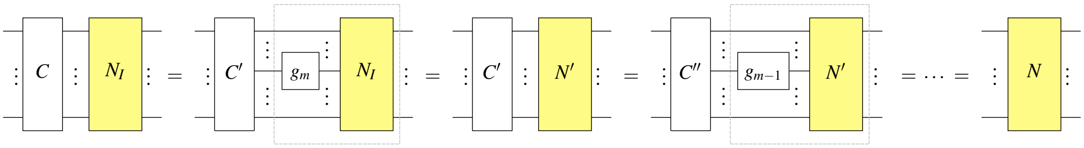
$$|\mathcal{C}_n| = d^{2n+1} \cdot |\mathrm{Sp}(2n, \mathbb{Z}_d)| = d^{2n+1} \prod_{k=1}^{n} \left(d^{2k} - 1\right) \cdot \left(d^{2k-1}\right).$$

4 A Complete Set of Multi-Qudit Clifford Relations

In this section, we derive a complete set of rewrite rules for multi-qudit Clifford circuits. In Section 4.1, we establish completeness at the symplectic level and then in Section 4.2, we lift these results to unitary Clifford circuits by accounting for Pauli corrections.

4.1 Establishing Symplectic Completeness

First, we show how to derive a complete set of relations that suffices to transform any multi-qudit Clifford circuits into its unique symplectic normal form. We call this the *Clifford* *normalisation* process, and it proceeds as follows. Let $C \in \mathfrak{C}_n$ be a circuit composed of gates $g_1, \ldots, g_m$. Append it on the right with the normal form of the identity circuit (N_I). Starting from the right-most gate g_m, iteratively 'push' each gate of C into the normal form, updating it at each step (e.g., from N_I to N'). This process continues until all gates have been absorbed into the updated normal form N:



Since the Pauli component plays no role in the present discussion, we suppress the distinction between the symplectic normal form N_S and the Pauli normal form N_P, and simply write N for the symplectic normal form.

Overview In (11) and (12), we present the symplectic normal form of the identity operator. Since a Clifford circuit is composed of H, S and CZ gates, it suffices to show how a normal form absorbs these gates and is updated to a new normal form. Locally, this reduces to demonstrating how to push a Clifford gate through each of the normal boxes specified in Figure 6. We call each such instance a *box relation*.

Figure 7.(a) gives an abstract illustration of *pushing through a normal box*, which results in an updated normal box M' and *residual dirty gates*, denoted as dir, appearing after M'. M' is uniquely determined by g and M which then uniquely fixes dir as some Clifford circuit. Figure 7.(b) shows a concrete example of pushing an H gate through a B box. In the appendix, Lemmas D.2 to D.14 show how we can find M' for all cases we need. These results are summarised in Figure 14.

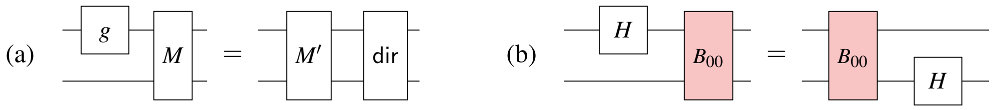


Figure 7: An abstract and concrete example of pushing a gate through a normal box M.

In total, we find 66 parametric box relations, which are listed in Section F. These relations cover every possible case of pushing through a normal box, making them sufficient to normalise any Clifford operator. Combined with Proposition 3.8, any two Clifford circuits that are equal as symplectic matrices can be rewritten into the same normal form using these box relations. This shows that the box relations are *complete* for multi-qudit Clifford circuits up to Pauli corrections.

Technical details We need to show that the Clifford normalisation terminates, and that when this happens there are no more gates left on the right-hand side of the normal form. Figure 8 provides a schematic summary of what a normal form could look like during the normalisation process. It accounts for all possible cases when pushing through a normal box.

Definition 4.1. *We say that a circuit is in* clean symplectic normal form *if it is of the form in Definition 3.2. A circuit is in* dirty symplectic normal form *if it is of the form in Figure 8, which has the structure of a normal form, but with any number of additional Clifford gates allowed in all the designated spots, subject to the following rules:*

- *H gates can be present at any wire labelled* ①*;*
- *S gates can be present at any wire labelled* ①, ②, *or* ③*;*
- *CZ gates can be present at any pair of adjacent wires, provided that the top wire is labelled* ①, ②, *or* ③, *and the bottom wire is labelled* ①.

In the context of a dirty normal form, we will refer to the normal boxes as clean, *and all the other H, S, X, Z, and CZ gates as* dirty.

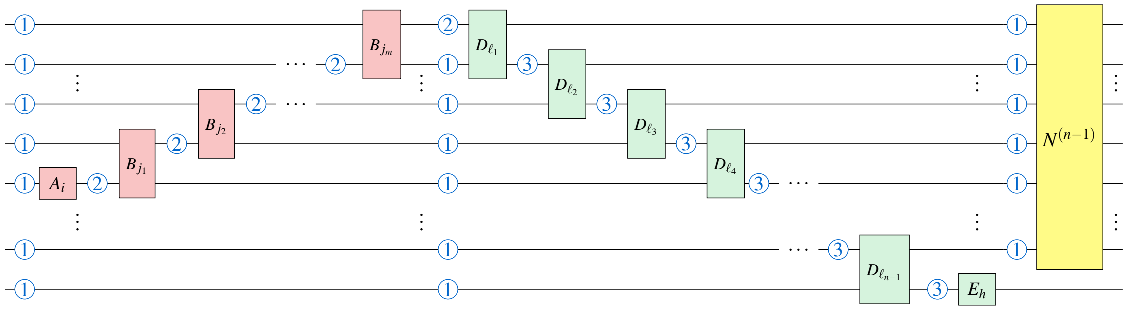


Figure 8: An example of an n-qudit dirty symplectic normal form. Here, $N^{(n-1)}$ is in dirty normal form subject to the same constraints as $N^{(n)}$.

The placement of dirty gates is restricted by which gates can be properly absorbed by each normal box, as this ultimately determines whether Clifford normalisation will terminate. In particular, the important point in Figure 8 is that there are no dirty gates allowed after the final E_h box or after the $N^{(n-1)}$ so that when the dirty gates get pushed through starting with the left-most, they will eventually get removed completely.

In order to ensure the restrictions of Definition 4.1 on the allowed dirty gates, we have the additional symmetries of the normal boxes in the third column of Figure 5. We can then confirm that all dirty gates follow the pattern described in Definition 4.1 by manually inspecting the box relations in Section F.

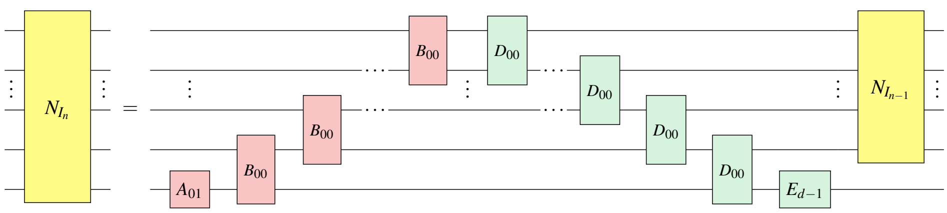
Lemma 4.2. *Any dirty normal form can be converted to its symplectic normal form by applying the box relations in Section F, in the left-to-right direction, a finite number of times.*

The proof is entirely analogous to that of [39, Lemma 4.2].

Proposition 4.3. *Consider a Clifford circuit composed of H, S, and CZ gates. Any such circuit can be converted to its symplectic normal form by using the box relations in Section F, together with the equations:*

(11)

Proof. For $1 \leq k \leq n$, let N_{I_k} be the normal form of the k-qudit identity operator. First note that



(12)

Indeed, an identity circuit can be converted to this normal form by applying (11) a finite number times. Appending the given Clifford circuit on the right with the normal form in (12), we obtain a dirty normal form, which can be converted to its normal form by Lemma 4.2. □

Propositions 3.8 and 4.3 show that qudit Clifford operators admit a presentation in terms of the generators H, S, and CZ, together with the derived generators from Figure 4, the normal boxes in Figure 6, and the 66 parametric box relations listed in Section F (up to Pauli corrections). This presentation, the number of rewrite rules, is highly redundant, so we extract a reduced set of relations, shown in Figure 9. The resulting rule set remains symplectically complete for multi-qudit Clifford circuits, while providing a more streamlined and intuitive account of qudit Clifford gate interactions.

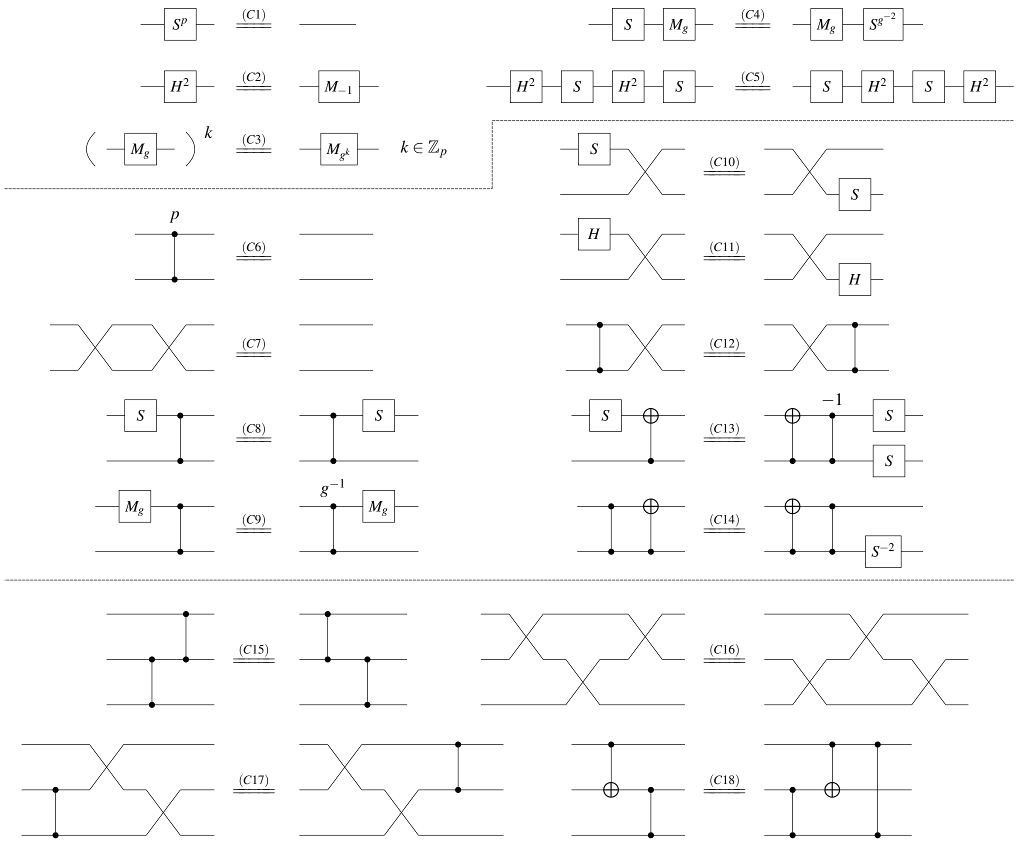


Figure 9: A complete set of symplectic relations for n-qudit Clifford circuits where d is an odd prime and $n \in \mathbb{N}$. Derived generators such as M_g, X, Z, SWAP, and CX are defined in Figure 2.

Theorem 4.4. *The* 18 *rewrite rules listed in Figure 9 are complete for qudit Clifford circuits up to Pauli corrections.*

Proof. It suffices to show that these 18 relations imply the full set of 66 parametric box relations. All proofs have been formalised, and the corresponding code is available in the GitHub repository [6]. □

4.2 From Symplectic to Unitary Completeness

In Section 4.1, we showed that Clifford circuits corresponding to the same symplectic matrix can be rewritten into the same normal form using the relations of Figure 9. To lift this result to unitary completeness, we account for the Pauli corrections that are not visible at the symplectic level. By combining symplectic completeness with the Pauli normal form from Section 3.2, we then obtain a complete and sound rule set of Clifford circuits for the unitary interpretation.

Our construction will follow a generic group-theoretic approach: given a group G with a normal subgroup N and a presentation of both N and the quotient group G/N, one can construct a presentation of G itself [32, Proposition 2.55]. This framework applies directly in our setting, since the Pauli group $\mathcal{P}_n$ is a normal subgroup of the Clifford group $\mathcal{C}_n$, and the quotient group $\mathcal{C}_n/\mathcal{P}_n$ is isomorphic to the symplectic group $\mathrm{Sp}(2n, \mathbb{Z}_d)$. Since $\mathcal{C}_n$ is isomorphic to the semidirect product of $\mathcal{P}_n$ by $\mathrm{Sp}(2n, \mathbb{Z}_d)$, lifting symplectic relations to unitary Clifford relations amounts to accounting for the associated Pauli relations, which leads to a particularly simple form of the general construction.

We first recall some notations from earlier. We will use $A; B$ to denote the sequential composition of gates in diagrammatic order, meaning that A is applied first and then B. For a Clifford circuit C, we write $[\![C]\!]_p$ for its projective unitary representation which ignores the global phase, and $[\![C]\!]_s$ for its interpretation as a symplectic matrix, i.e. where we ignore Paulis. In particular, in the symplectic representation, Paulis are all equal to the identity: $[\![P]\!]_s = \mathrm{id}$. Recall that if two circuits are equal in the unitary interpretation that they are then also equal in the symplectic interpretation.

Lemma 4.5. *Let C_1 and C_2 be two qudit Clifford circuits with equal symplectic representation: $[\![C_1]\!]_s = [\![C_2]\!]_s$. Then there exists a unique Pauli P (unique up to a global phase) such that $[\![C_1]\!]_p = [\![C_2; P]\!]_p$.*

In other words: given two Cliffords implementing the same symplectic map, we can find a unique "Pauli correction" that makes the Clifford unitaries exactly equal up to a global phase.

Definition 4.6. *Let $\mathcal{R}_s := \{C_l^{(i)} = C_r^{(i)}\}$ be a set of rewrite rules of Clifford circuits. We say it is* symplectically complete *when they are* sound *with respect to the symplectic interpretation, $[\![C_l^{(i)}]\!]_s = [\![C_r^{(i)}]\!]_s$ for all i, and* complete *for this interpretation: given circuits C_1 and C_2 such that $[\![C_1]\!]_s = [\![C_2]\!]_s$, we can find a sequence of rewrites from $\mathcal{R}_s$ that rewrites C_1 to C_2.*

Definition 4.7. *Let $\mathcal{R}_p$ be a set of rewrite rules of Clifford circuits. We say it is* Pauli complete *if they are sound for the projective interpretation, and they suffice to prove the commutation and cancellation identities for Paulis as well as how to push Paulis through all the Clifford generators. That is, $\mathcal{R}_p$ must prove the following equations:*

- $Z; X = X; Z$
- $Z^d = X^d = 1$
- $Z; H = H; X^{-1}$ *and* $X; H = H; Z$
- $Z; S = S; Z$ *and* $X; S = S; Z; X$
- $(Z \otimes I); CZ = CZ; (Z \otimes I)$
- $(X \otimes I); CZ = CZ; (X \otimes Z)$

Proposition 4.8. *The rewrite rules of Figure 1 are Pauli complete.*

Proof. There are only two non-trivial equations, $Z; X = X; Z$ and $X; S = S; Z; X$. We first prove all the other ones.

Define $S' := H^2; S; H^2$ so that $Z = S^{-1}; S'$ and (C5) simplifies to $S'; S = S; S'$. Note that combining (C1) and $H^4 = \mathrm{id}$ we get $(S')^d = \mathrm{id}$.

$Z^d = 1$ follows from repeated application of (C5) to rewrite $Z^d = (S'; S)^d = (S'^d); S^d = \mathrm{id}$. Since we have $X = H; Z; H^3$ we then also get $X^d = \mathrm{id}$, $Z; H = H; X^{-1}$ and $X; H = H; Z$.

$Z; S = S; Z$ follows directly from (C5). To prove $(Z \otimes I); CZ = CZ; (Z \otimes I)$ first note that $(S' \otimes I); CZ = CZ; (S' \otimes I)$ by using (C2) to convert the H^2 in S' to M_{-1}, (C10) to pushing this through the CZ, and then (C9) to push the S, then pushing another M_{-1} through CZ using (C10), and finally converting these back to H^2 using (C2). Then $(Z \otimes I); CZ = CZ; (Z \otimes I)$ easily follows as well. That $(X \otimes I); CZ = CZ; (X \otimes Z)$ follows similarly, but using (C14) instead.

Now, to prove $X; S = S; Z; X$. From (C3) with $k = 0$ we get $M_1 = \mathrm{id}$. Combining this with definition (T1) for $g = 1$ we get $\mathrm{id} = H; S; H; S; H; S$ which we can rewrite to $H; S; H; S = S^{-1}; H^3$. We can then write $X; S = H; S^{-1}; H; (H; S; H; S) = H; S^{-1}; H; S^{-1}; H^3$. On the other hand, using $S; Z = S'$ we get

$$\begin{aligned} S; Z; X &= S'; X = H^2; S; H^2; H; S^{-1}; H^2; S; H \\ &= H^2; S; H; (S')^{-1}; S; H \\ &= H^2; S; H; S; (S')^{-1}; H \\ &= H; (H; S; H; S); H^2; S^{-1}; H^3 \\ &= H; S^{-1}; H^3; H^2; S^{-1}; H^3 \\ &= H; S^{-1}; H; S^{-1}; H^3, \end{aligned}$$

which matches the expression we found for $X; S$.

Finally, to prove $Z; X = X; Z$ we will actually prove $Z; X; Z^{-1}; X^{-1} = \mathrm{id}$ which suffices. Using (T1) with $g = -1$ and (C2) gives us $H^2 = M_{-1} = H; S^{-1}; H; S^{-1}; H; S^{-1}; X; Z^{-1}$ where for the Pauli we get $(1 - g)/2 = 1$ and $(1 - g)/(2g) = -1$. Bringing everything to the other side gives us $X; Z^{-1} = S; H^3; S; H^3; S; H$. Further using $Z = H^2; S; H^2; S^{-1}$ and $X^{-1} = H^3; S^{-1}; H^2; S; H^3$ we calculate:

$$\begin{aligned} Z; (X; Z^{-1}); X^{-1} &= H^2; S; H^2; S^{-1}; S; H^3; S; H^3; S; H; H^3; S^{-1}; H^2; S; H^3 \\ &= H^2; S; H^2; H^3; S; H^3; S; S^{-1}; H^2; S; H^3 \\ &= H^2; S; H; S; H^3; H^2; S; H^3 \\ &= H^2; S; H; S; H; S; H^3 \\ &= H; (H; S; H; S; H; S); H^3 \\ &= H; H^3 \\ &= \mathrm{id} \end{aligned}$$

□

Proposition 4.9. *Let $\mathcal{R}_s := \{C_l^{(i)} = C_r^{(i)}\}$ be a symplectically complete set of rewrite rules and $\mathcal{R}_p$ a Pauli complete set of rewrite rules. Then we can efficiently find a modified set of rewrite rules $\mathcal{R}'_s := \{C_l^{(i)} = C_r^{(i)}; P^{(i)}\}$ where the $P^{(i)}$ are uniquely defined Paulis such that $\mathcal{R}'_s$ is sound in the projective interpretation and $\mathcal{R}'_S \cup \mathcal{R}_p$ is sound and complete for the projective interpretation.*

Proof. By Lemma 4.5 we can find unique Paulis to make the equations of $\mathcal{R}_s$ sound in the projective interpretation by changing them to $\mathcal{R}'_s := \{C_l^{(i)} = C_r^{(i)}; P^{(i)}\}$. Hence the equations $\mathcal{R}'_S \cup \mathcal{R}_p$ are indeed all sound for the projective interpretation. Now, to prove completeness let us assume we have two circuits C_1 and C_2 that are equal in the projective interpretation: $[\![C_1]\!]_p = [\![C_2]\!]_p$. We need to find a set of rewrites from $\mathcal{R}'_S \cup \mathcal{R}_p$ that rewrites C_1 into C_2.

Note that C_1 and C_2 are also necessarily equal in the symplectic interpretation. Hence, there is a series of rewrite rules from $\mathcal{R}_s$ that rewrites C_1 to C_2 in a symplectically sound way. Let's consider the first two rewrites in this sequence: $C_1 \rightsquigarrow C_1' \rightsquigarrow C_1''$ and suppose the rewrites used were $C_l^{(i)} = C_r^{(i)}$ and $C_l^{(j)} = C_r^{(j)}$. These rewrites were potentially applied locally in some larger context.

Now instead of $C_l^{(i)} = C_r^{(i)}$ we must use the projectively sound modification $C_l^{(i)} = C_r^{(i)}; P^{(i)}$. The LHS is still the same, and hence the rule still matches to C_1, but now we don't end up with C_1', but instead a modified circuit that has $P^{(i)}$ appearing at the end of the application of $C_r^{(i)}$. Using the rules from the Pauli complete $\mathcal{R}_p$ we can now push $P^{(i)}$ further to the right in the modified C_1' past all the generators. This changes $P^{(i)}$ to a potentially different Pauli, but otherwise does not change any of the structure of C_1'. At the end we hence end up with a circuit $C_1'; Q^{(i)}$ where the $Q^{(i)}$ is uniquely determined by $P^{(i)}$. We hence see that our rewrite rules prove $C_1 \rightsquigarrow C_1'; Q^{(i)}$.

Now, the second rewrite rule $C_l^{(j)} = C_r^{(j)}$ is modified to the projectively sound $C_l^{(j)} = C_r^{(j)}; P^{(j)}$. The LHS is still the same, and we still have C_1' (the only change is that it is composed with $Q^{(i)}$, but this does not change the fact whether the LHS matches or not). Hence, we can apply this modified rewrite rule to $C_1'; Q^{(i)}$. This then results in a modified C_1'' where there is the Pauli $P^{(j)}$ composed after the matching $C_r^{(j)}$. In the same way as above we can push this to the end of the circuit to end up with $C_1''; Q^{(j)}; Q^{(i)}$.

Repeating this procedure for all the rewrite rules from $\mathcal{R}_s$ that bring us from C_1 to C_2 gives us a set of now projectively sound rewrites from $\mathcal{R}'_S \cup \mathcal{R}_p$ that rewrites C_1 to $C_2; P$ where P is some big collection of Paulis. Now, because we know that C_1 and C_2 are projectively equal, we must have $[\![P]\!]_u = \mathrm{id}$, and so $\mathcal{R}_p$ must be able to rewrite P to the identity (empty) circuit. This then gives us a rewrite from C_1 to $C_2; P$ and finally to C_2. □

This statement allows us to extend a symplectically complete set of rewrites, like that of Theorem 4.4 to a projectively complete set of rewrite rules. We can apply a very similar procedure to boost this to a unitarily complete set of rewrite rules: We have $\widehat{\mathcal{C}}_n \cong \mathcal{C}_n/\langle\omega\rangle$ where $\langle\omega\rangle$ is a normal subgroup of $\mathcal{C}_n$. Hence, it suffices to find a complete set of rewrite rules for the group $\langle\omega\rangle$ and for "pushing" through the generators of $\mathcal{C}_n$ and then we can boost a complete set of rewrite rules of $\widehat{\mathcal{C}}_n$ to a complete set of rewrite rules of $\mathcal{C}_n$. A complete set of rewrite rules for $\langle\omega\rangle$ is easily found: we only need $\omega^d = 1$. Pushing ω through generators is equally trivial since it commutes with everything.

Theorem 4.10. *The rewrite rules in Figure 1 are* complete *for n-qudit Clifford circuits over any qudit dimension d that is an odd prime. That is, if two n-qudit unitary Clifford circuits C_1 and C_2 implement the same linear map, then the rewrite rules in Figure 1 can be used to derive their equality.*

Proof. Theorem 4.4 shows the rules are symplectically complete and Proposition 4.8 shows they are Pauli complete. Hence, by Proposition 4.9 they are projectively complete. □

5 Conclusion and Open Problems

We found a complete and finite equational theory for n-qudit Clifford circuits in every odd prime dimension, by presenting a small set of local rewrite rules that suffices to determine equality of Clifford unitaries. Our approach proceeds by introducing a symplectic normal form for Clifford circuits, deriving relations at the symplectic level, and then lifting these relations to circuit identities by accounting for Pauli corrections.

An immediate open question is to extend this completeness framework beyond odd prime dimensions, in particular to prime-power or arbitrary dimensions, where the arithmetic over $\mathbb{Z}_d$ and the behaviour of Clifford phases are more intricate. More broadly, it would be interesting to develop compact circuit-level equational theories for larger circuit fragments, such as Clifford–permutation–dihedral circuits. In the meantime, it will be fruitful to extend our software infrastructure as a verified backend for automated optimisation and synthesis of qudit circuit families.

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A Proofs from Section 2

In this appendix, we provide proofs that were omitted from Section 2 and document algebraic results related to the qudit Clifford group. Recall the summation of a geometric sequence for $a_n = a_1 q^{n-1}$ when $q \neq 1$ is

$$S_n = \frac{a_1(1-q^n)}{1-q}. \tag{13}$$

Lemma A.1. *Let ω be the primitive d-th root of unity and $k \in \mathbb{Z}_d$.*

$$\sum_{\ell=0}^{d-1} \omega^{k\ell} = \begin{cases} 0, & k \not\equiv_d 0 \\ p, & k \equiv_d 0. \end{cases} \tag{14}$$

Proof. When $k \equiv_d 0$, $\sum_{\ell=0}^{d-1} \omega^{k\ell} = \sum_{\ell=0}^{d-1} \omega^0 = p$. When $k \not\equiv_d 0$, according to (13),

$$\sum_{\ell=0}^{d-1} \omega^{k\ell} = \frac{1\left(1-(\omega^k)^d\right)}{1-\omega^k} = 0. \qquad \square$$

Lemma A.2. $\det(H) = 1$.

Recall that $H = \frac{1}{\lambda_d \sqrt{d}} F$, where $F = \sum_{j,\ell=0}^{d-1} \omega^{j\ell} |j\rangle\langle \ell|$ is a discrete Fourier transform. The determinant of F is a standard computation (see e.g., [42]), and the phase λ_d is chosen so that $\det(H) = 1$ (see also [47]). For completeness, we give a short elementary proof.

Proof. We first compute

$$\begin{aligned} \det(F) &= \prod_{0 \le \ell < j \le d-1} (\omega^j - \omega^\ell) = \prod_{0 \le \ell < j \le d-1} \omega^\ell(\omega^{j-\ell} - 1) \\ &= \left(\prod_{t=1}^{d-1} \prod_{\ell=1}^{d-1-t} \omega^\ell\right) \cdot \left(\prod_{t=1}^{d-1} (\omega^t - 1)^{d-t}\right) \\ &= \omega^{(d-1)d(d-2)/6} \left(\prod_{t=1}^{d-1} (\omega^t - 1)^{d-t}\right). \end{aligned}$$

Let $\xi := e^{\pi i/d}$. Since $\omega^t - 1 = 2i\xi^t \sin(\pi t/d)$ for all t, we obtain that

$$\begin{aligned} \prod_{t=1}^{d-1} (\omega^t - 1)^{d-t} &= (2i)^{\sum_{t=1}^{d-1}(d-t)} \cdot \xi^{\sum_{t=1}^{d-1} t(d-t)} \cdot \prod_{t=1}^{d-1} (\sin(\pi t/d))^{d-t} \\ &= (2i)^{d(d-1)/2} \cdot \xi^{(d-1)d(d-2)/6} \cdot \left(\prod_{t=1}^{(d-1)/2} \sin(\pi t/d)\right)^d, \end{aligned}$$

where the last equality pairs $(\sin(\pi t/d))^{d-t}$ with $(\sin(\pi(d-t)/d))^t$. Differentiating the cyclotomic polynomial $x^d - 1 = \prod_{t=0}^{d-1}(x - \omega^t)$ and evaluating at $x = 1$ yields $\prod_{t=1}^{d-1}(1-\omega^t) = p$. Note that $|1 - e^{i\theta}| = 2\sin(\theta/2)$ for all $\theta \in (0, 2\pi)$. This implies $\prod_{t=1}^{d-1} 2\sin(\pi t/d) = p$. Since $\sin(\pi t/d) > 0$ for all $1 \le t \le d-1$, by pairing t with $d-t$, we see that $\prod_{t=1}^{(d-1)/2} \sin(\pi t/d) = \left(\prod_{t=1}^{(d-1)} \sin(\pi t/d)\right)^{1/2} = 2^{-(d-1)/2} \cdot d^{1/2}$. Hence

$$\prod_{t=1}^{d-1} (\omega^t - 1)^{d-t} = i^{d(d-1)/2} \cdot \xi^{(d-1)d(d-2)/6} \cdot d^{d/2}.$$

It follows that

$$\det(F) = d^{d/2} e^{i\pi(d-1)/4}.$$

Since λ_d satisfies $\lambda_d^d = e^{i\pi(d-1)/4}$ for all odd prime d, the lemma follows. $\square$

Lemma A.3. $X^d = Z^d = 1$.

Proof. Note that, for any $0 \le j \le d-1$, we have $X^d|j\rangle = |j+d\rangle = |j\rangle$ and $Z^d|j\rangle = \omega^d|j\rangle = |j\rangle$. $\square$

Lemma A.4. $ZXZ^\dagger X^\dagger = \omega$.

Proof. Note that, for $0 \le j \le d-1$, we have

$$ZXZ^\dagger X^\dagger|j\rangle = ZXZ^\dagger|j-1\rangle = ZX\omega^{-(j-1)}|j-1\rangle = Z\omega^{-(j-1)}|j\rangle = \omega^{-(j-1)+j}|j\rangle = \omega|j\rangle.$$

$\square$

Proposition 2.5. $\frac{1}{\lambda_d\sqrt{d}} \in \mathbb{Z}[1/d, \omega]$.

Proof. This follows from the results in [47]. $\square$

Proposition 2.6. *The roots of unity in $\mathbb{Z}[1/d, \omega]$ are $\pm\omega^t$, $t \in \mathbb{Z}_d$.*

Proof. By Theorem 5.3 in [21], the roots of unity in $\mathbb{Q}(\omega)$ are exactly the complex numbers of the form $\pm\omega^t$, $t \in \mathbb{Z}_d$. Since $\mathbb{Z}[1/d, \omega] \subset \mathbb{Q}(\omega)$, this completes the proof. $\square$

Proposition 2.12. *For $0 \le j \le d-1$, $H^2|j\rangle = \lambda_d^{-2}|-j\rangle$, where the negation $-j$ is performed modulo d.*

Proof. We have

$$\begin{aligned} H^2|j\rangle &= H\left(\frac{1}{\lambda_d\sqrt{d}}\sum_{k=0}^{d-1}\omega^{kj}|k\rangle\right) \\ &= \frac{1}{\lambda_d\sqrt{d}}\sum_{k=0}^{d-1}\omega^{kj}H|k\rangle \\ &= \frac{1}{\lambda_d\sqrt{d}}\sum_{k=0}^{d-1}\omega^{kj}\left(\frac{1}{\lambda_d\sqrt{d}}\sum_{\ell=0}^{d-1}\omega^{\ell k}|\ell\rangle\right) \\ &= \frac{1}{\lambda_d^2 d}\sum_{\ell=0}^{d-1}\left(\sum_{k=0}^{d-1}\omega^{k(j+\ell)}\right)|\ell\rangle. \end{aligned}$$

Now consider the sum $\sum_{k=0}^{d-i}\omega^{k(j+\ell)}$. When $\ell \equiv -j \pmod d$, then $w^{k(j+\ell)} = 1$, so that the sum is equal to d. When $\ell \not\equiv -j \pmod d$, then $\omega^{j+\ell}$ is a primitive d-th root of unity, so that the sum is equal to 0. Hence,

$$H^2|j\rangle = \frac{1}{\lambda_d^2 d}\sum_{\ell=0}^{d-1}\left(\sum_{k=0}^{d-1}\omega^{k(j+\ell)}\right)|\ell\rangle = \frac{1}{\lambda_d^2}|-j\rangle,$$

as desired. $\square$

Definition A.5. *Let $k \in \mathbb{Z}_d$ and $f : \mathbb{Z}_d \mapsto \mathbb{Z}_d$ be an invertible function. A* generalised permutation matrix *is a matrix that maps $|k\rangle$ to $a_k|f(k)\rangle$, for some $a_k \in \mathbb{C}$ and $\|a_k\|^2 = 1$.*

Corollary A.6. *H^2 is a generalised permutation matrix.*

Lemma 2.18. $\det(S) = \det(H) = \det(CZ) = 1$.

Proof. By Lemma A.2, $\det(H) = 1$. Next, we show that $\det(S) = \det(\mathrm{CZ}) = 1$. Using straightforward properties of sums of integers, we have

$$\sum_{j=0}^{d-1} j^2 = \left(\sum_{j=1}^{d} j^2\right) - d^2 = \frac{d(d+1)(2d+1)}{6} - d^2. \tag{15}$$

$$\sum_{j=0}^{d-1} j = \frac{(0+(d-1))\,d}{2} = \frac{d(d-1)}{2}. \tag{16}$$

$$\sum_{j,\ell=0}^{d-1} j\ell = \sum_{j=0}^{d-1} j\left(\sum_{\ell=0}^{d-1}\ell\right) \overset{(16)}{=\!=} \sum_{j=0}^{d-1} j\left(\frac{d(d-1)}{2}\right) \overset{(16)}{=\!=} \left(\frac{d(d-1)}{2}\right)^2 = \frac{d^2(d-1)^2}{4}. \tag{17}$$

Then we have

$$\sum_{j=0}^{d-1} j^2 - \sum_{j=0}^{d-1} j \ \overset{(15)}{\underset{(16)}{=\!=}} \ \frac{d(d+1)(2d+1)}{6} - d^2 - \frac{d(d-1)}{2} = \frac{d(d^2-3d+2)}{3}. \tag{18}$$

$$\sum_{j=0}^{d-1} j^2 + \sum_{j=0}^{d-1} j \ \overset{(15)}{\underset{(16)}{=\!=}} \ \frac{d(d+1)(2d+1)}{6} - d^2 + \frac{d(d-1)}{2} = \frac{d(d^2-1)}{3}. \tag{19}$$

Since $\omega^d = 1$, it then follows that

$$\begin{aligned}
\det(S) &\overset{2.10}{=\!=} \prod_{j=0}^{d-1} \omega^{\frac{j(j-1)}{2}} = \omega^{\sum_{j=0}^{d-1}\frac{j(j-1)}{2}} = \omega^{\frac{1}{2}\left(\sum_{j=0}^{d-1} j^2 - \sum_{j=0}^{d-1} j\right)} \\
&\overset{(18)}{=\!=} \omega^{\frac{1}{2}\left(\frac{d(d^2-3d+2)}{3}\right)} = \omega^{d(d^2-3d+2)/6} = 1. \\
\det(\mathrm{CZ}) &= \prod_{j,\ell=0}^{d-1} \omega^{j\ell} = \omega^{\sum_{j,\ell=0}^{d-1} j\ell} \overset{(17)}{=\!=} \omega^{d^2(d-1)^2/4} = 1.
\end{aligned}$$

□

Lemma 2.25. *$H, S \in \widetilde{\mathcal{C}_1}$. Specifically, $HXH^\dagger = Z$, $HZH^\dagger = X^{-1}$, $SXS^\dagger = XZ$, $SZS^\dagger = Z$.*

Proof. One can check that $\lambda_d \lambda_d^\dagger = 1$. By Definitions 2.8 and 2.10, we have

$$\begin{aligned}
HXH^\dagger|j\rangle &= HX\left(\frac{1}{\lambda_d^\dagger\sqrt{d}}\sum_{k=0}^{d-1}\omega^{-kj}|k\rangle\right) \\
&= H\left(\frac{1}{\lambda_d^\dagger\sqrt{d}}\sum_{k=0}^{d-1}\omega^{-kj}|k+1\rangle\right) \\
&= \frac{1}{\lambda_d^\dagger\sqrt{d}}\sum_{k=0}^{d-1}\omega^{-kj}\left(\frac{1}{\lambda_d\sqrt{d}}\sum_{\ell=0}^{d-1}\omega^{(k+1)\ell}|\ell\rangle\right) \\
&= \frac{1}{d}\sum_{\ell=0}^{d-1}\omega^{\ell}\left(\sum_{k=0}^{d-1}\omega^{k(\ell-j)}\right)|\ell\rangle.
\end{aligned}$$

Now consider the sum $\sum_{k=0}^{d-i} \omega^{k(\ell-j)}$. When $\ell \equiv j \pmod{d}$, then $w^{k(\ell-j)} = 1$, so that the sum is equal to d. When $\ell \not\equiv j \pmod{d}$, then $\omega^{\ell-j}$ is a primitive d-th root of unity, so that the sum is equal to 0. Hence,

$$HXH^\dagger|j\rangle \;=\; \frac{1}{d}\sum_{\ell=0}^{d-1}\omega^{\ell}\left(\sum_{k=0}^{d-1}\omega^{k(\ell-j)}\right)|\ell\rangle \;=\; \omega^{j}|j\rangle \;=\; Z|j\rangle,$$

as desired. Note that, by Corollary 2.15, we have $H^2 = (H^\dagger)^2$. Hence, $HZH^\dagger = H^2X(H^\dagger)^2 = H^2XH^2$. Thus

$$HZH^\dagger|j\rangle \;=\; H^2XH^2|j\rangle \;=\; H^2X|{-j}\rangle \;=\; H^2|{-j+1}\rangle \;=\; |{-(-j+1)}\rangle \;=\; |j-1\rangle \;=\; X^\dagger|j\rangle.$$

Similarly, we have

$$\begin{aligned} S'XS'^\dagger|j\rangle \;&=\; S'X\omega^{-\frac{j(j+1)}{2}}|j\rangle \;=\; S'\omega^{-\frac{(j+1)(j+2)}{2}}|j+1\rangle \\ &=\; \omega^{-\frac{j(j+1)}{2}+\frac{(j+1)(j+2)}{2}}|j+1\rangle \;=\; \omega^{j+1}|j+1\rangle \;=\; \omega XZ|j\rangle. \end{aligned}$$

Finally, we have

$$\begin{aligned} SZS^\dagger|j\rangle \;&=\; SZ\omega-\frac{j(j-1)}{2}|j\rangle \;=\; S\omega-\frac{j(j-1)}{2}+1|j\rangle \\ &=\; \omega-\frac{j(j-1)}{2}+1+\frac{j(j-1)}{2}|j\rangle \;=\; \omega|j\rangle \;=\; Z|j\rangle, \end{aligned}$$

and

$$\begin{aligned} S'ZS'^\dagger|j\rangle \;&=\; SZ\omega-\frac{j(j+1)}{2}|j\rangle \;=\; S\omega-\frac{j(j+1)}{2}+1|j\rangle \\ &=\; \omega-\frac{j(j+1)}{2}+1+\frac{j(j+1)}{2}|j\rangle \;=\; \omega|j\rangle \;=\; Z|j\rangle. \end{aligned}$$

□

Lemma 2.14. *For every $g \in \mathbb{Z}_d^*$, M_g defined below satisfies $M_g|x\rangle = |gx\rangle$ for all $x \in \mathbb{Z}_d$.*

$$-\boxed{M_g}- \quad = \quad -\boxed{H}-\boxed{S^{g^{-1}}}-\boxed{H}-\boxed{S^{g}}-\boxed{H}-\boxed{S^{g^{-1}}}-\boxed{X^{(1-g)/2}}-\boxed{Z^{(1-g)/2g}}- \quad \cdot\lambda_d^2\left(\tfrac{g}{d}\right)_L \omega^{\frac{-g^2+4g+2}{8g}}$$

Proof. Fix a $g \in \mathbb{Z}_d^*$. By tracking the Pauli computation (and ignoring the global phase) we can check that $M_gZM_g^\dagger = Z^{g^{-1}}$ and $M_gXM_g^\dagger = X^g$. We must then have $M_g|0\rangle = M_gZ^g|0\rangle = Z^{g^{-1}g}M_g|0\rangle$, so that $M_g|0\rangle$ is a $+1$ eigenstate of Z and hence must be proportional to $|0\rangle$. In addition, $M_g|a\rangle = M_gX^a|0\rangle = X^{ga}M_g|0\rangle \propto X^{ga}|0\rangle = |ga\rangle$. Hence, there is a $c_g \in \mathbb{C}$ such that $M_g|x\rangle = c_g|gx\rangle$ for all $x \in \mathbb{Z}_d$ and in particular, $M_g|0\rangle = c_g|0\rangle$.

By direct computation we can also see that

$$M_g|0\rangle \;=\; \frac{1}{\lambda_d d^{3/2}}\omega^{-\frac{g^2-4g-2}{8g}}\left(\frac{g}{d}\right)_L\sum_h \omega^{t_h}\mu_h|h+\frac{1-g}{2}\rangle,$$

where

$$t_h \;=\; \frac{h(h-1)}{2g}+\frac{1-g}{2g}\left(h+\frac{1-g}{2}\right)$$

and

$$\mu_h \;=\; \sum_{k,j}\omega^{\frac{j(j-1)}{2g}+kj+\frac{k(k-1)}{2g}+kh}.$$

As we also have $M_g|0\rangle = c_g|0\rangle$ we must then have $\mu_h = 0$ for all $h \neq \frac{g-1}{2}$ so that $\sum_h \mu_h = \mu_{\frac{g-1}{2}}$. It follows that

$$\mu_{\frac{g-1}{2}} = \sum_h \mu_h = \sum_{k,j} \omega^{\frac{j(j-1)}{2g}+kj+\frac{k(k-1)}{2g}} \cdot \sum_h \omega^{kh} = d\left(\sum_j \omega^{\frac{j(j-1)}{2g}}\right) = \lambda_d d^{3/2}\left(\frac{g}{d}\right)_L \omega^{-\frac{1}{8g}}.$$

The second last equality uses that $\sum_h \omega^{kh} = \begin{cases} d & \text{if } k=0 \\ 0 & \text{otherwise} \end{cases}$. The last equality follows from the property of the quadratic Gauss sum $\sum_j \omega^{\frac{j^2}{2g}} = \lambda_d \sqrt{d}\left(\frac{g}{d}\right)_L$, where $\left(\frac{\cdot}{\cdot}\right)_L$ is the Legendre symbol. It follows that

$$c_g = \left(\frac{g}{d}\right)_L \omega^{-\frac{g^2-4g-2}{8g}} \frac{1}{\lambda_d d^{3/2}} \omega^{t_{\frac{g-1}{2}}} \mu_{\frac{g-1}{2}} = 1.$$

We conclude that $M_g|x\rangle = |gx\rangle$ for all $x \in \mathbb{Z}_d$. ∎

B Clifford Conjugation

Lemma B.1. *For all $a \in \mathbb{Z}_d$,*

$$X^a -\boxed{H}- Z^a \qquad\qquad Z^a -\boxed{H}- X^{-a} \qquad\qquad Z^{-a} -\boxed{H}- X^a \tag{20}$$

Proof. Based on Lemma 2.25, we proceed by repeatedly applying the action of H on a copies of Pauli X and Z.

$$\begin{aligned} HX^aH^\dagger &= (HXH^\dagger)^a \overset{(9)}{=\!=} Z^a \\ HZ^aH^\dagger &= (HZH^\dagger)^a \overset{(9)}{=\!=} \left(X^{-1}\right)^a = X^{-a} \\ H^\dagger X^a H &= (H^\dagger XH)^a \overset{(9)}{=\!=} \left(Z^{-1}\right)^a = Z^{-a} \end{aligned}$$ ∎

Lemma B.2.

$$\begin{array}{cccc} X -\boxed{H^2}- X^{-1} & Z -\boxed{H^2}- Z^{-1} & X^{-1} -\boxed{H^2}- X & Z^{-1} -\boxed{H^2}- Z \\ X -\boxed{H^3}- Z^{-1} & Z -\boxed{H^3}- X & X^{-1} -\boxed{H^3}- Z & \end{array} \tag{21}$$

Proof. Based on Lemma 2.25, we proceed by repeatedly applying the action of H on Pauli X and Z.

$$X -\boxed{H} \overset{Z}{\text{———}} \boxed{H}- X^{-1} \qquad\qquad X -\boxed{H} \overset{Z}{\text{———}} \boxed{H} \overset{X^{-1}}{\text{———}} \boxed{H}- Z^{-1}$$

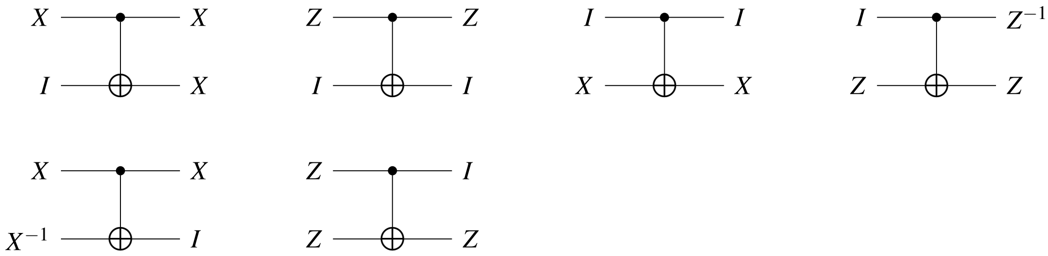
$$Z -\boxed{H} \overset{X^{-1}}{\text{———}} \boxed{H}- Z^{-1} \qquad\qquad Z -\boxed{H} \overset{X^{-1}}{\text{———}} \boxed{H} \overset{Z^{-1}}{\text{———}} \boxed{H}- X$$

$$X^{-1} -\boxed{H} \overset{Z^{-1}}{\text{———}} \boxed{H}- X \qquad\qquad X^{-1} -\boxed{H} \overset{Z^{-1}}{\text{———}} \boxed{H} \overset{X}{\text{———}} \boxed{H}- Z$$

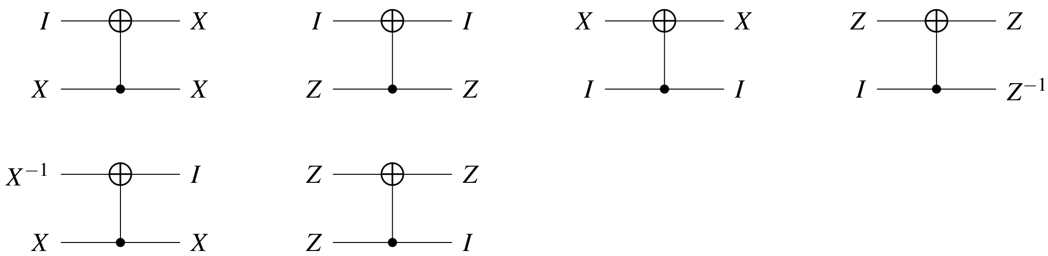
$$Z^{-1} -\boxed{H} \overset{X}{\text{———}} \boxed{H}- Z$$

∎

Lemma B.3.

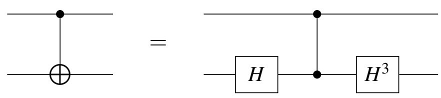


(22)

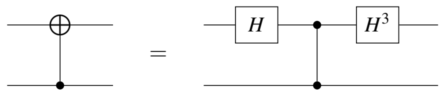


(23)

Proof. Recall that

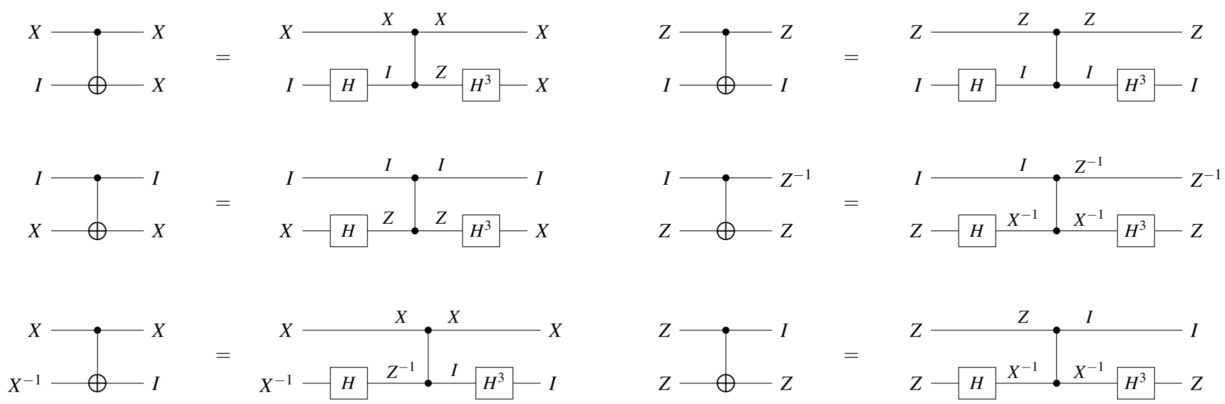


(*D*7)



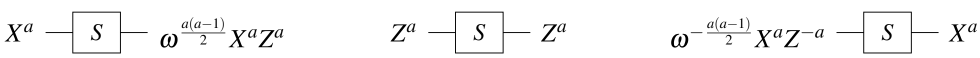
(*D*8)

Up to symmetry, it suffices to show (22). By (9), (10), and (21), we therefore have



∎

Lemma B.4. *For all $a \in \mathbb{Z}_d$,*



(24)

Proof. By Lemma 2.25, $SZ^aS^\dagger = \left(SZS^\dagger\right)^a = Z^a$. For the actions of S on X^a, we proceed by induction on a.

Base Case: When $a = 1$, the statement holds by Lemma 2.25.

Induction Hypothesis (IH): Suppose the statement holds for $a \geq 1$. That is,

$$SX^aS^\dagger = \omega^{\frac{a(a-1)}{2}} X^aZ^a, \qquad S^\dagger X^aS = \omega^{-\frac{a(a-1)}{2}} X^aZ^{-a}. \tag{25}$$

Induction Step: We want to show that the statement holds for $a+1$.

$$\begin{aligned} SX^{a+1}S^\dagger &= S\left(X^aX\right)S^\dagger = \left(SX^aS^\dagger\right)\left(SXS^\dagger\right) \overset{\text{IH}}{\underset{(9)}{=\!=}} \left(\omega^{\frac{a(a-1)}{2}}X^aZ^a\right)(XZ) \\ &\overset{(6)}{=\!=} \omega^{\frac{a(a-1)}{2}}\omega^aX^{a+1}Z^{a+1} = \omega^{\frac{(a+1)a}{2}}X^{a+1}Z^{a+1}. \end{aligned}$$

$$\begin{aligned} S^\dagger X^{a+1}S &= S^\dagger\left(X^aX\right)S = \left(S^\dagger X^aS\right)\left(S^\dagger XS\right) \overset{\text{IH}}{\underset{(9)}{=\!=}} \left(\omega^{-\frac{a(a-1)}{2}}X^aZ^{-a}\right)\left(XZ^{-1}\right) \\ &\overset{(6)}{=\!=} \omega^{-\frac{a(a-1)}{2}}\omega^{-a}X^{a+1}Z^{-(a+1)} = \omega^{-\frac{(a+1)a}{2}}X^{a+1}Z^{-(a+1)}. \end{aligned}$$

□

Lemma B.5. *For all $a \in \mathbb{Z}_d$,*

$$X \;-\!\boxed{S^a}\!-\; XZ^a \qquad\qquad Z \;-\!\boxed{S^a}\!-\; Z \qquad\qquad XZ^{-a} \;-\!\boxed{S^a}\!-\; X \tag{26}$$

Proof. By Lemma 2.25, $S^aZ\left(S^\dagger\right)^a = Z$. For the actions of S^a on X, we proceed by induction on a.

Base Case: When $a = 1$, the statement holds by Lemma 2.25.

Induction Hypothesis (IH): Suppose the statement holds for $a \geq 1$. That is,

$$S^aX\left(S^\dagger\right)^a = XZ^a, \qquad \left(S^\dagger\right)^aXS^a = XZ^{-a}. \tag{27}$$

Induction Step: We want to show that the statement holds for $a+1$.

$$\begin{aligned} S^{a+1}X\left(S^\dagger\right)^{a+1} &= S\left(S^aX\left(S^\dagger\right)^a\right)S^\dagger \overset{\text{IH}}{=\!=} S\left(XZ^a\right)S^\dagger \\ &= \left(SXS^\dagger\right)\left(SZ^aS^\dagger\right) \overset{(9)}{=\!=} XZZ^a = XZ^{a+1} \\ \left(S^\dagger\right)^{a+1}XS^{a+1} &= S^\dagger\left(\left(S^\dagger\right)^aXS^a\right)S \overset{\text{IH}}{=\!=} S^\dagger\left(XZ^{-a}\right)S \\ &= \left(S^\dagger XS\right)\left(S^\dagger Z^{-a}S\right) \overset{(9)}{=\!=} XZ^{-1}Z^{-a} = XZ^{-(a+1)}. \end{aligned}$$

□

Lemma B.6. *For all $a, b \in \mathbb{Z}_d$,*

$$X^a \;-\!\boxed{S^b}\!-\; \omega^{\frac{ab(a-1)}{2}}X^aZ^{ab} \qquad\qquad Z^a \;-\!\boxed{S^b}\!-\; Z^a \qquad\qquad \omega^{-\frac{ab(a-1)}{2}}X^aZ^{-ab} \;-\!\boxed{S^b}\!-\; X^a \tag{28}$$

Proof. By Lemma 2.25, $S^b Z^a \left(S^\dagger\right)^b = Z^a$. For the actions of S^b on X^a, we proceed by induction on b.

Base Case: When $b = 1$, the statement holds by Lemma B.4.

$$X^a \;-\!\boxed{S}\!-\; \omega^{\frac{a(a-1)}{2}} X^a Z^a \qquad\qquad \omega^{-\frac{a(a-1)}{2}} X^a Z^{-a} \;-\!\boxed{S}\!-\; X^a$$

Induction Hypothesis (IH): Suppose the statement holds for $b \geq 1$. That is,

$$S^b X^a \left(S^\dagger\right)^b = \omega^{\frac{ab(a-1)}{2}} X^a Z^{ab}, \qquad \left(S^\dagger\right)^b X^a S^b = \omega^{-\frac{ab(a-1)}{2}} X^a Z^{-ab}. \tag{29}$$

Induction Step: We want to show that the statement holds for $b + 1$.

$$\begin{aligned}
S^{b+1} X^a \left(S^\dagger\right)^{b+1} &= S\left(S^b X^a \left(S^\dagger\right)^b\right) S^\dagger \overset{\text{IH}}{=\!=} S\left(\omega^{\frac{ab(a-1)}{2}} X^a Z^{ab}\right) S^\dagger \\
&= \omega^{\frac{ab(a-1)}{2}} \left(S X^a S^\dagger\right)\left(S Z^{ab} S^\dagger\right) \overset{(24)}{=\!=} \omega^{\frac{ab(a-1)}{2}} \left(\omega^{\frac{a(a-1)}{2}} X^a Z^a\right) Z^{ab} \\
&= \omega^{\frac{a(b+1)(a-1)}{2}} X^a Z^{a(b+1)}. \\
\left(S^\dagger\right)^{b+1} X^a S^{b+1} &= S^\dagger \left(\left(S^\dagger\right)^b X^a S^b\right) S \overset{\text{IH}}{=\!=} S^\dagger \left(\omega^{-\frac{ab(a-1)}{2}} X^a Z^{-ab}\right) S \\
&= \omega^{-\frac{ab(a-1)}{2}} \left(S^\dagger X^a S\right)\left(S^\dagger Z^{-ab} S\right) \\
&\overset{(24)}{=\!=} \omega^{-\frac{ab(a-1)}{2}} \left(\omega^{-\frac{a(a-1)}{2}} X^a Z^{-a}\right) Z^{-ab} \\
&= \omega^{-\frac{a(b+1)(a-1)}{2}} X^a Z^{-a(b+1)}.
\end{aligned}$$

$\square$

Lemma B.7. *For all $a, b \in \mathbb{Z}_d$,*

$$\left(X Z^{-a}\right)^b = \omega^{-\frac{ab(b-1)}{2}} X^b Z^{-ab}. \tag{30}$$

Proof. We proceed by induction on b.

Base Case: When $b = 1$, the statement holds trivially.

Induction Hypothesis (IH): Suppose the statement holds for $b \geq 1$. That is,

$$\left(X Z^{-a}\right)^b = \omega^{-\frac{ab(b-1)}{2}} X^b Z^{-ab}. \tag{31}$$

Induction Step: We want to show that the statement holds for $b + 1$.

$$\begin{aligned}
\left(X Z^{-a}\right)^{b+1} &= \left(X Z^{-a}\right)^b \left(X Z^{-a}\right) \overset{\text{IH}}{=\!=} \left(\omega^{-\frac{ab(b-1)}{2}} X^b Z^{-ab}\right)\left(X Z^{-a}\right) \\
&\overset{(6)}{=\!=} \omega^{-\frac{ab(b-1)}{2}} \omega^{-ab} X^{b+1} Z^{-ab} Z^{-a} = \omega^{-\frac{a(b+1)b}{2}} X^{b+1} Z^{-a(b+1)}.
\end{aligned}$$

$\square$

Lemma B.8. *Let p be an odd prime and $\omega = e^{2\pi i/p}$.*

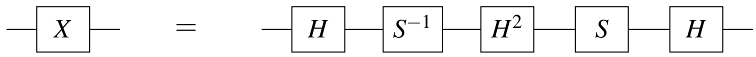
$$-1 = (-\omega)^p \tag{D1}$$
$$\omega = (-\omega)^{p+1} \tag{D2}$$

Proof.

$$(D1).\text{RHS} = (-\omega)^p = -\omega^p \overset{(5)}{=\!=} -1 = (D1).\text{LHS}.$$

$$(D2).\text{RHS} = (-\omega)^{p+1} = (-\omega)^p(-\omega) \overset{(D1)}{=\!=} (-1)(-\omega) = \omega = (D2).\text{LHS}. \qquad \square$$

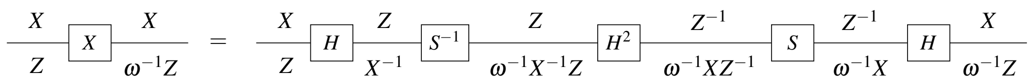
Lemma B.9.



(D4)

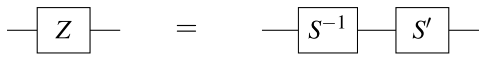
Proof. By (6), it suffices to check that RHS has the following action on X and Z.

Based on (9) and Lemmas B.1, B.2 and B.4 to B.6, we have



$\square$

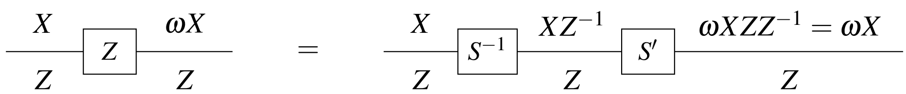
Lemma B.10.



(D5)

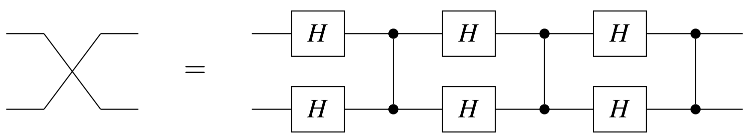
Proof. By (6), it suffices to check that RHS has the following action on X and Z.

Based on (9) and Lemma B.5, we have



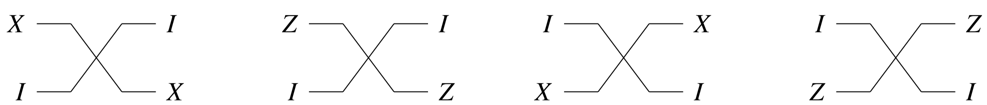
$\square$

Lemma B.11.



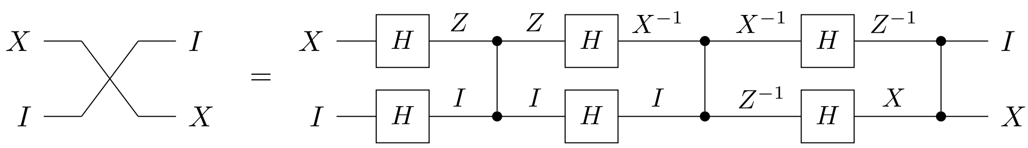
(D6)

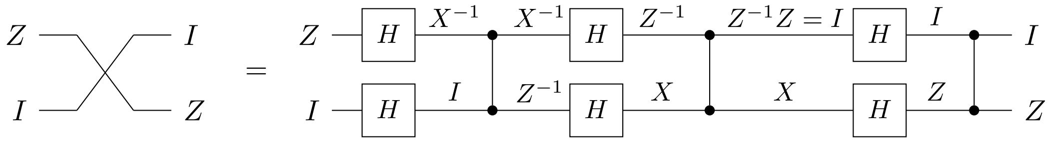
Proof. We need to show that the RHS of (D6) acts on the Pauli generators as specified by (32).



(32)

Up to symmetry, it suffices to show the action of RHS on $X \otimes I$ and $Z \otimes I$. By (9) and (10), we can show that





$\square$

C Actions of the Normal Boxes

Figures 10, 11, 12, 13 list the actions of all the normal boxes on all Pauli generators up to global phase.

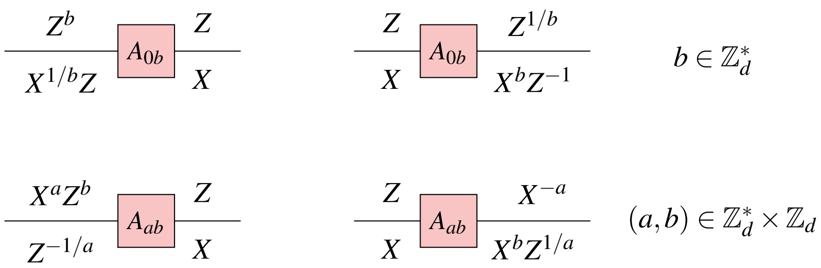


Figure 10: Up to a global phase, the action of A boxes on the Pauli generators.

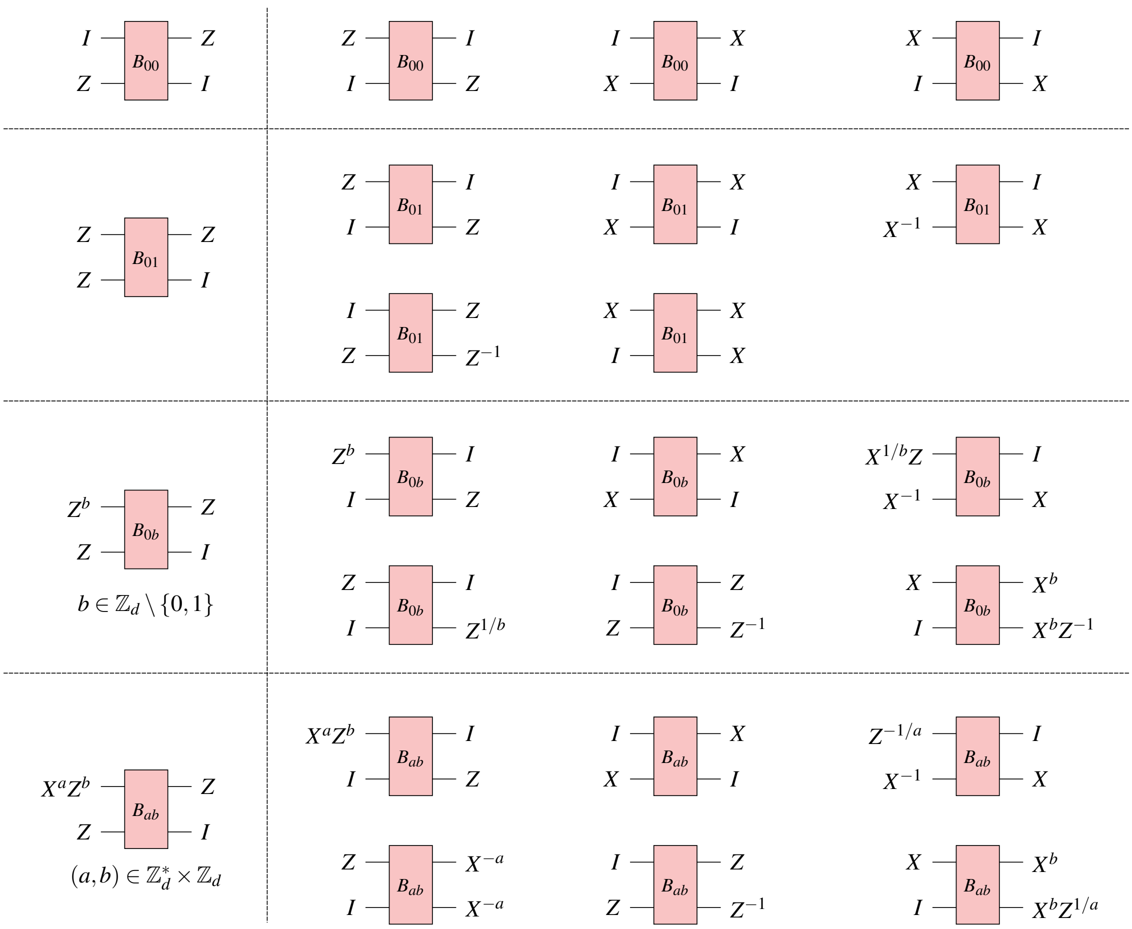


Figure 11: The action of B boxes on the Pauli generators up to a global phase,.

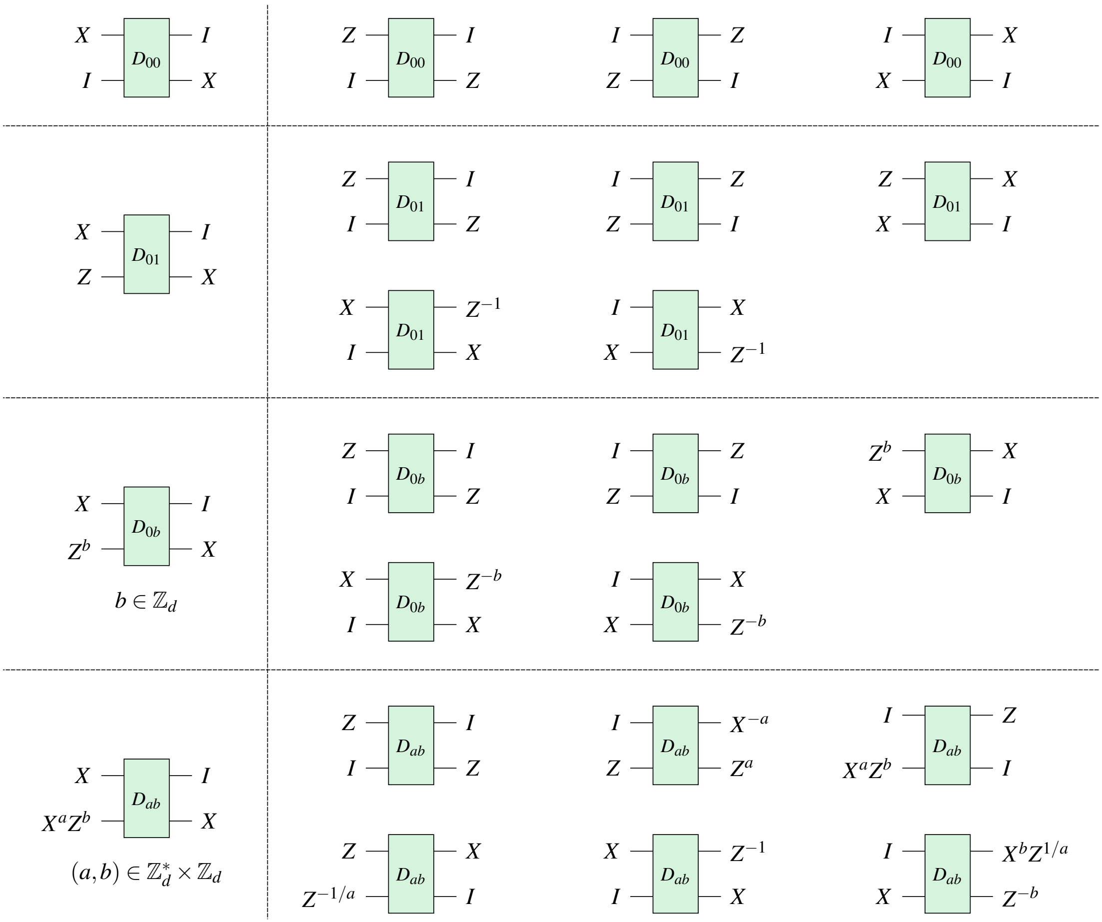


Figure 12: The action of D boxes on the Pauli generators up to a global phase.

Figure 13: The action of E boxes on the Pauli generators up to a global phase.

D Pushing Through a Normal Box

The goal of this section is to establish what we call the proof-of-principle box relations as shown in Figure 14. I.e. that the dirty gates resulting from pushing the generators through the boxes that occur in the normal form have the shape that we want them to have.

Recall that we use $\mathcal{C}_n$ to denote the set of n-qudit Clifford circuits, where p is an odd prime, and we use $\mathcal{B}_n$ to denote the set of n-qudit Pauli operators which generate the n-qudit Pauli group $\mathcal{P}_n$.

Proposition D.1. *Let $k_1, k_2 \in \mathbb{N}$, and $C \in \mathcal{C}_{k_1+k_2}$. Suppose that for every $B \in \mathcal{B}_{k_1}$ there exists a $B' \in \mathcal{P}_{k_1}$ such that*

$$C(B \otimes I)C^\dagger = B' \otimes I. \tag{33}$$

Then there exist $C_1 \in \mathcal{C}_{k_1}$ and $C_2 \in \mathcal{C}_{k_2}$ such that $C = C_1 \otimes C_2$, up to scalar. Here, I is the identity map on $\mathbb{C}^{d^{k_2}}$.

Proof. By Equation (33), the mapping $\Phi : \mathcal{B}_{k_1} \to \mathcal{M}_{d^{k_1}}(\mathbb{C})$ sending $B \mapsto \mathrm{tr}_2\left(C(B \otimes I)C^\dagger\right)$ extends to an automorphism Φ of $\mathcal{P}_{k_1}$ such that $C(B \otimes I)C^\dagger = \Phi(B) \otimes I$, where tr_2 is the

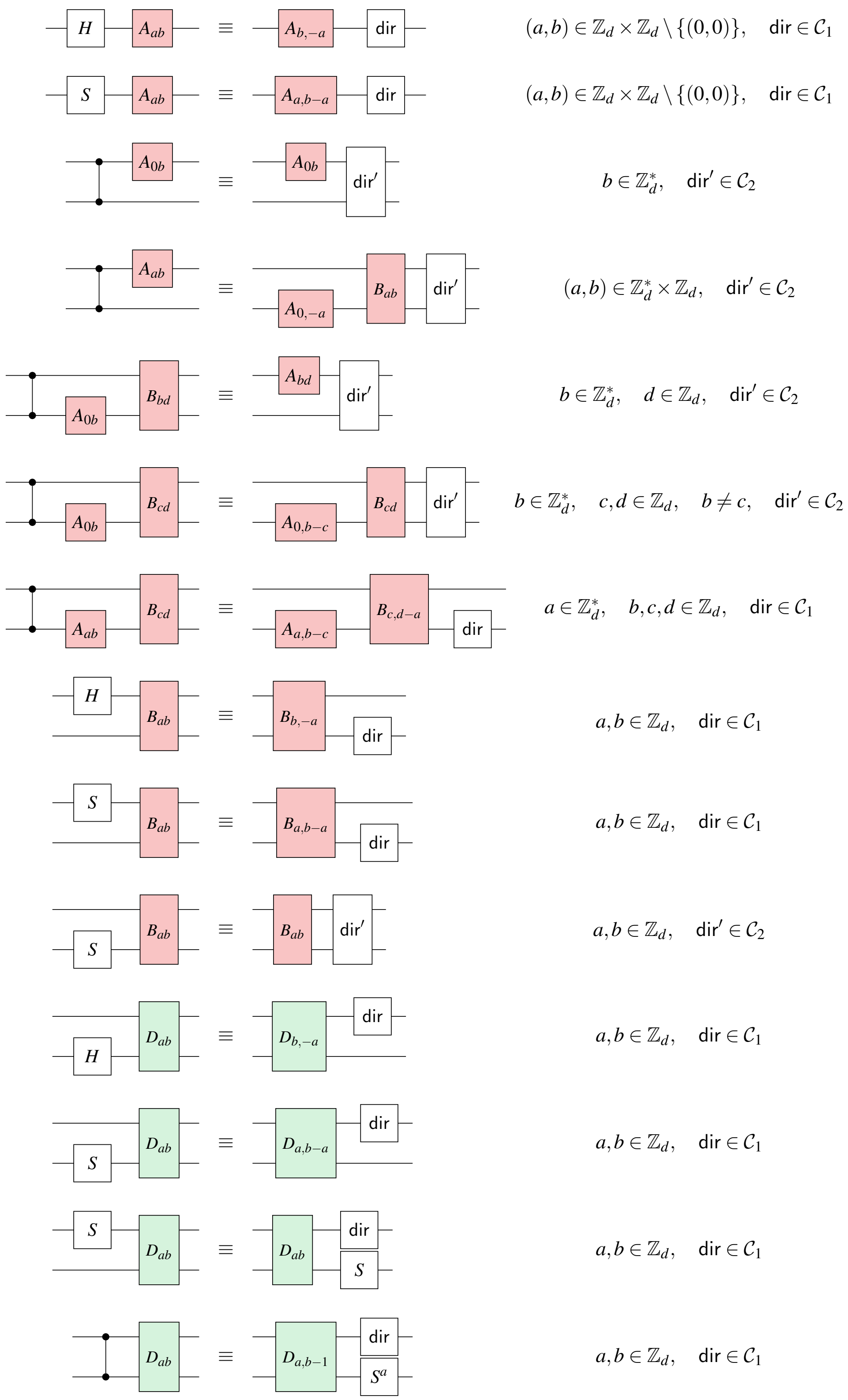
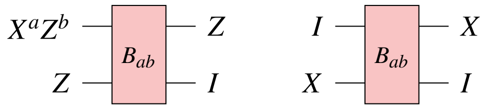


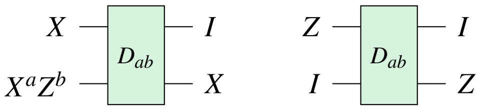
Figure 14: Proof-of-principle single- and two-qudit box relations. They demonstrate the structure of residual dirty gates in all scenarios when we normalise a two-qudit Clifford operator through "pushing-through-a-normal box".

normalized partical trace $\frac{1}{d^{k_2}} id \otimes \mathrm{Tr}$ that traces out the second system $\mathcal{M}_{d^{k_2}}(\mathbb{C})$. Hence, there exists a $C_1 \in \mathcal{C}_{k_1}$ such that $C_1 B C_1^\dagger = \Phi(B)$. The rest follows from [39, Lemma E.6]. $\square$

Firstly, note that (34) and (35) follow from Figures 11 and 12. Based on this, we can derive proof-of-principle box relations as shown in Figure 14.

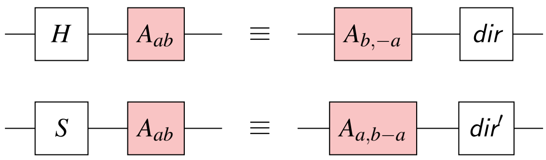


(34)



(35)

Lemma D.2. *For all $(a,b) \in \mathbb{Z}_d \times \mathbb{Z}_d \setminus \{(0,0)\}$, there exist* dir, dir$'$ $\in \mathfrak{C}_1$ *such that*

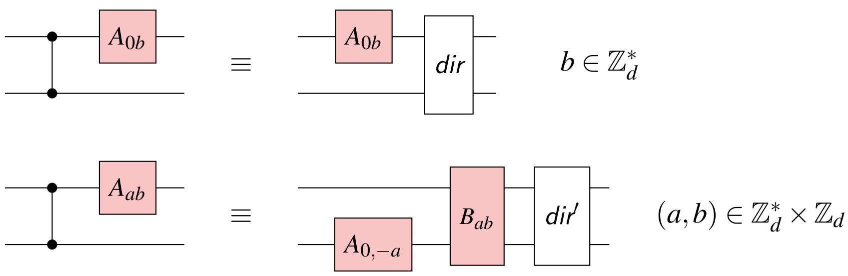


(36)

Proof. Computing the preimage of Z in the LHS of (36) yields

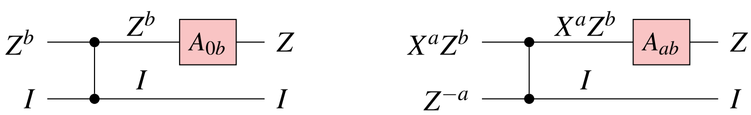
By the uniqueness of the symplectic normal form, X^bZ^{-a} and X^aZ^{b-a} imply that the updated normal boxes should be those in the RHS of (36). $\square$

Lemma D.3. *For all $(a,b) \in \mathbb{Z}_d \times \mathbb{Z}_d \setminus \{(0,0)\}$, there exist* dir, dir$'$ $\in \mathfrak{C}_2$ *such that*



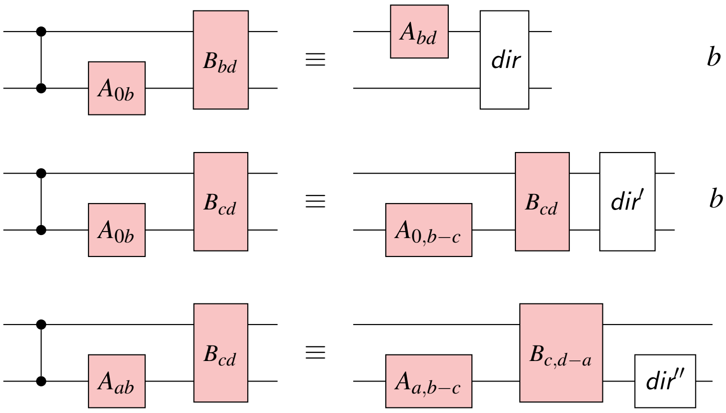
(37)

Proof. Computing the preimage of $Z \otimes I$ in the LHS of (37) yields



By the uniqueness of the symplectic normal form, $Z^b \otimes I$ and $X^aZ^b \otimes Z^{-a}$ imply that the updated normal boxes should be those in the RHS of (37). $\square$

Lemma D.4. *For all* $(a,b) \in \mathbb{Z}_d \times \mathbb{Z}_d \setminus \{(0,0)\}$ *and* $(c,d) \in \mathbb{Z}_d \times \mathbb{Z}_d$, *there exist dir, dir′* $\in \mathfrak{C}_2$ *and dir″* $\in \mathfrak{C}_1$ *such that*

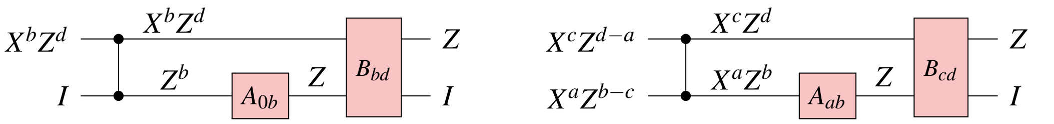


$b \in \mathbb{Z}_d^*, \quad d \in \mathbb{Z}_d$

$b \in \mathbb{Z}_d^*, \quad c,d \in \mathbb{Z}_d, \quad b \neq c \quad (38)$

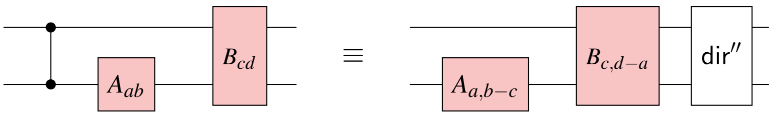
$a \in \mathbb{Z}_d^*, \quad b,c,d \in \mathbb{Z}_d$

Proof. Computing the preimage of $Z \otimes I$ in the LHS of (38) yields

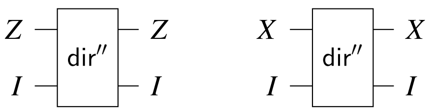


By the uniqueness of the symplectic normal form, $X^bZ^d \otimes I$ and $X^cZ^{d-a} \otimes X^aZ^{b-c}$ imply that the updated normal boxes should be those in the RHS of (38).

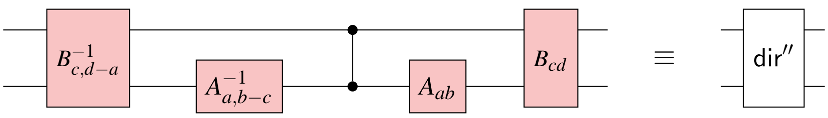
Next, we consider the case when $a \neq 0$. Suppose that for some dir″ $\in \mathfrak{C}_2$,



By Proposition D.1, it remains to show that

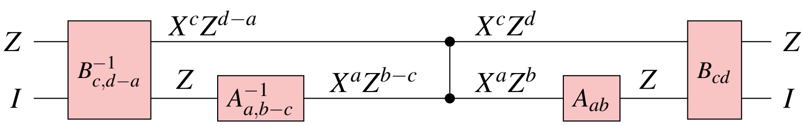


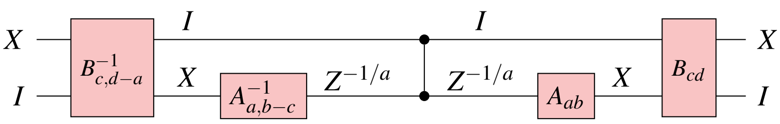
Equivalently, we can compute the preimage of $Z \otimes I$ and $X \otimes I$ in the LHS of (39).



(39)

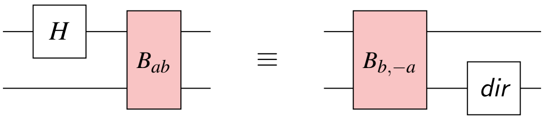
By (34), we have





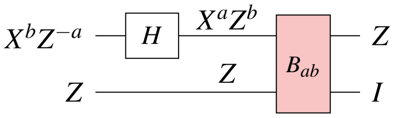
This completes the proof. □

Lemma D.5. *For all $a, b \in \mathbb{Z}_d$, there exists* $\mathsf{dir} \in \mathfrak{C}_1$ *such that*

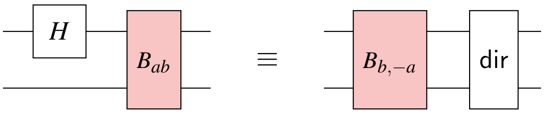


(40)

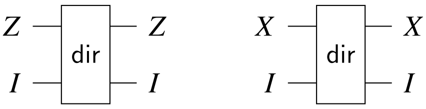
Proof. Computing the preimage of $Z \otimes I$ in the LHS of (40) yields



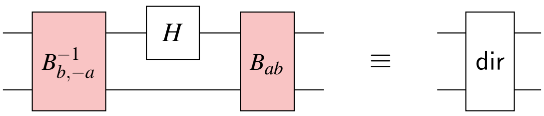
By the uniqueness of the symplectic normal form, $X^bZ^{-a} \otimes Z$ implies that the updated normal box should be the one in the RHS of (40). Suppose that for some $\mathsf{dir} \in \mathfrak{C}_2$,



By Proposition D.1, it remains to show that

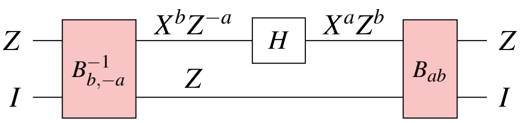


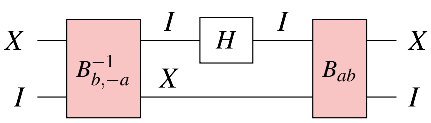
Equivalently, we can compute the preimage of $Z \otimes I$ and $X \otimes I$ in the LHS of (41).



(41)

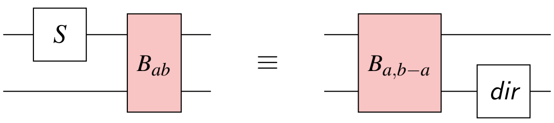
By (34), we have





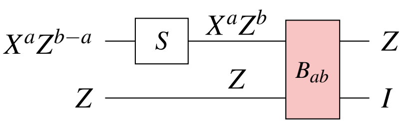
This completes the proof. □

Lemma D.6. *For all $a, b \in \mathbb{Z}_d$, there exists* $\mathsf{dir} \in \mathfrak{C}_1$ *such that*

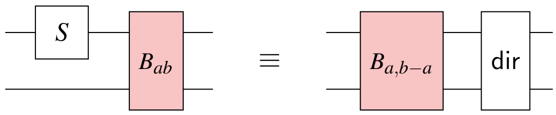


(42)

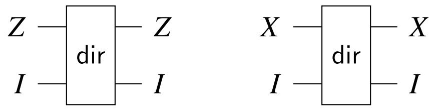
Proof. Computing the preimage of $Z \otimes I$ in the LHS of (42) yields



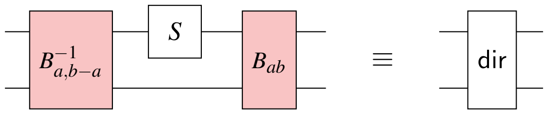
By the uniqueness of the symplectic normal form, $X^a Z^{b-a} \otimes Z$ implies that the updated normal box should be the one in the RHS of (42). Suppose that for some $\mathsf{dir} \in \mathfrak{C}_2$,



By Proposition D.1, it remains to show that

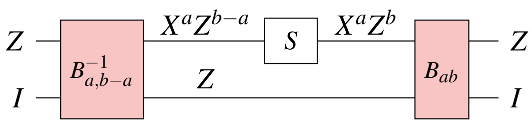


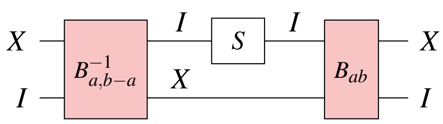
Equivalently, we can compute the preimage of $Z \otimes I$ and $X \otimes I$ in the LHS of (43).



(43)

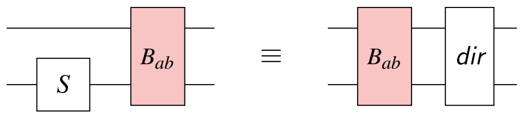
By (34), we have





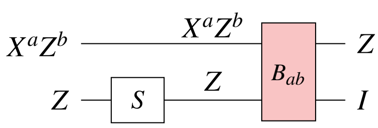
This completes the proof. ☐

Lemma D.7. *For all $a, b \in \mathbb{Z}_d$, there exists $\mathsf{dir} \in \mathfrak{C}_2$ such that*



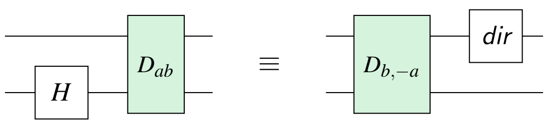
(44)

Proof. Computing the preimage of $Z \otimes I$ in the LHS of (44) yields



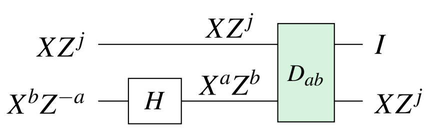
By the uniqueness of the symplectic normal form, $X^a Z^b \otimes Z$ implies that the updated normal box should be the one in the RHS of (44). ☐

Lemma D.8. *For all $a, b \in \mathbb{Z}_d$, there exists $\mathsf{dir} \in \mathfrak{C}_1$ such that*

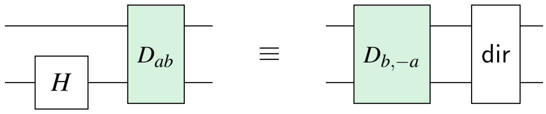


(45)

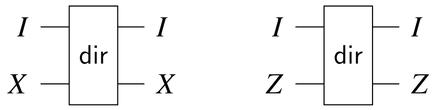
Proof. For any $j \in \mathbb{Z}_d$, computing the preimage of $I \otimes XZ^j$ in the LHS of (45) yields



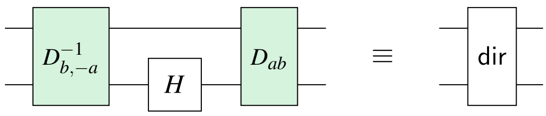
By the uniqueness of the symplectic normal form, $XZ^j \otimes X^bZ^{-a}$ implies that the updated normal box should be the one in the RHS of (45). Suppose that for some $\mathsf{dir} \in \mathfrak{C}_2$,



By Proposition D.1, it remains to show that

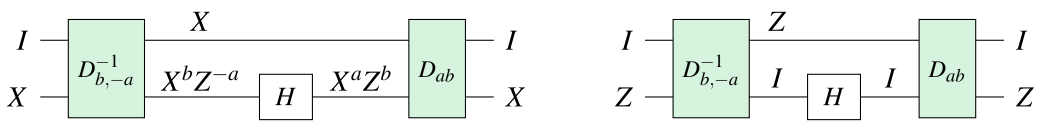


Equivalently, we can compute the preimage of $I \otimes X$ and $I \otimes Z$ in the LHS of (46).



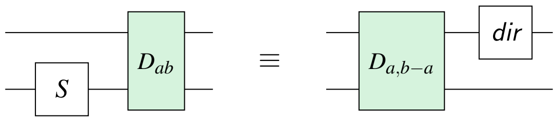
(46)

By (35), we have



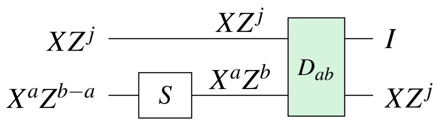
This completes the proof. $\square$

Lemma D.9. *For all $a, b \in \mathbb{Z}_d$, there exists $\mathsf{dir} \in \mathfrak{C}_1$ such that*

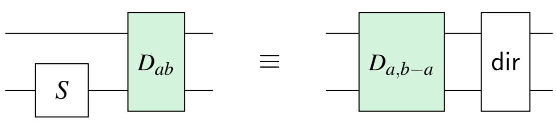


(47)

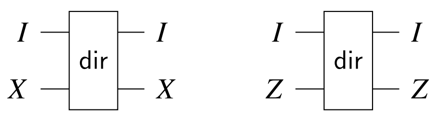
Proof. For any $j \in \mathbb{Z}_d$, computing the preimage of $I \otimes XZ^j$ in the LHS of (47) yields



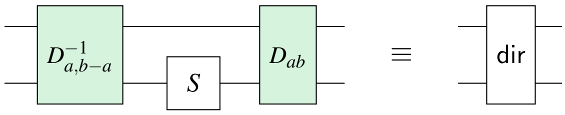
By the uniqueness of the symplectic normal form, $XZ^j \otimes X^aZ^{b-a}$ implies that the updated normal box should be the one in the RHS of (47). Suppose that for some $\mathsf{dir} \in \mathfrak{C}_2$,



By Proposition D.1, it remains to show that

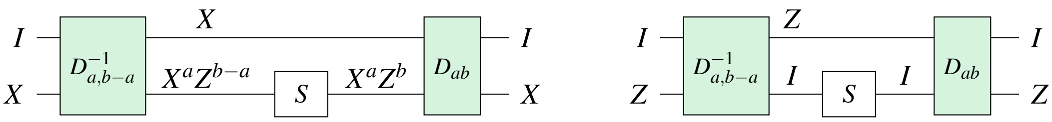


Equivalently, we can compute the preimage of $I \otimes X$ and $I \otimes Z$ in the LHS of (48).



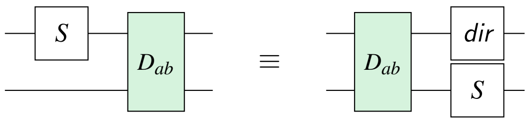
(48)

By (35), we have



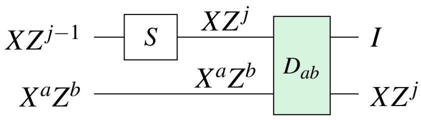
This completes the proof. □

Lemma D.10. *For all $a, b \in \mathbb{Z}_d$, there exists $\mathsf{dir} \in \mathfrak{C}_1$ such that*

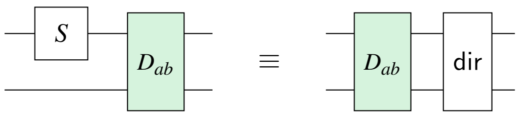


(49)

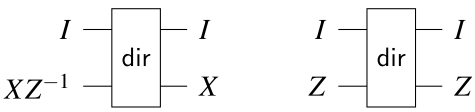
Proof. For any $j \in \mathbb{Z}_d$, computing the preimage of $I \otimes XZ^j$ in the LHS of (49) yields



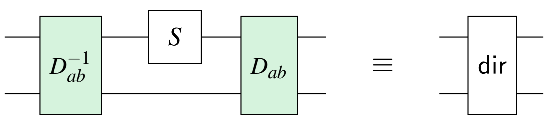
By the uniqueness of the symplectic normal form, $XZ^{j-1} \otimes X^aZ^b$ implies that the updated normal box should be the one in the RHS of (49). Suppose that for some $\mathsf{dir} \in \mathfrak{C}_2$,



By Proposition D.1, it remains to show that

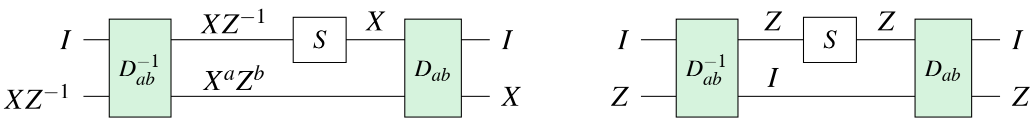


Equivalently, we can compute the preimage of $I \otimes X$ and $I \otimes Z$ in the LHS of (50).



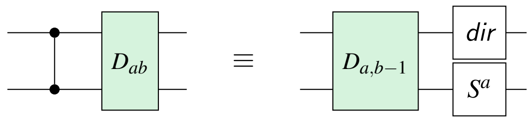
(50)

By (35), we have



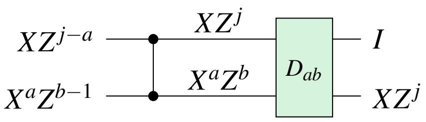
Since S sends XZ^{-1} to X and Z to Z, this completes the proof. □

Lemma D.11. *For all $a, b \in \mathbb{Z}_d$, there exists $\mathit{dir} \in \mathfrak{C}_1$ such that*

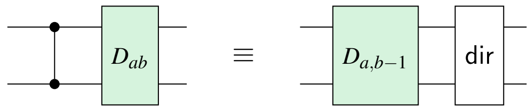


(51)

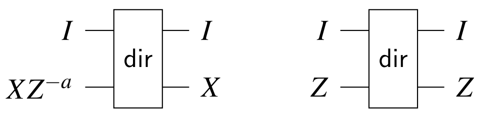
Proof. For any $j \in \mathbb{Z}_d$, computing the preimage of $I \otimes XZ^j$ in the LHS of (51) yields



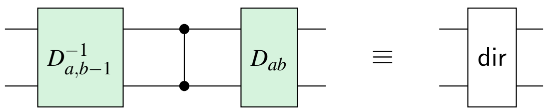
By the uniqueness of the symplectic normal form, $XZ^{j-a} \otimes X^aZ^{b-1}$ implies that the updated normal box should be the one in the RHS of (51). Suppose that for some $\mathsf{dir} \in \mathfrak{C}_2$,



By Proposition D.1, it remains to show that

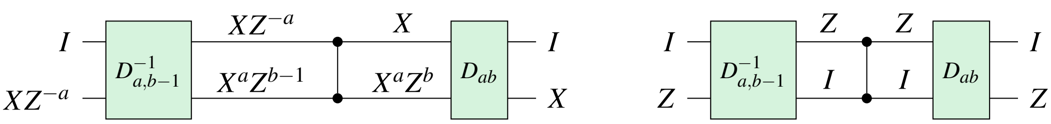


Equivalently, we can compute the preimage of $I \otimes X$ and $I \otimes Z$ in the LHS of (52).



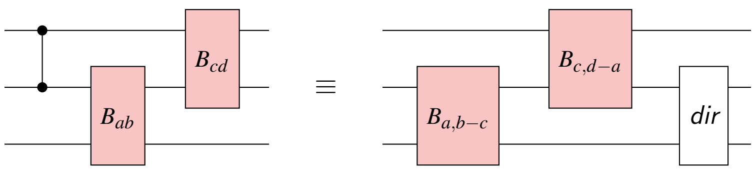
(52)

By (35), we have



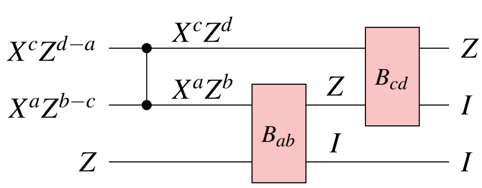
Since S^a sends XZ^{-a} to X and Z to Z, this completes the proof. $\square$

Lemma D.12. *For all $a, b \in \mathbb{Z}_d$, there exists $\mathit{dir} \in \mathfrak{C}_2$ such that*

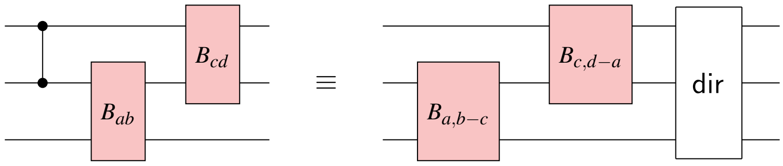


(53)

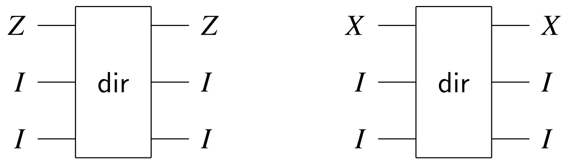
Proof. Firstly, find preimage of $Z \otimes I \otimes I$ in the LHS of (53).



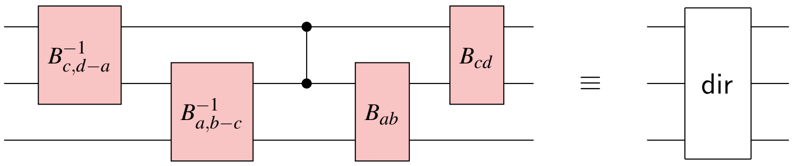
By the uniqueness of the symplectic normal form, $X^cZ^{d-a} \otimes X^aZ^{b-c} \otimes Z$ implies that the updated normal boxes should be the ones in the RHS of (53). Suppose that for some $\mathsf{dir} \in \mathfrak{C}_3$,



By Proposition D.1, it remains to show that



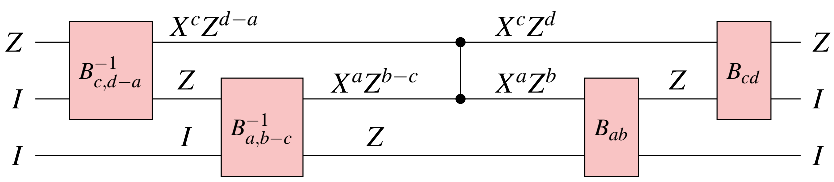
Equivalently, we can compute the preimage of $Z \otimes I \otimes I$ and $X \otimes I \otimes I$ in the LHS of (54).

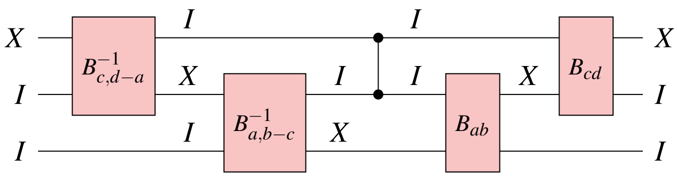


(54)

According to Figure 11,

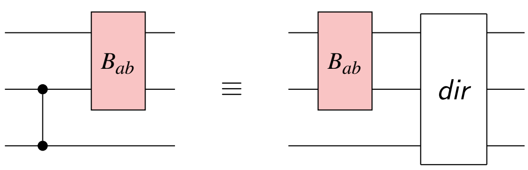
we have





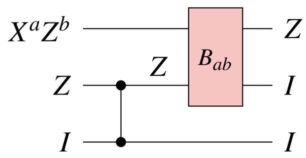
This completes the proof. ☐

Lemma D.13. *For all $a, b \in \mathbb{Z}_d$, there exists $\mathsf{dir} \in \mathfrak{C}_3$ such that*



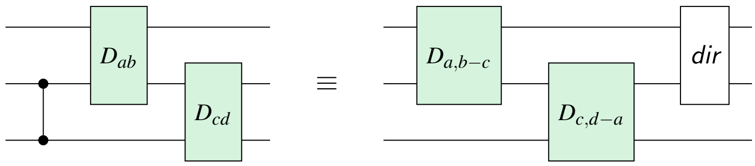
(55)

Proof. Computing the preimage of $Z \otimes I \otimes I$ in the LHS of (55) yields



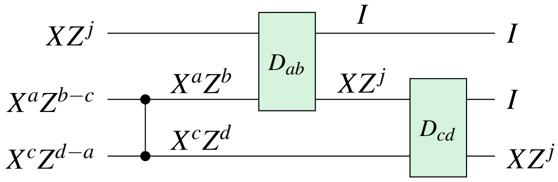
By the uniqueness of the symplectic normal form, $X^aZ^b \otimes Z \otimes I$ implies that the updated normal box should be the one in the RHS of (55). ☐

Lemma D.14. *For all $a, b \in \mathbb{Z}_d$, there exists dir $\in \mathfrak{C}_2$ such that*

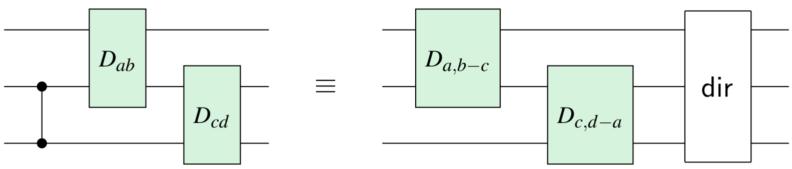


(56)

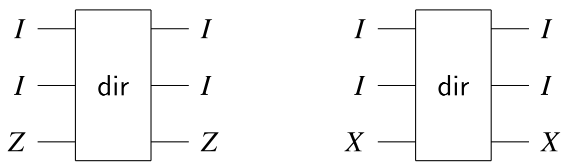
Proof. For any $j \in \mathbb{Z}_d$, computing the preimage of $I \otimes I \otimes XZ^j$ in the LHS of (56) yields



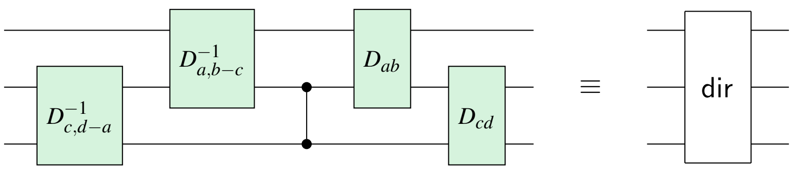
By the uniqueness of the symplectic normal form, $XZ^j \otimes X^aZ^{b-c} \otimes X^cZ^{d-a}$ implies that the updated normal boxes should be the ones in the RHS of (56). Suppose that for some dir $\in \mathfrak{C}_3$,



By Proposition D.1, it remains to show that



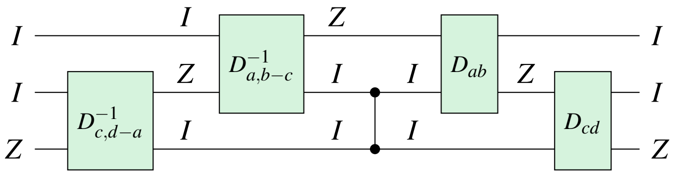
Equivalently, we can compute the preimage of $I \otimes I \otimes Z$ and $I \otimes I \otimes X$ in the LHS of (57).

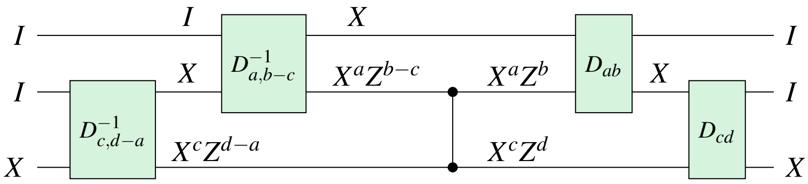


(57)

According to Figure 12,

we have





This completes the proof. $\square$

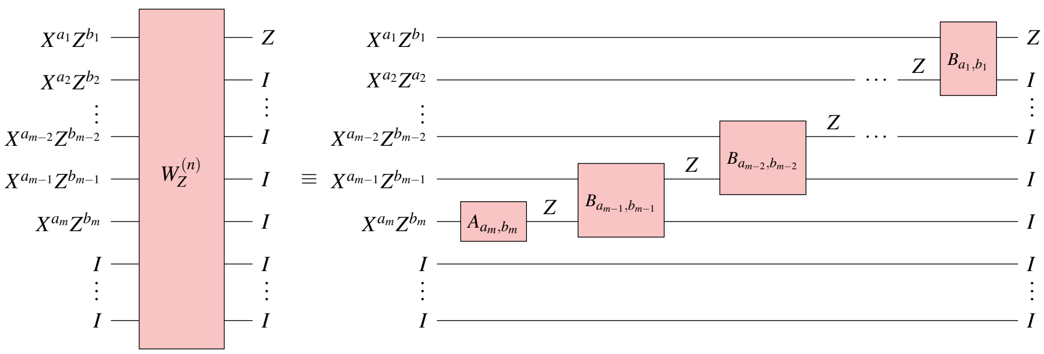
E Proofs from Section 3.1

Lemma 3.4. *For every non-scalar n-qudit Pauli operator $P \in \mathcal{P}_n \setminus \{\omega^t I;\ t \in \mathbb{Z}_d\}$, there exists a unique Z-normal circuit W_Z such that $[\![W_Z]\!]_u \bullet P \equiv_p Z \otimes I \otimes \cdots \otimes I$.*

Proof. There are unique $a_j, b_j \in \mathbb{Z}_d, 1 \leq j \leq n$ such that $P \equiv_p \bigotimes_{j=1}^n X^{a_j} Z^{b_j}$. Since P is not a scalar, there exists j such that $X^{a_j} Z^{b_j}$ is not an identity. Let m be the largest such index, then we can write

$$P \equiv_p \left(X^{a_1} Z^{b_1}\right) \otimes \cdots \otimes \left(X^{a_m} Z^{b_m}\right) \otimes I \otimes \cdots \otimes I. \tag{58}$$

Next, we show how to construct a unique Z-normal circuit W_Z based on (58) and the unique normal box actions specified in the second column of Figure 5. Starting from the qudit wire m, A_{a_m,b_m} is uniquely determined by $X^{a_m} Z^{b_m}$. As a result, the unitary implementation of this A box maps $X^{a_m} Z^{b_m}$ to Z up to a global phase.Consequently, $B_{a_{m-1},b_{m-1}}$ is uniquely determined by $X^{a_{m-1}} Z^{b_{m-1}} \otimes Z$. Then, this B box maps $X^{a_{m-1}} Z^{b_{m-1}} \otimes Z$ to $Z \otimes I$. Continue this process by concatenating B normal boxes upward, until reaching the top two qudit wires, where B_{a_1,b_1} is uniquely determined by $X^{a_1} Z^{b_1} \otimes Z$. As a result, $X^{a_1} Z^{b_1} \otimes Z$ is mapped to $Z \otimes I$. This gives us a Z-normal circuit W_Z that is displayed below, which is uniquely determined by P. Since all the Pauli propagations are tracked up to global phases, we conclude that $[\![W_Z]\!]_u \bullet P \equiv_p Z \otimes I \otimes \cdots \otimes I$.



$\square$

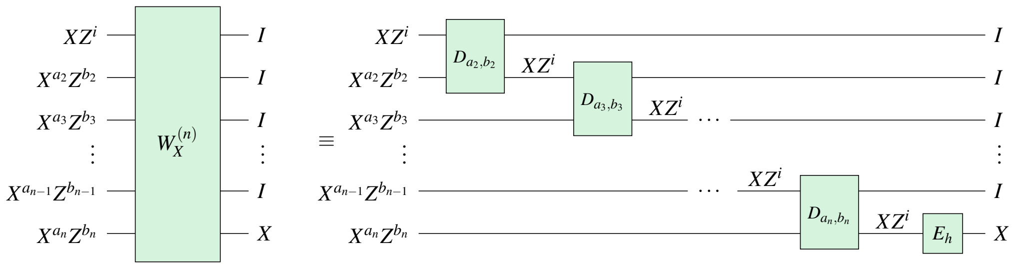
Note that from this proof it is clear that given a Z-normal circuit W_Z, we can read off its preimage of $Z \otimes I \otimes \cdots \otimes I$ by reading the indices of each normal box in W_Z from right to left.

Lemma 3.5. *For every $Q \in \mathcal{P}_n$ such that $(Z \otimes I \otimes \cdots \otimes I)Q = \omega Q(Z \otimes I \otimes \cdots \otimes I)$, there exists a unique X-normal circuit W_X such that $[\![W_X]\!]_u \bullet Q \equiv_p I \otimes \cdots \otimes I \otimes X$.*

Proof. There are unique $a_j, b_j \in \mathbb{Z}_d, 1 \le j \le n$ such that $Q \equiv_p \bigotimes_{j=1}^n X^{a_j} Z^{b_j}$. Since $(Z \otimes I \otimes \cdots \otimes I)Q = \omega Q(Z \otimes I \otimes \cdots \otimes I)$, we must have

$$Q \equiv_p \left(XZ^i\right) \otimes \left(X^{a_2} Z^{b_2}\right) \otimes \cdots \otimes \left(X^{a_n} Z^{b_n}\right). \tag{59}$$

Next, we show how to construct a unique X-normal circuit W_X based on (59) and the unique normal box actions specified in the second column of Figure 5. Starting from qudit wires 1 and 2, D_{a_2,b_2} is uniquely determined by $(XZ^i) \otimes (X^{a_2} Z^{b_2})$. As a result, this D box maps $(XZ^i) \otimes (X^{a_2} Z^{b_2})$ to $I \otimes (XZ^i)$. Consequently, D_{a_3,b_3} is uniquely determined by $(XZ^i) \otimes (X^{a_3} Z^{b_3})$. Again, this D box maps $(XZ^i) \otimes (X^{a_3} Z^{b_3})$ to $I \otimes (XZ^i)$. Continue this process by concatenating D normal boxes downward, until reaching the bottom two qudit wires, where D_{a_n,b_n} is uniquely determined by $(XZ^i) \otimes (X^{a_n} Z^{b_n})$. Thus, $(XZ^i) \otimes (X^{a_n} Z^{b_n})$ is mapped to $I \otimes (XZ^i)$ by this last D box. Then, E_i is uniquely determined by XZ^i, and it maps XZ^i to X. This gives us an X-normal circuit W_X that is displayed below. It is uniquely determined by Q and $[\![W_X]\!]_u \bullet Q \equiv_p I \otimes \cdots \otimes I \otimes X$.

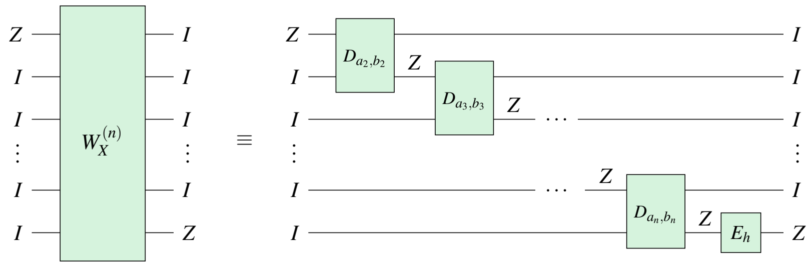


□

Given an X-normal circuit W_X, we can read off its preimage of $I \otimes \cdots \otimes I \otimes X$ by reading the indices of each normal box in W_X from right to left.

Lemma 3.6. *Every X-normal circuit W_X satisfies $[\![W_X]\!]_u \bullet (Z \otimes I \otimes \cdots \otimes I \otimes I) \equiv_p I \otimes I \otimes \cdots \otimes I \otimes Z$.*

Proof. This follows from the additional actions of D and E boxes on certain Pauli operators (see the third column of Figure 5). In particular, every D box maps $Z \otimes I$ to $I \otimes Z$, every E box maps Z to Z. It is visualized below.



□

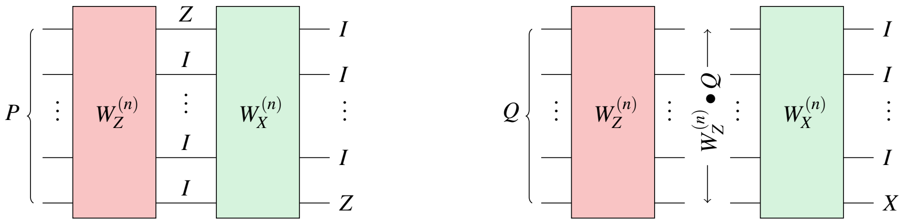
Lemma 3.7. *Let $P, Q \in \mathcal{P}_n$ such that $PQ = \omega QP$. Then there exist unique normal circuits W_Z and W_X such that $[\![W_X W_Z]\!]_u \bullet P \equiv_p I \otimes \cdots \otimes I \otimes Z$ and $[\![W_X W_Z]\!]_u \bullet Q \equiv_p I \otimes \cdots \otimes I \otimes X$.*

Proof. Since $PQ = \omega QP$, we must have $P \notin \langle \omega \rangle$. By Lemma 3.4, there exists a unique Z-normal circuit W_Z such that $[\![W_Z]\!]_u \bullet P \equiv_p Z \otimes I \otimes \cdots \otimes I$. Moreover, $([\![W_Z]\!]_u \bullet P)([\![W_Z]\!]_u \bullet Q) = [\![W_Z]\!]_u \bullet (PQ) = [\![W_Z]\!]_u \bullet (\omega QP) = \omega([\![W_Z]\!]_u \bullet Q)([\![W_Z]\!]_u \bullet P)$. Hence $(Z \otimes I \otimes \cdots \otimes I)([\![W_Z]\!]_u \bullet Q) = \omega([\![W_Z]\!]_u \bullet Q)(Z \otimes I \otimes \cdots \otimes I)$. By Lemma 3.5, there exists a unique X-normal circuit W_X such that $[\![W_X W_Z]\!]_u \bullet Q = [\![W_X]\!]_u \bullet ([\![W_Z]\!]_u \bullet Q) \equiv_p I \otimes \cdots \otimes I \otimes X$.

By Lemma 3.6, $[\![W_X W_Z]\!]_u \bullet P = [\![W_X]\!]_u \bullet ([\![W_Z]\!]_u \bullet P) \equiv_p [\![W_X]\!]_u \bullet (Z \otimes I \otimes \cdots \otimes I) \equiv_p I \otimes \cdots \otimes I \otimes Z$. It follows that

$$[\![W_X W_Z]\!]_u \bullet P \equiv_p I \otimes \cdots \otimes I \otimes Z, \qquad [\![W_X W_Z]\!]_u \bullet Q \equiv_p I \otimes \cdots \otimes I \otimes X. \tag{60}$$

Diagrammatically, we can express this process:



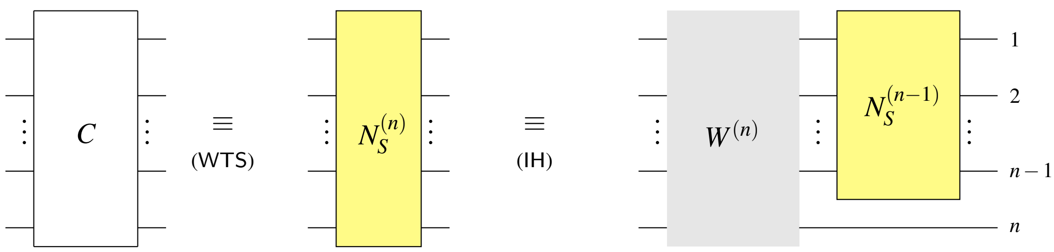
This proves the existence of the desired Z-normal circuit W_Z and X-normal circuit W_X. Now suppose towards contradiction that there exist another Z-normal circuit W'_Z and X-normal circuit W'_X satisfying (60). Since $[\![W'_X]\!]_u \bullet ([\![W'_Z]\!]_u \bullet P) \equiv_p I \otimes \cdots \otimes I \otimes Z$, Lemma 3.6 implies that $[\![W'_X]\!]_u \bullet P \equiv_p Z \otimes I \cdots \otimes I$. By the uniqueness of the Z-normal circuit in Lemma 3.4, $W_Z \equiv W'_Z$. Then

$$[\![W'_X]\!]_u \bullet ([\![W'_Z]\!]_u \bullet Q) \equiv_p [\![W'_X]\!]_u \bullet ([\![W_Z]\!]_u \bullet Q) \equiv_p I \otimes \cdots \otimes I \otimes X.$$

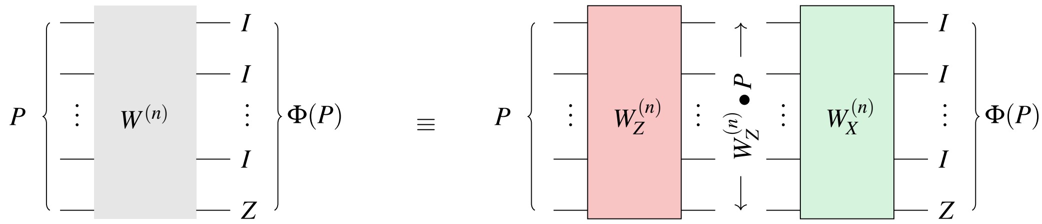
By the uniqueness of the X-normal circuit in Lemma 3.5, $W'_X \equiv W_X$. ☐

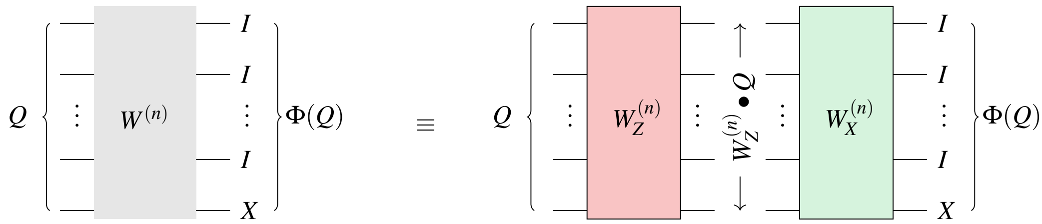
Proposition 3.8. *Let $\phi : \mathcal{P}_n \to \mathcal{P}_n$ be an automorphism of the Pauli group such that ϕ preserves scalars ($\phi(\omega) = \omega$). There exists a unique Clifford circuit $C \in \mathfrak{C}_n$ in symplectic normal form such that for all $P \in \mathcal{P}_n$, $[\![C]\!]_u \bullet P \equiv_p \phi(P)$.*

Proof. We proceed by induction on n. When $n = 0$, the Pauli operators are scalars of the form ω^t, $t \in \mathbb{Z}_d$. In this case, ϕ is the identity. So C must be an empty wire without any gates. Now suppose that our claim is true for $n - 1$ and consider the case of n. First we prove that the symplectic normal circuit $W^{(n)}$ on the n-th layer exists.



Since ϕ is bijective, there exist $P, Q \in \mathcal{P}_n$ such that $\phi(P) = I \otimes \cdots \otimes I \otimes Z$ and $\phi(Q) = I \otimes \cdots \otimes I \otimes X$. That is, $P = \phi^{-1}(I \otimes \cdots \otimes I \otimes Z)$ and $Q = \phi^{-1}(I \otimes \cdots \otimes I \otimes X)$. Then $PQ = \phi^{-1}(I \otimes \ldots \otimes I \otimes ZX)$. Since $I \otimes \ldots \otimes I \otimes Z$ ω-anticommutes with $I \otimes \ldots \otimes I \otimes X$, $PQ = \omega QP$. By Lemma 3.7, there exist a unique X-circuit W_X and a unique Z-circuit W_Z such that $[\![W_X W_Z]\!]_u \bullet P \equiv_p I \otimes \ldots \otimes I \otimes Z = \phi(P)$ and $[\![W_X W_Z]\!]_u \bullet Q \equiv_p I \otimes \ldots \otimes I \otimes X = \phi(Q)$.





Let $\phi' : \mathcal{P}_n \to \mathcal{P}_n$ be the new automorphism defined as

$$\phi'(U) = \phi\left(\llbracket W_X W_Z \rrbracket_u^{-1} \bullet U\right). \tag{61}$$

Then $I \otimes \cdots \otimes I \otimes Z$ and $I \otimes \cdots \otimes I \otimes X$ are fixed points of ϕ', since for $P \in \{X, Z\}$,

$$\begin{aligned}\phi'(I \otimes \cdots \otimes I \otimes P) &= \phi\left(\llbracket W_X W_Z \rrbracket_u^{-1} \bullet (I \otimes \cdots \otimes I \otimes P)\right) \\ &= \phi\left(\llbracket W_X W_Z \rrbracket_u^{-1} \bullet \llbracket W_X W_Z \rrbracket_u \bullet P\right) \equiv_p \phi(P) = I \otimes \cdots \otimes I \otimes P.\end{aligned}$$

For any $R \in \mathcal{P}_{n-1}$, since $R \otimes I$ commutes with $I \otimes \cdots \otimes I \otimes Z$ and $I \otimes \cdots \otimes I \otimes X$, $\phi'(R \otimes I)$ commutes with $\phi'(I \otimes \cdots \otimes I \otimes Z) = I \otimes \cdots \otimes I \otimes Z$ and $\phi'(I \otimes \cdots \otimes I \otimes X) = I \otimes \cdots \otimes I \otimes X$. This implies that $\phi'(R \otimes I) = S \otimes I$ for some $S \in \mathcal{P}_{n-1}$. Then there exists an automorphism $\phi'' : \mathcal{P}_{n-1} \to \mathcal{P}_{n-1}$ such that $\phi''(R) = S$. Since ϕ' fixes $I \otimes \cdots \otimes I \otimes Z$ and $I \otimes \cdots \otimes I \otimes X$,

$$\phi' = \phi'' \otimes I. \tag{62}$$

By the induction hypothesis, for all $R \in \mathcal{P}_{n-1}$, there exists $C' \in \mathfrak{C}_{n-1}$ in normal form such that

$$\llbracket C' \rrbracket_u \bullet R \equiv_p \phi''(R), \tag{63}$$

Then $C = (C' \otimes I) W_X W_Z$ is in normal form. Next, we show that $\llbracket C \rrbracket_u \bullet U \equiv_p \phi(U)$ for all $U \in \mathcal{P}_n$. By (61),

$$\llbracket W_X W_Z \rrbracket_u^{-1} \bullet U = \phi^{-1} \phi'(U). \tag{64}$$

It follows that

$$\begin{aligned}\llbracket C \rrbracket_u \bullet U = \llbracket (C' \otimes I) W_X W_Z \rrbracket_u \bullet U &\overset{(64)}{=\!=} \llbracket C' \otimes I \rrbracket_u \bullet \left(\phi'^{-1}\left(\phi(U)\right)\right) \\ &\overset{(62)}{=\!=} \llbracket C' \otimes I \rrbracket_u \bullet (\phi''^{-1} \otimes I)\left(\phi(U)\right) \overset{(63)}{\equiv_p} \phi(U).\end{aligned}$$

To prove uniqueness, suppose towards contradiction that $D \in \mathfrak{C}_n$ is another Clifford circuit in symplectic normal form such that $\llbracket D \rrbracket_u \bullet U \equiv_p \phi(U)$ for all $U \in \mathcal{P}_n$. By induction,

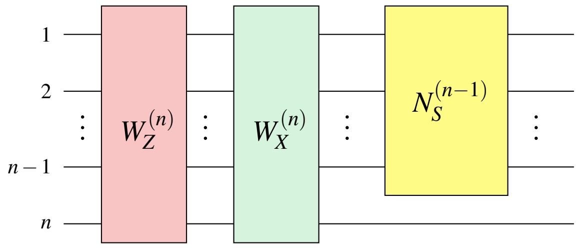
$D = (D' \otimes I)\,(W'_X W'_Z)$, where W'_X is an X-normal circuit, W'_Z is a Z-normal circuit, and D' is a symplectic normal Clifford circuit on $n-1$ qudits. Since $[\![D]\!]_u \bullet P \equiv_p \phi(P) = I \otimes \cdots \otimes I \otimes Z$, $([\![D' \otimes I]\!]_u [\![W'_X W'_Z]\!]_u) \bullet P \equiv_p I \otimes \cdots \otimes I \otimes Z$. Then

$$[\![W'_X W'_Z]\!]_u \bullet P \equiv_p [\![D' \otimes I]\!]_u^{-1} \bullet (I \otimes \cdots \otimes I \otimes Z) = ([\![D']\!]_u^{-1} \otimes I) \bullet (I \otimes \cdots \otimes I \otimes Z) = I \otimes \cdots \otimes I \otimes Z.$$

Similarly, $[\![D]\!]_u \bullet Q \equiv_p I \otimes \cdots \otimes I \otimes X$ implies that $[\![W'_X W'_Z]\!]_u \bullet Q \equiv_p I \otimes \cdots \otimes I \otimes X$. By the uniqueness of the X- and Z-normal circuits in Lemma 3.7, $W'_X \equiv W_X$ and $W'_Z \equiv W_Z$. By induction hypothesis, $C' \equiv D'$. Thus, the same is true for C and D. This completes the proof. $\square$

Lemma 3.9. $|\mathrm{Sp}(2n, \mathbb{Z}_d)| = \prod_{k=1}^{n} \left(d^{2k} - 1\right) \cdot \left(d^{2k-1}\right)$.

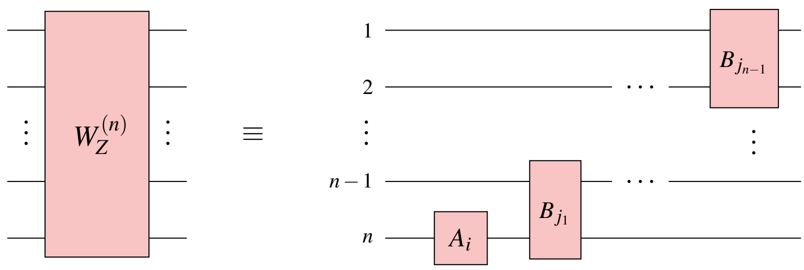
Proof. Since $\mathrm{Sp}(2n, \mathbb{Z}_d) \cong \mathcal{C}_n / \mathcal{P}_n$, by Lemma 4.5 and proposition 3.8, it is sufficient to count the total number of distinct symplectic normal forms. Recall the normal form of an arbitrary n-qudit Clifford operator up to a global base ω^t, $t \in \mathbb{Z}_d$.



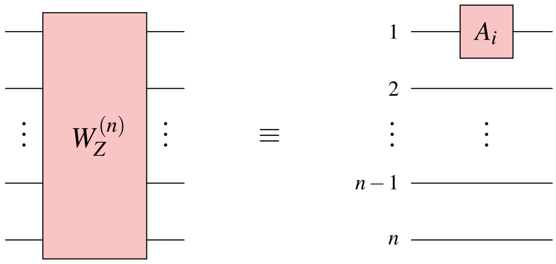
By induction, our problem is reduced to counting different ways of constructing $W_Z^{(n)}$ and $W_X^{(n)}$.

- For $W_Z^{(n)}$, according to Figure 6, there are $d^2 - 1$ choices for an A box. For B boxes, we can start concatenating them upward from any qudit wire. According to Definition 3.1, we proceed by cases.

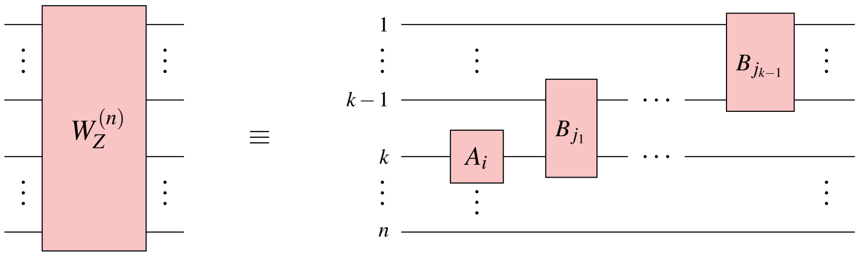
  In one extreme case: The ladder of B boxes starts from the bottom qudit wire, as shown below. There are $n-1$ B boxes in $W_Z^{(n)}$. According to Figure 6, there are d^2 choices for a B box. Since each choice of a normal box is independent of each other, there are $(d^2 - 1) \cdot d^{2(n-1)} = d^{2n} - d^{2(n-1)}$ distinct $W_Z^{(n)}$ when it expands over all n qudits.



  In the other extreme case: The ladder of B boxes starts from the top qudit wire, as shown below. There is no B box in $W_Z^{(n)}$. Since each choice of a normal box is independent of each other, there are $d^2 - 1$ distinct $W_Z^{(n)}$ when it expands over 1 qudit.



In all other remaining cases: The ladder of B boxes starts from qudit wire k, $1 < k < n$. In the illustration below, there are $(k-1)$ B boxes in $W_Z^{(n)}$. Reasoning analogously as before, there are $(d^2-1)\cdot d^{2(k-1)} = d^{2k} - d^{2(k-1)}$ distinct $W_Z^{(n)}$ when it expands over k qudits.



Considering all cases of concatenating B boxes in $W_Z^{(n)}$, the number of distinct $W_Z^{(n)}$ is

$$\sum_{k=1}^{n}\left(d^{2k}-d^{2(k-1)}\right) = \left(\sum_{k=1}^{n} d^{2k}\right) - \left(\sum_{k=1}^{n} d^{2(k-1)}\right) = \frac{\left(d^2-1\right)\left(1-d^{2n}\right)}{1-d^2} = d^{2n}-1.$$

- For $W_X^{(n)}$, we consider all possible ways of concatenating D and E boxes. According to Figure 6, there are p choices for E boxes respectively. For each D box, there are d^2 choices. According to Definition 3.1, there are $n-1$ D boxes in $W_X^{(n)}$. Since each choice of a normal box is independent of each other, there are $p \cdot \left(d^2\right)^{(n-1)} = d^{2n-1}$ distinct $W_X^{(n)}$.

Since the choice of $W_Z^{(n)}$ and $W_X^{(n)}$ is independent of each other, there are $\left(d^{2n}-1\right)\cdot\left(d^{2n-1}\right)$ distinct $W_X^{(n)}W_Z^{(n)}$. According to the symplectic normal form outlined in Definition 3.2, we can calculate the total number of distinct $\widetilde{N^{(n)}}$:

$$\prod_{k=1}^{n}\left(d^{2k}-1\right)\cdot\left(d^{2k-1}\right).$$

Since they are in one-to-one correspondence with the elements in $\mathcal{C}_n/\mathcal{P}_n$, this completes the proof. $\square$

F A Complete Set of Box Relations

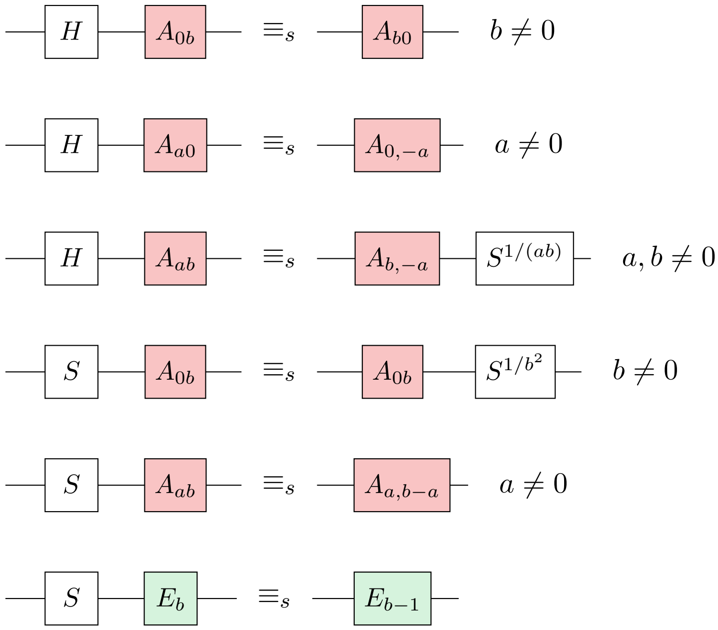


Figure 15: Single-qudit box relations: Up to Pauli correction, pushing H (or S) through an A box, pushing S through an E box.

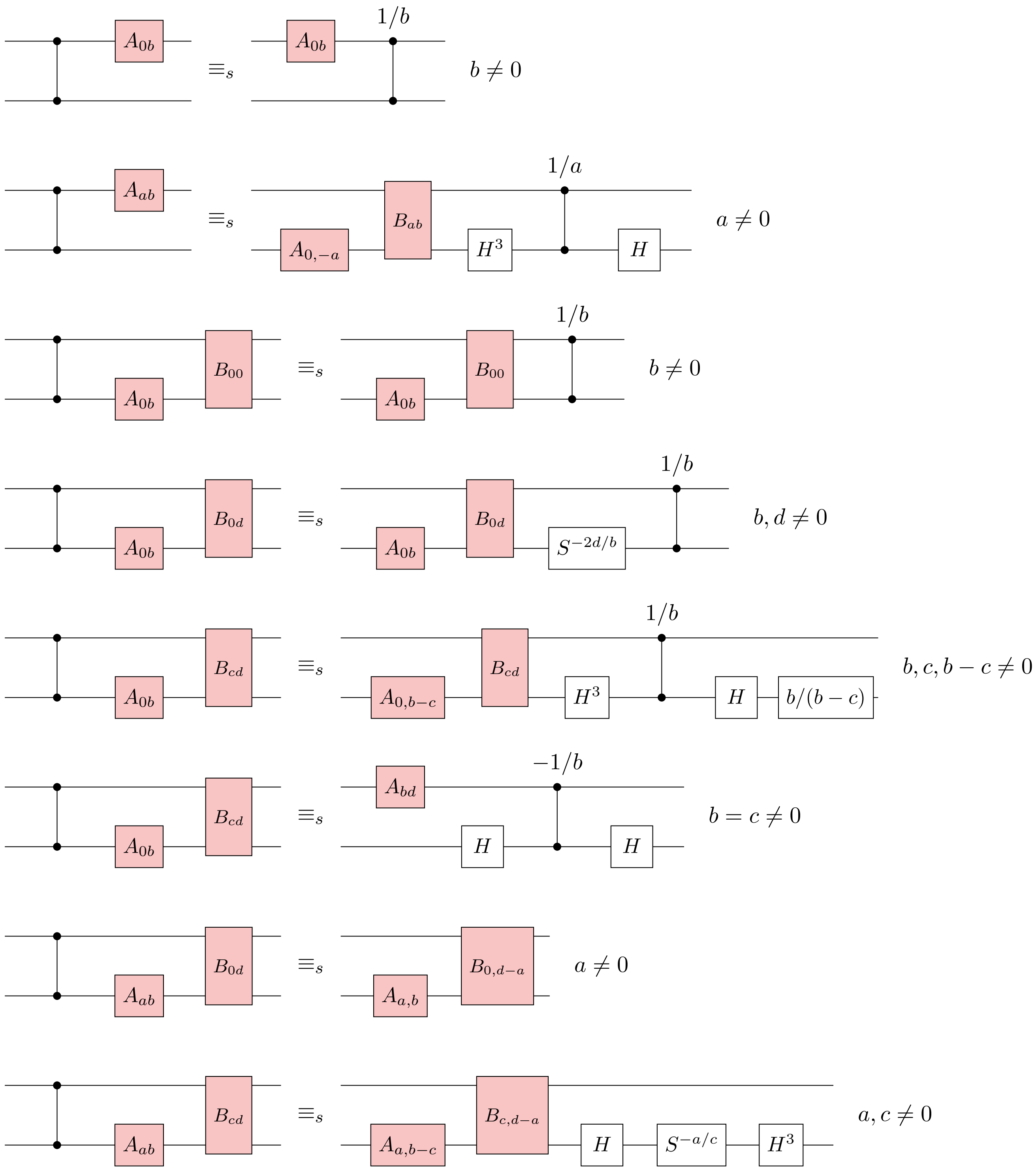


Figure 16: Two-qudit box relations: Up to Pauli correction, pushing CZ through an A box or $A.B$ boxes.

G Clifford Simplification Algorithm

G.1 Automated simplification: single-qudit circuits

We present here an algorithm that automatically simplifies any single-qudit Clifford circuit using just local rewrites. We show that the strategy suffices to prove any single-qudit identity. That is, if a circuit is equal to the identity, then this rewrite strategy rewrites the circuit to the identity. In this section and the following whenever we say we "apply" a lemma or proposition, we mean we apply it from the

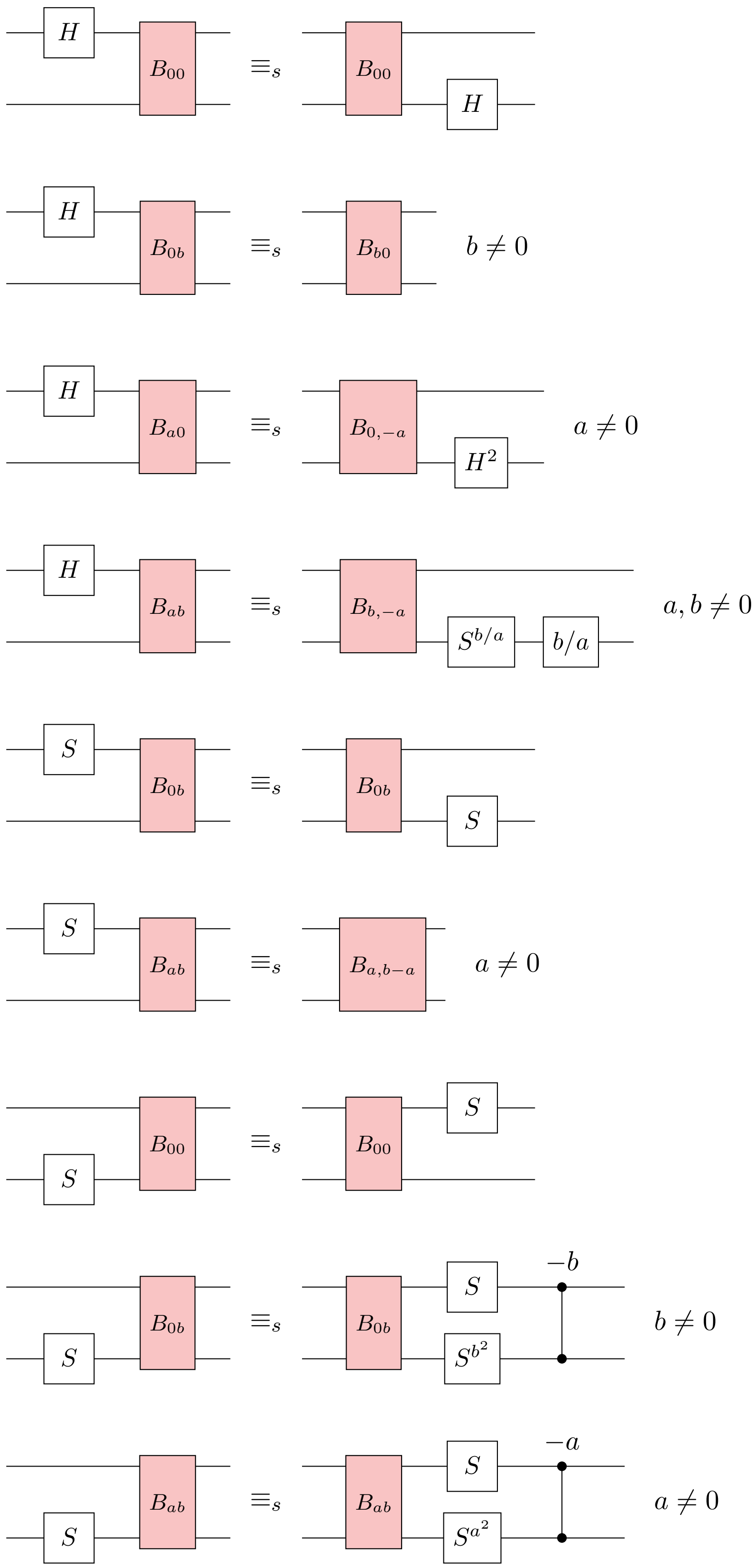


Figure 17: Two-qudit box relations: Up to Pauli correction, pushing H ↑, S ↑, and S ↓ through a B box.

left-hand side to the right-hand side, and when the rewrite "matches" we mean there is a place in the circuit where the left-hand side matches.

First, we can always push Pauli gates to the end of the circuit.

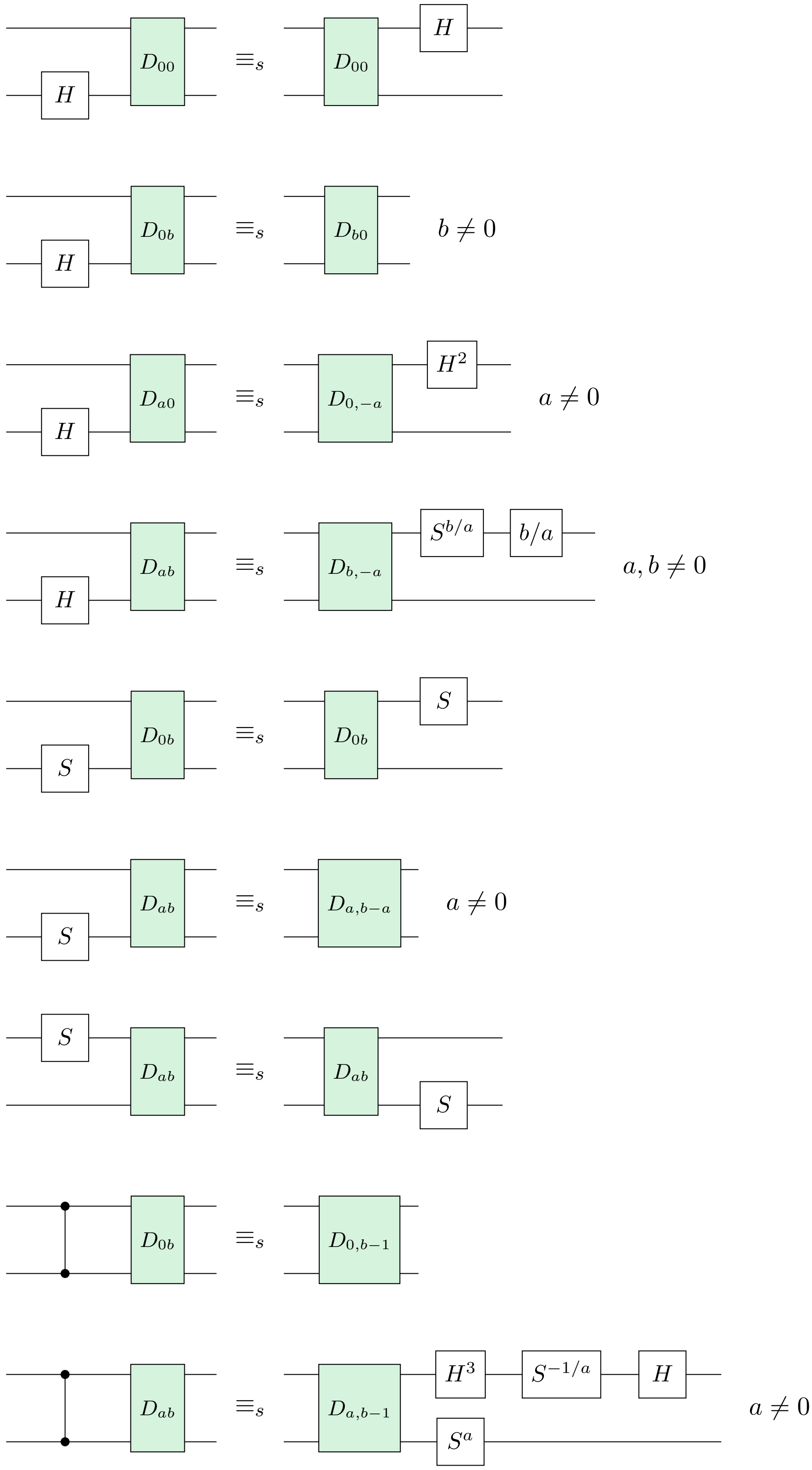


Figure 18: Two-qudit box relations: Up to Pauli correction, pushing H ↓, S ↑, S ↓, and CZ through a D box.

Proposition G.1. *We can push Pauli gates through any generator.*

Second, whenever we have an H^2, we can always push it to the end of the circuit (but just before the Pauli gates).

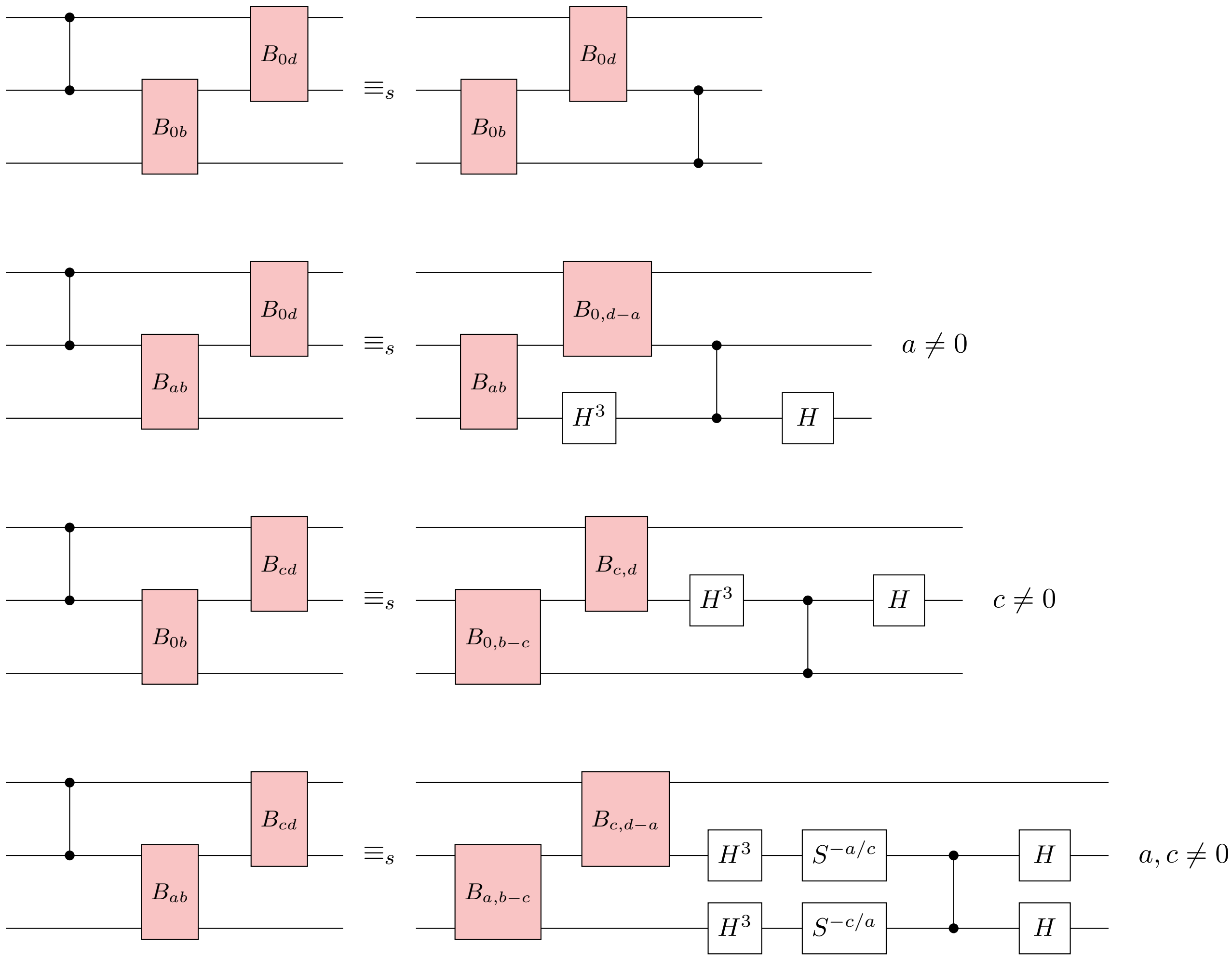


Figure 19: Three-qudit box relations: Up to Pauli correction, pushing CZ ↑ through two B boxes.

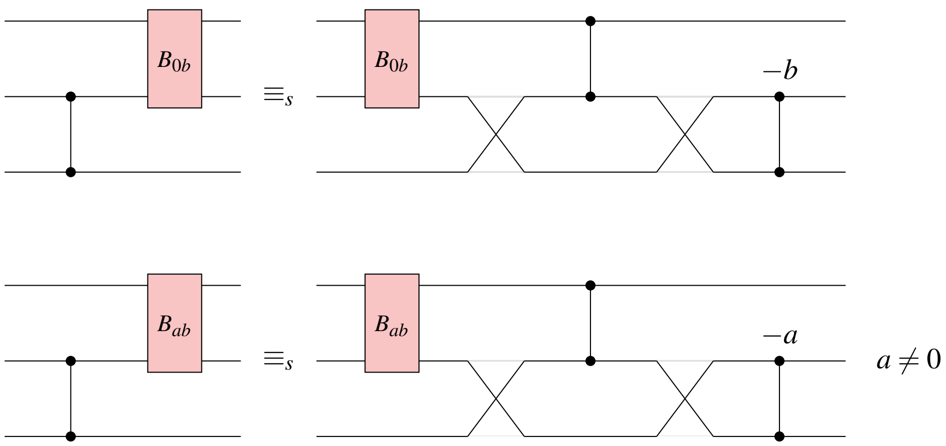


Figure 20: Three-qudit box relations: Up to Pauli correction, pushing CZ ↓ through a B box.

Proposition G.2. *H^2 commutes with H and S (up to a Pauli), and when commuted through CZ^a or CX^a turns them into CZ^{-a}, respectively CX^{-a}.*

For this reason we only need consider single H gates in the middle of a circuit, and never H^3, since we can treat that as HH^2, and just push the H^2 to the end of the circuit.

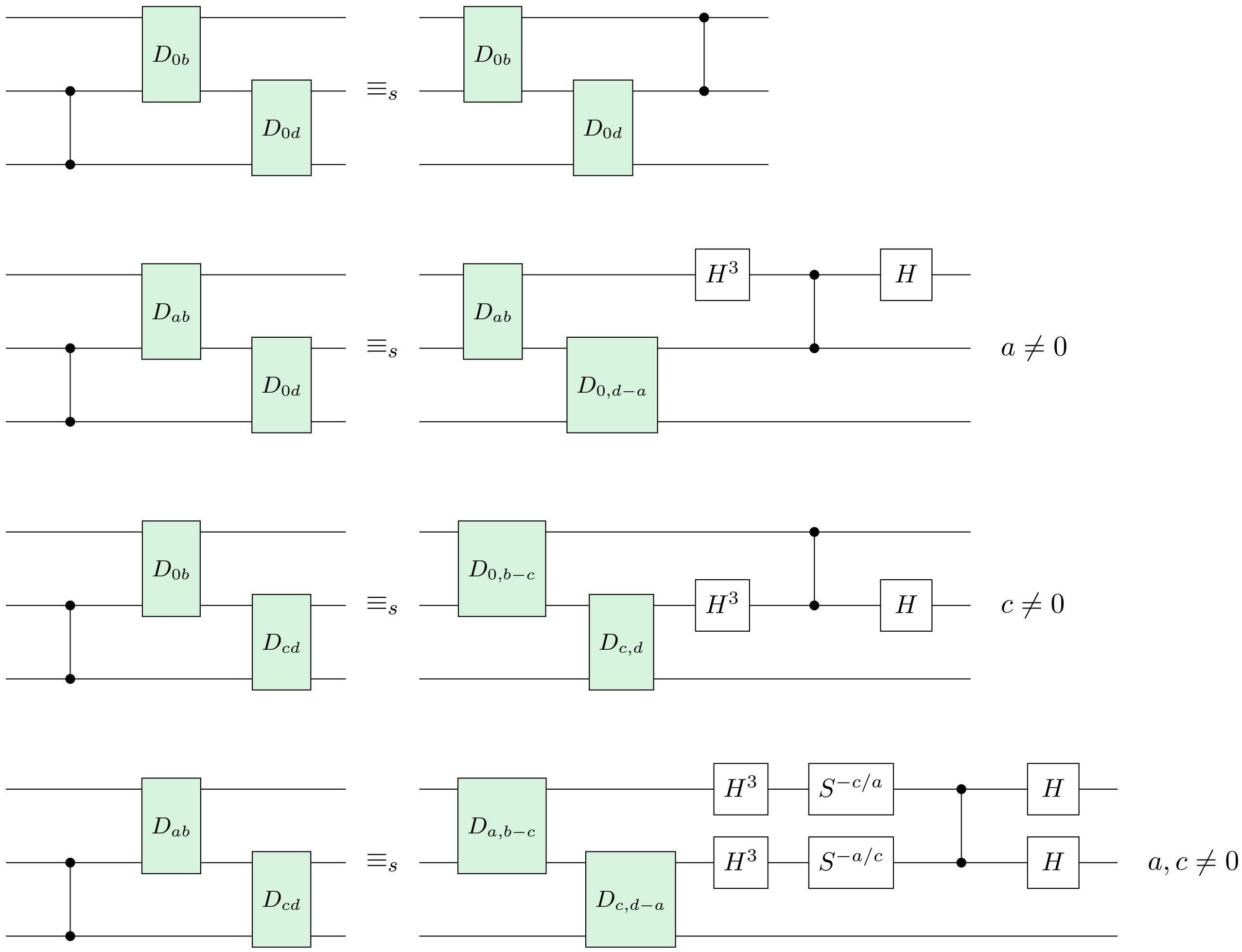


Figure 21: Three-qudit box relations: Up to Pauli correction, pushing CZ $\downarrow$ through two D boxes.

The H^2 is actually a special case of a *multiplier*.

Definition G.3. *Let* $a \in 1, \ldots, p-1$. *We define the* multiplier by a *as the Clifford unitary* M_a *that acts on computational basis states as* $M_a|x\rangle = |ax\rangle$.

It can be easily verified that $M_a^{-1} = M_{a^{-1}}$ where a^{-1} is the inverse of a modulo p, and since it sends computational basis states to computational basis states it must be unitary so that $M_a^\dagger = M_a^{-1}$. It is Clifford since $M_a Z M_{a^{-1}} = Z^{a^{-1}}$ and $M_a X M_{a^{-1}} = X^a$.

Proposition G.4. *The multiplier* M_a *can be pushed through any generator as follows:* $H; M_a = M_{a^{-1}}; H$, $S; M_a = M_a; S^{a^2}$ *(up to a Pauli),* $M_a; CZ = CZ^a; M_a$, *and when put on the target of a CX we have* $M_a; CX = CX^{a^{-1}}; M_a$.

These simple pushing rules mean we can always deal with Paulis, H^2 and multipliers by pushing them to the end of the circuit. It then remains to show how we simplify a sequence of S and H gates. We will do this using what we call Euler decomposition rules.

We have the rule $(S; H)^3 = \mathrm{id}$ up to global phase. By bringing a part of this to the other side of the equality we get $S; H; S = H^{-1}; S^{-1}; H^{-1}$. We call this an **Euler decomposition** rule, as in the qubit case this rule relates to the Euler decomposition of the Hadamard in terms of Z and X rotations. Using the fact that $H^{-1} = HH^2$, and that H^2 can be pushed through an S at the cost of introducing a Pauli Z that we can ignore, we see that this Euler rule can be written as $H; S^{-1}; H = S; H; S$, up to Paulis. Taking the adjoint and getting rid of the H^{-1} we can also write this as $H; S; H = S^{-1}; H^{-1}; S^{-1}$

We can generalize these rules to allow for any power of S between the H gates.

Lemma G.5. *The multiplier can be implemented up to a Pauli by H and S gates as follows:*

$$M_b \;=\; H; S^b; H; S^{b^{-1}}; H; S^b \tag{65}$$

TODO: This might actually be the multiplier for b^{-1}.

Proof. Check by constructing the symplectic matrices on both sides. □

We can then rewrite this to the following equation that generalizes the Euler decomposition:

$$H; S^b; H; \;=\; M_b; S^{-b}; H^{-1}; S^{-b^{-1}} \;=\; S^{-b^{-1}}; H; S^{-b}; M_{-b^{-1}} \tag{66}$$

In our rewrite strategy we apply this rule from left-to-right in our circuit whenever we have a sequence $S^a; H; S^b; H$. We see then that this simplifies to $S^a; S^{-b^{-1}}; H; S^{-b}; M_{-b^{-1}} = S^{a-b^{-1}}; H; S^{-b}; M_{-b^{-1}}$, which has one fewer H gate. The resulting multiplier can be pushed to the end of the circuit. Hence, by applying this rule in this setting, in combination with the other pushing rule, we can always decrease the number of the Hadamard gates in the circuit, until we have at most one. We see then that any single-qudit circuit can be rewritten to one of the following forms:

$$S^a; H; S^b; M_c; Z^d; X^e$$
$$S^a; M_c; Z^d; X^e$$

By inspecting the symplectic matrices it can be checked that these circuits are only equal to the identity matrix when $a = b = c = d = e = 0$, and hence the circuit itself is also equal to the identity circuit. We hence conclude the following.

Proposition G.6. *Given a single-qudit Clifford circuit C that is equal as a linear map to the identity, the above rewrite strategy simplifies it to the empty circuit.*

Any set of rewrites that allow us to prove all the steps in this rewrite strategy will then necessarily be complete.

G.2 Automated simplification: two-qudit circuits

We now extend the previous algorithm to also work for two-qudit circuits. The general strategy of the algorithm is to identify useful derived generators—CX, SWAP, Multipliers—and then try to push these (derived) generators through each other in such a way as to separate out the entangling part of the circuit from the single-qudit gates.

First, we can always push Pauli gates to the end of the circuit.

Proposition G.7. *We can push Pauli gates through any generator.*

Second, whenever we have an H^2, we can always push it to the end of the circuit (but just before the Pauli gates).

Proposition G.8. *H^2 commutes with H and S (up to a Pauli), and when commuted through CZ^a or CX^a turns them into CZ^{-a}, respectively CX^{-a}.*

For this reason we only need consider single H gates in the middle of a circuit, and never H^3, since we can treat that as HH^2, and just push the H^2 to the end of the circuit.

We can do something similar with multipliers.

Proposition G.9. *The multiplier M_a can be pushed through any generator as follows: $H; M_a = M_{a^{-1}}; H$, $S; M_a = M_a; S^{a^2}$ (up to a Pauli), $M_a; CZ = CZ^a; M_a$, and when put on the target of a CX we have $M_a; CX = CX^{a^{-1}}; M_a$.*

We can commute SWAP gates to the end of a circuit (but in the context of the algorithm, to just before the H^2 and Pauli gates).

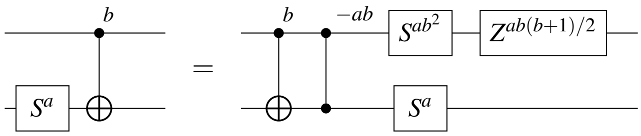
Proposition G.10. *SWAP gates can be pushed through all generators, interchanging the controls and targets of the generators in the obvious ways.*

Lemma G.11. *We can commute S gates through CZ gates and the controls of CX gates.*

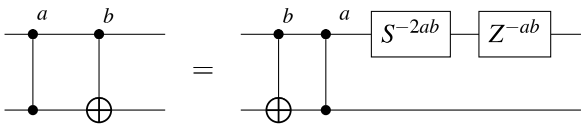
Lemma G.12. *We can push an H through a CZ, turning it into a CX, or through the target of a CX, turning it into a CZ (up to H^2 gates).*

These are the "easy" commutations, as they don't introduce any new gates (beyond Paulis and other gates that can readily be pushed to the end of the circuit) and hence can be treated as "free". In what follows we introduce more complicated rewrites that do introduce new gates, and hence more care must be taken to avoid infinite loops when applied in an automated setting.

Lemma G.13. *Any number of S gates can be pushed through the target of any number of CX gates, producing CZ gates in the process.*

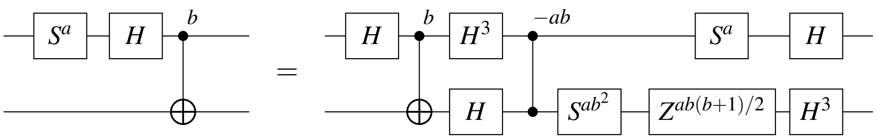


Lemma G.14. *Any number of CZ gates can be pushed through any number of CX gates, producing S gates in the process.*

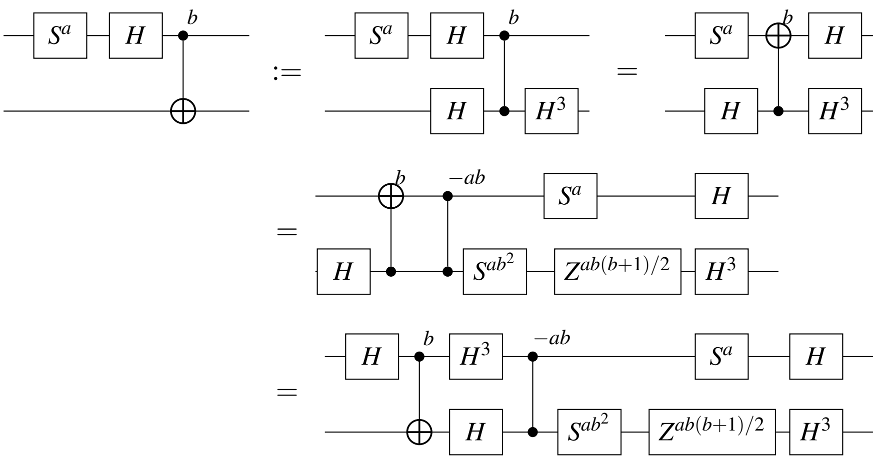


Combining these last four lemmas we see that we can always rewrite a circuit to one where an S gate is never behind a CZ or CX, since it can always be pushed through, a CZ is never directly behind a CX, and an H is never behind a CZ or the target of a CX. Hence, the only place for an S to be "stuck" is behind an H, and the only place for H to be stuck is for the H to be behind another S, or to be on the control wire of a CX. It turns out that in this last case we can also push the S to the other side of the entangling gates.

Lemma G.15. *If some power of S is stuck behind an H, which is stuck behind a power of CX, then the S gates can be pushed past the entangling gate.*



Proof.



□

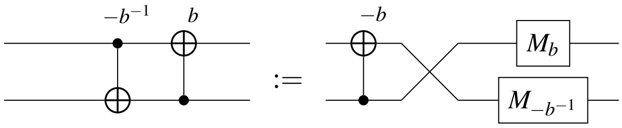
Now consider the rightmost S gate in the circuit that is stuck behind at least one entangling gate, then the H that S is stuck behind cannot be stuck behind another S (as then it wouldn't have been the rightmost S gate), and hence the above rewrite applies.

What this means is that as long as there is one S gate that is to the left of at least one CZ or CX gate, that we can apply one of the above five lemmas to make progress. After none of those lemmas apply any more that means that our circuit looks like an entangling circuit containing only CZ, CX and H gates, and a non-entangling circuit of single-qudit gates and SWAP gates to the right of the entangling part.

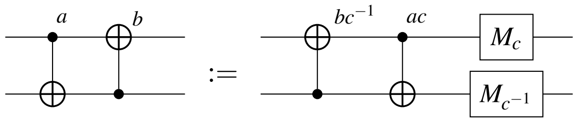
The next steps are to simplify the entangling part even further. Note that each H always appears by itself, never as an H^3. Furthermore, CZ gates can always be pushed to the right of a CX. The entangling part hence consists of a repeating series of blocks, each of which consists of a layer of Hadamards, followed by CX gates, followed by CZ gates. Our goal is to analyse when an H gate can be pushed through blocks, but before we do that we need to simplify the CX part.

Whenever we have CX gates with matching control and target we treat them as a single "unit" CX^a. The only complication then is when we have CX units with opposing controls and targets.

Lemma G.16. *Let $b \in \{1, \ldots, p-1\}$. Two opposing CX gates with repetitions equal to b and $-b^{-1}$ (modulo p) can be changed into a single CX unit followed by a SWAP and multipliers.*



Lemma G.17. *Let $a, b \in \{1, \ldots, p-1\}$ such that $ab + 1 \neq 0$ modulo p. Define $c = ab + 1$. Then we can interchange two CX units with opposing controls and targets at the cost of introducing multipliers.*

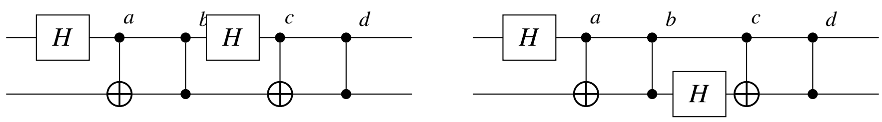


Note that when we have two CX gates with opposing controls and targets that exactly one of these two cases applies, since if $ab + 1 = 0$ modulo p, then $a = -b^{-1}$ so that Lemma G.16 applies. Applying Lemma G.16 directly reduces the number of CX gates, and the resulting SWAP and multipliers can be pushed to the end of the circuit. Lemma G.17 we wish to apply whenever there is a third CX unit directly after the first two. The swapping of the control and target then allows us to merge this third unit into the second one, also reducing the number of distinct CX units.

Proposition G.18. *Any two-qudit CX circuit can be rewritten to a circuit containing at most two distinct CX units, a potential SWAP gate and some multipliers.*

Hence, considering the ways an H can be stuck behind a layer of CX gates we see that we only need to consider the cases where it is stuck behind one CX unit, or two CX units with opposing controls and target. If Lemma G.16 matches, we can reduce the two CX units to just one. otherwise, we know that H is stuck behind the control, so if we then apply Lemma G.17 it will then be stuck behind the target and can be pushed through it using Lemma G.12, transforming the first CX unit into CZs so that it is also stuck behind a single CX unit.

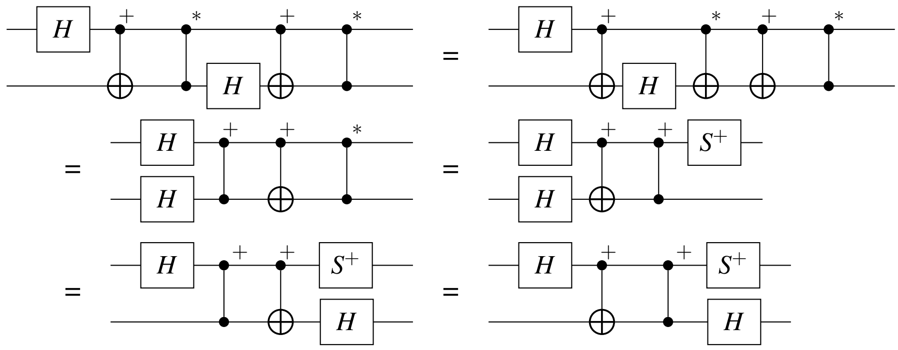
Now suppose there are at least two H gates to the left of at least one entangling gate. Consider then the two rightmost such H gates. We then distinguish two cases depending on whether the H gates are on the same qudit or a different one:



(67)

Here $a, c \neq 0$ and b and d can be arbitrary. We will show that there is a sequence of rewrites that rewrites these circuit fragments to ones where there is at most 1 H gate to the left of an entangling gate. For one of the case distinctions we require at least one more Hadamard to be to the left of this fragment. Hence we then see that if there are 3 Hadamards in the entangling part, that we can always rewrite to a circuit where there is one fewer Hadamard in the entangling part Since the specific number of copies of gates determined by the variables a, b, c, d, and how they update throughout the rewrites is not important for this argument, we will just write the number of copies using a $*$ or $+$ (following regular expression conventions) to denote any number of copies, respectively any non-zero number of copies.

The second case is the easiest, so we do that one first:

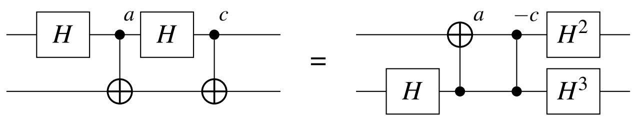


(68)

We see that we end up with the same number of H gates, but now one of them has "escaped" to the entanglement-free portion. Note that in the new pattern we now have a CX gate behind

For the first case we must make a couple of case distinctions. For let's consider the case where there are no CZ gates ($b = d = 0$)

Lemma G.19.

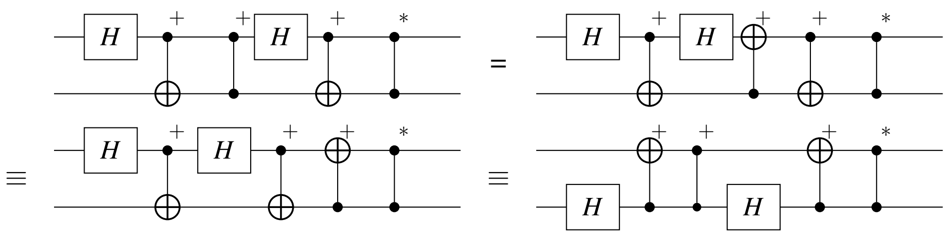


Proof. Transform CX gates back into CZ gates according to their definition, and then reabsorb Hadamards on the other qudit to transform the first one back into a CX with the target on the other qudit. □

We can also prove something similar for when $b = 0$, but $d \neq 0$, and hence only have to consider the case where $b \neq 0$ (d can still be arbitrary).

We can't just yet prove the general case. Instead we will first show that if the two Hadamards appear on the same qudit, that we can choose which qudit they appear on. For these rewrites we will again write $*$ and $+$ to denote that we don't care about the exact number of gates in a CX or CZ unit. In addition we will write $\equiv$ instead of $=$ to denote that we are ignoring potential multipliers, Paulis, H^2 or SWAPs, as these can just be pushed to the end of the circuit to the irrelevant part.

So, note the following:

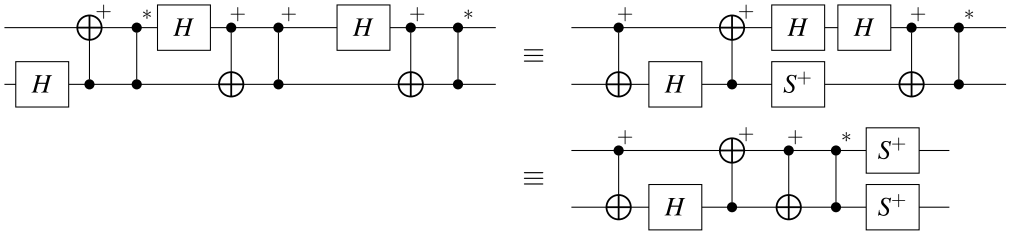


(69)

Note that here we used the rewrite Lemma G.17 which requires the multiples of the gate to satisfy a certain condition. However, if this is not satisfied then we can apply Lemma G.16 instead, in which case the final CX in this equation would have been a SWAP instead, so that we can in fact push out the final Hadamard past the last CZ gate and we are done. Hence, without loss of generality we may assume Lemma G.17 applies, as it is the hardest case.

Now consider the third-to-last Hadamard in the circuit. Either it appears on the same qudit as the last two Hadamards, or it appears on a different qudit. If it appears on the same, we apply Eq. (69)

to change the target of the last two qudits. In both cases we hence can make it so the third-to-last Hadamard appears on a different qudit. We can then use Eq. (68) to simplify the resulting circuit:



(70)

Here in the last step we combined the H gates and pushed them to the end, followed by pushing the S gate past the CX gate.

We hence see that if we have at least 3 H gates in the entangling part, that we can always reduce the number by one. After we are done the entangling part then only contains at most 2 H gates. These must necessarily occur on the same qudit, since otherwise Eq. (68) would apply.

Proposition G.20. *Let C be a two-qudit Clifford circuit that is semantically equal to the identity. Then the above rewrite strategy together with rules sufficing to show single-qudit Clifford completeness rewrites C to the identity circuit.*

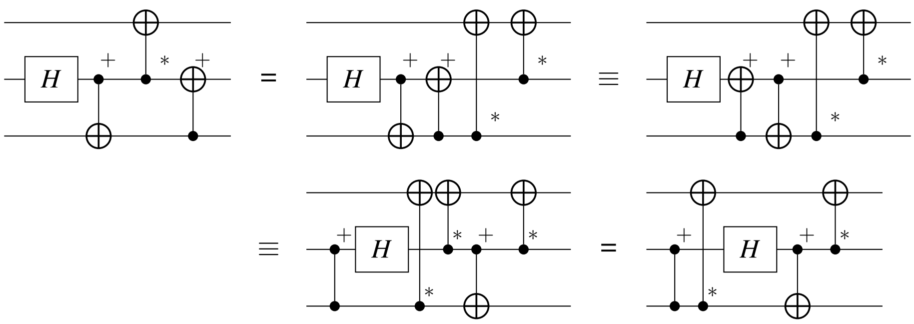
Proof. We follow all the steps, pushing Pauli gates, multipliers, SWAPs and S gates to the end of the circuit out of the entangling portion of the circuit. We then rewrite the entangling portion until it contains at most two H gates. As C implements the identity and the only entangling between the qudits happens in this now small reduced entangling portion of the circuit we can by analysing the possible symplectic matrices of the entangling circuits see that the only way it can not be entangling (which is required since C does not entangle) is for it to not contain any entangling gates at all. We have then reduced the circuit to consists solely of single-qudit gates and SWAPs, the SWAPs must cancel out since C is the identity, and hence the circuit consists only of single-qudit gates which can be rewritten to the identity circuit by assumption of single-qudit completeness. □

To phrase this another way: any set of rewrites that satisfy to show the validity of all the rewrites used in this algorithm is complete for two-qudit circuits.

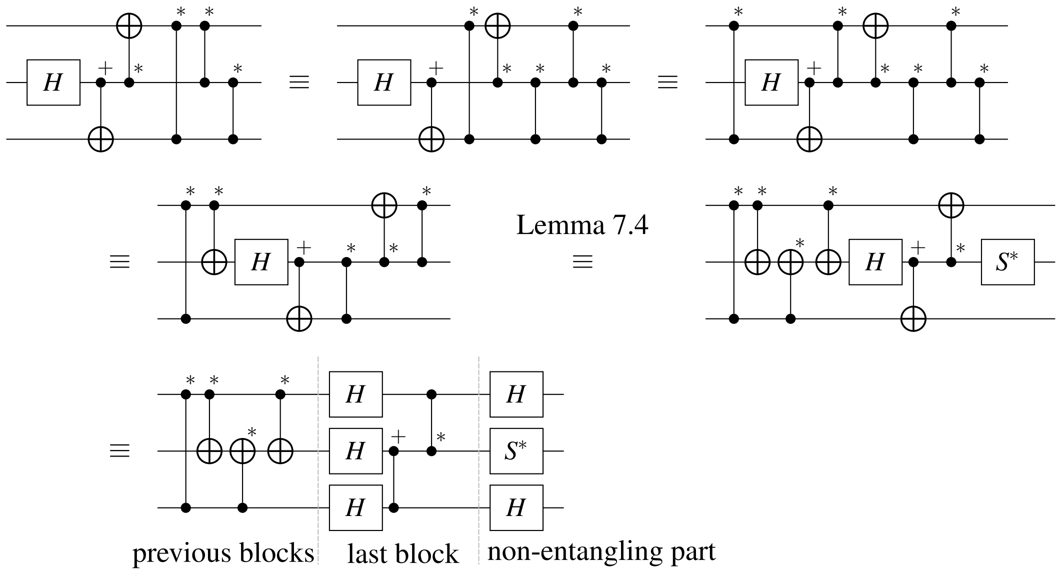
G.3 Automated simplification: three-qudit circuits

We can use a similar strategy to simplify three-qudit circuits and prove completeness there. The first steps are identical, moving all the Pauli gates, multipliers, S gates and SWAPs to the end. We can also still ensure CZ gates always appear after CX gates, though now requiring a couple more cases to deal with the additional qudit. The entangling portion of the circuit that we wish to simplify hence still consists of blocks consisting of Hadamard gates, stuck behind the control of at least one CX gate, followed by some CZ gates.

Our goal will again be to reduce the number of H gates in the entangling portion, but our strategy for this will work somewhat differently. First, let us consider the last H gate stuck behind at least one entangling gate. We may assume there is a CX gate after it, and that the first CX gate after it has the control on the same qudit as the H. Another CX gate we cannot push through H is the other possible CX with the control on the qudit of the H. All other CX gates that can appear after can be pushed through. Let's demonstrate with an example:



The other cases are proven similarly. Hence, while in the two-qudit case we only had to consider a single CX after each H, now we can have two of them. Note that in this process of pushing past the H that we introduced new CX past behind the H. To regularize the pattern, we have to potentially push these past some CZ gates as well, which introduce new S gates. We will show how to deal with those momentarily. Before we do that, we first rewrite this last block containing an H to a different form:



(71)

While this contains more Hadamards now, this more symmetric form allows us to more easily consume gates coming from the left. First, we show that we can push S gates past this last block without increasing the number of entangling units or H gates (this is necessary as S gates will appear because we pushed CX gates to the left which will have to be commuted past CZ gates which introduces S gates in the process).